

SOLIDWORKS API (.NET VBA C#) — Examples (32/50):

- Get_Preselected_Object_Example_VBNET
- Get_Sketch_Contours_Example_VB

SOLIDWORKS API Documentation (converted from CHM)

Get Preselected Object Example (VB.NET)

This example shows how to get a preselected object when a preselection event is fired.

```

'-----
' Preconditions:
' 1. Open a part, assembly, or drawing.
' 2. Click Tools > Options > System Options > top VSTA debugger
'    on macro exit if the check box is selected.
' 3. If a part or assembly document is active, then
'    preselect (mouse over) a face.
'    - or -
'    If a drawing document is active, then preselect
'    an edge.
' 4. Open the Immediate window.
'
' Postconditions:
' 1. When a face is preselected in a part or assembly
'    document, or an edge is pre-selected in a drawing document,
'    then that interface's UserSelectionPreNotify event is fired.
' 2. To stop running the macro, click Debug > Stop Debugging
'    in the SolidWorks Visual Studio Tools for Applications IDE.
' 3. Examine the Immediate window.
' 4. Click Tools > Options > System Options > Stop VSTA debugger
'    on macro exit to select the check box.
'-----
Imports SolidWorks.Interop.sldworks
Imports SolidWorks.Interop.swconst
Imports System
Imports System.Collections
Imports System.Diagnostics

Partial Class SolidWorksMacro

    Public WithEvents pDoc As PartDoc
    Public WithEvents aDoc As AssemblyDoc
    Public WithEvents dDoc As DrawingDoc

    Public Sub main()

        Dim swModel As ModelDoc2
        Dim openPart As Hashtable
        Dim openAssembly As Hashtable
        Dim openDrawing As Hashtable

        swModel = swApp.ActiveDoc

        ' Determine the document type
        ' and set up event handlers
        If swModel.GetType = swDocumentTypes_e.swDocPART Then
            pDoc = swModel
            openPart = New Hashtable
        ElseIf swModel.GetType = swDocumentTypes_e.swDocASSEMBLY Then
            aDoc = swModel
            openAssembly = New Hashtable
        ElseIf swModel.GetType = swDocumentTypes_e.swDocDRAWING Then
            dDoc = swModel
            openDrawing = New Hashtable
        End If
        AttachEventHandlers()

    End Sub

    Sub AttachEventHandlers()
        AttachSWEvents()
    End Sub

    Sub AttachSWEvents()
        If Not pDoc Is Nothing Then
            AddHandler pDoc.UserSelectionPreNotify, AddressOf Me.pDoc_UserSelectionPreNotify
        End If
        If Not aDoc Is Nothing Then
            AddHandler aDoc.UserSelectionPreNotify, AddressOf Me.aDoc_UserSelectionPreNotify
        End If
        If Not dDoc Is Nothing Then
            AddHandler dDoc.UserSelectionPreNotify, AddressOf Me.dDoc_UserSelectionPreNotify
        End If
    End Sub

    Private Function pDoc_UserSelectionPreNotify(ByVal SelectionType As Integer) As Integer
        If SelectionType = swSelectType_e.swSelFACES Then
            Dim swModel As ModelDoc2
            Dim swSelMgr As SelectionMgr
            Dim SelectedObject As Object
            Dim swFace As Face2
            Dim swFeature As Feature

            swModel = swApp.ActiveDoc
            swSelMgr = swModel.SelectionManager
            SelectedObject = swSelMgr.GetPreSelectedObject()
            If SelectionType = swSelectType_e.swSelFACES Then
                swFace = SelectedObject
                swFeature = swFace.GetFeature
                Debug.Print("Name of feature whose face you preselected: " & swFeature.Name)
                Debug.Print("    Mouse over a different face, or click Debug > Stop Debugging to stop running macro")
                Debug.Print(" ")
            End If
        End If
    End Function

```

```

Private Function aDoc_UserSelectionPreNotify(ByVal SelectionType As Integer) As Integer
    If SelectionType = swSelectType_e.swSelFACES Then
        Dim swModel As ModelDoc2
        Dim swSelMgr As SelectionMgr
        Dim SelectedObject As Object
        Dim swFace As Face2
        Dim swFeature As Feature

        swModel = swApp.ActiveDoc
        swSelMgr = swModel.SelectionManager
        SelectedObject = swSelMgr.GetPreSelectedObject()
        If SelectionType = swSelectType_e.swSelFACES Then
            swFace = SelectedObject
            swFeature = swFace.GetFeature
            Debug.Print("Name of feature whose face you preselected: " & swFeature.Name)
            Debug.Print("  Mouse over a different face, or click Debug > Stop Debugging to stop running macro")
            Debug.Print(" ")
        End If
    End If
End Function

Private Function dDoc_UserSelectionPreNotify(ByVal SelectionType As Integer) As Integer
    If SelectionType = swSelectType_e.swSelEDGES Then
        Dim swModel As ModelDoc2
        Dim swSelMgr As SelectionMgr
        Dim SelectedObject As Object
        Dim swEdge As Edge
        Dim swBody As Body2

        swModel = swApp.ActiveDoc
        swSelMgr = swModel.SelectionManager
        SelectedObject = swSelMgr.GetPreSelectedObject()
        If SelectionType = swSelectType_e.swSelEDGES Then
            swEdge = SelectedObject
            swBody = swEdge.GetBody
            Debug.Print("Name of body whose edge you preselected: " & swBody.Name)
            Debug.Print("  Mouse over a different edge, or click Debug > Stop Debugging to stop running macro")
            Debug.Print(" ")
        End If
    End If
End Function

''' <summary>
''' The SldWorks swApp variable is pre-assigned for you.
''' </summary>

Public swApp As SldWorks
End Class

```

Get Preview Bitmap Example (VBA)

This example shows how to get the preview bitmap in a SOLIDWORKS Design document.

```
'-----  
' Preconditions:  
' 1. Verify that the specified part document exists.  
' 2. Open the Immediate window.  
'  
' Postconditions: Examine the Immediate window.  
'-----
```

Option Explicit

Dim swApp As SldWorks.SldWorks

Sub main()

```
    Set swApp = Application.SldWorks  
    Dim p As stdole.StdPicture 'See MSDN for details about the StdPicture object  
    Dim fileName As String
```

```
    fileName = "C:\Users\Public\Documents\SOLIDWORKS\SOLIDWORKS 2018\samples\tutorial\api\centerlink.sldprt"  
    Set p = swApp.GetPreviewBitmap(fileName, "Default")  
    Debug.Print "Handle: " & p.Handle  
    Debug.Print "Height: " & p.Height  
    Debug.Print "Width: " & p.Width  
    Debug.Print "Type (1 = bitmap): " & p.Type
```

End Sub

Get Process ID of SOLIDWORKS Design Session Example (C#)

This example shows how to get and verify the process ID of the current SOLIDWORKS Design session.

```
//-----
// Preconditions:
// 1. Verify that only one SOLIDWORKS Design session is running.
// 2. Open the Immediate window.
//
// Postconditions:
// 1. Gets the process ID of the SOLIDWORKS Design session.
// 2. Examine the Immediate window.
// 3. To verify the process ID, open a Command Prompt window,
//    type TASKLIST, and locate sldworks.exe in the Image Name
//    column and its process ID in the corresponding PID column.
// 4. Exit the Command Prompt window.
//-----
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System.Runtime.InteropServices;
using System;
using System.Diagnostics;

namespace GetProcessIDCSharp.csproj
{
    public partial class SolidWorksMacro
    {
        public void Main()
        {
            // Print the current SOLIDWORKS Design session's PID to the
            // Immediate window
            Debug.Print("SOLIDWORKS Design process ID: " + swApp.GetProcessID());
        }

        /// <summary>
        /// The SldWorks swApp variable is pre-assigned for you.
        /// </summary>
        public SldWorks swApp;
    }
}
```

Get Process ID of SOLIDWORKS Design Session Example (VBA)

This example shows how to get and verify the process ID of the current SOLIDWORKS Design session.

```
'-----  
' Preconditions:  
' 1. Verify that only one SOLIDWORKS Design session is running.  
' 2. Open the Immediate window.  
'  
' Postconditions:  
' 1. Gets the process ID of the SOLIDWORKS Design session.  
' 2. Examine the Immediate window.  
' 3. To verify the process ID, open a Command Prompt window,  
'    type TASKLIST, and locate sldworks.exe in the Image Name  
'    column and its process ID in the corresponding PID column.  
' 4. Exit the Command Prompt window.  
'-----  
  
Dim swApp As SldWorks.SldWorks  
  
Sub main()  
    Set swApp = Application.SldWorks  
    ' Print the current SOLIDWORKS Design session's PID to the  
    ' Immediate window  
    Debug.Print "SOLIDWORKS Design process ID: " & swApp.GetProcessID  
  
End Sub
```

Get Process ID of SOLIDWORKS Design Session Example (VB.NET)

This example shows how to get and verify the process ID of the current SOLIDWORKS Design session.

```
-----
' Preconditions:
' 1. Verify that only one SOLIDWORKS Design session is running.
' 2. Open the Immediate window.
'
' Postconditions:
' 1. Gets the process ID of the SOLIDWORKS Design session.
' 2. Examine the Immediate window.
' 3. To verify the process ID, open a Command Prompt window,
'    type TASKLIST, and locate sldworks.exe in the Image Name
'    column and its process ID in the corresponding PID column.
' 4. Exit the Command Prompt window.
-----
Imports SolidWorks.Interop.sldworks
Imports SolidWorks.Interop.swconst
Imports System.Runtime.InteropServices
Imports System
Imports System.Diagnostics

Partial Class SolidWorksMacro

    Public Sub Main()

        ' Print the current SOLIDWORKS Design session's PID to the
        ' Immediate window
        Debug.Print("SOLIDWORKS Design process ID: " & swApp.GetProcessID)

    End Sub

    ''' <summary>
    ''' The SldWorks swApp variable is pre-assigned for you.
    ''' </summary>
    Public swApp As SldWorks

End Class
```

Get Projected Curve Example (VBA)

This example shows how to bidirectionally project a sketch onto a face.

```

'-----
' Preconditions: Open the Immediate window.
'
' Postconditions:
' 1. Creates a thin extrusion of a circle.
' 2. Sketches a corner rectangle on the Top Plane.
' 3. Selects the sketch and the inner face of the extrusion.
' 4. Bidirectionally projects the sketch onto the inner face, creating Curve1.
' 5. Examine the FeatureManager design tree and the graphics area.
'-----

Dim swApp As SldWorks.SldWorks
Dim Part As SldWorks.ModelDoc2
Dim swPart As PartDoc
Dim swSheetWidth As Double
Dim swSheetHeight As Double
Dim defTemplate As String
Dim skSegment As SldWorks.SketchSegment
Dim myFeature As SldWorks.Feature
Dim swFeat As SldWorks.Feature
Dim swFeatMgr As SldWorks.FeatureManager
Dim swFeatData As SldWorks.ProjectionCurveFeatureData
Dim vSkLines As Variant
Dim boolstatus As Boolean
Dim longstatus As Long, longwarnings As Long
Option Explicit
Sub main()

    Set swApp = Application.SldWorks

    swSheetWidth = 0
    swSheetHeight = 0

    defTemplate = swApp.GetUserPreferenceStringValue(swDefaultTemplatePart)

    Set Part = swApp.NewDocument(defTemplate, 0, swSheetWidth, swSheetHeight)
    Set swPart = Part
    swApp.ActivateDoc2 "Part1", False, longstatus
    Set Part = swApp.ActiveDoc

    Part.SketchManager.InsertSketch True
    Part.ClearSelection2 True

    Set skSegment = Part.SketchManager.CreateCircle(0#, 0#, 0#, 0#, 0.075937, 0#)

    boolstatus = Part.Extension.SelectByID2("Arc1", "SKETCHSEGMENT", 0, 0, 0, False, 0, Nothing, 0)

    Set myFeature = Part.FeatureManager.FeatureExtrusionThin2(True, False, False, 0, 0, 0.05, 0.01, False, False, False, False,
1.74532925199433E-02, 1.74532925199433E-02, False, False, False, False, True, 0.01, 0.01, 0.01, 0, 0, False, 0.005, True, True, 0,
0, False)
    Part.SelectionManager.EnableContourSelection = False
    boolstatus = Part.Extension.SelectByID2("Top Plane", "PLANE", 0, 0, 0, False, 0, Nothing, 0)
    Part.SketchManager.InsertSketch True
    boolstatus = Part.Extension.SelectByID2("Sketch2", "SKETCH", 0, 0, 0, False, 0, Nothing, 0)
    Part.ClearSelection2 True
    Set skSegment = Part.SketchManager.CreateLine(-0.015062, -0.011566, 0#, -0.020354, -0.032124, 0#)
    Part.SetPickMode
    Part.ClearSelection2 True
    vSkLines = Part.SketchManager.CreateCornerRectangle(0, -1.27873930746226E-02, 0, 9.77008554030195E-03, -3.21240207064702E-02,
0)
    Part.SetPickMode
    Part.ClearSelection2 True
    Part.SketchManager.InsertSketch True
    Part.ClearSelection2 True

    boolstatus = Part.Extension.SelectByID2("Sketch2", "SKETCH", 4.47795587263838E-03, -1.27873930746226E-02, 0, True, 2, Nothing,
0)
    boolstatus = Part.Extension.SelectByRay(8.5488248477642E-03, -0.075454504318941, 2.93545895954139E-02, 0, -0.707106781186541,
0.707106781186554, 6.92047725771388E-04, 2, True, 1, 0)

    Set swFeatMgr = Part.FeatureManager

    Set swFeatData = swFeatMgr.CreateDefinition(swFmRefCurve)
    swFeatData.Bidirectional = True
    swFeatData.Reverse = False
    Set swFeat = swFeatMgr.CreateFeature(swFeatData)
    Part.ClearSelection2 True

End Sub

```

Get Projected Curve Feature Data Example (C#)

This example shows how to get data for a projected curve feature.

```
//-----
// Preconditions:
// 1. Verify that the part document exists.
// 2. Open the Immediate window.
//
// Postconditions:
// 1. Opens the specified part document.
// 2. Selects a face and sketches a spline on that face.
// 3. Selects the sketch of the spline and a face.
// 4. Inserts a projected curve feature.
// 5. Gets some projected curve feature data and prints it
//    to the Immediate window.
// 6. Examines the Immediate window, FeatureManager design tree, and
//    the graphics area.
//
// NOTE: Because the part is used elsewhere, do not save changes.
//-----
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System.Runtime.InteropServices;
using System;
using System.Diagnostics;

namespace ProjectCurveCSharp.csproj
{
    public partial class SolidWorksMacro
    {
        public void Main()
        {
            ModelDoc2 swModel = default(ModelDoc2);
            ModelDocExtension swModelDocExt = default(ModelDocExtension);
            SketchManager swSketchManager = default(SketchManager);
            SketchSegment swSketchSegment = default(SketchSegment);
            Feature swFeature = default(Feature);
            SelectionMgr swSelectionManager = default(SelectionMgr);
            ProjectionCurveFeatureData swProjectionCurveFeatureData = default(ProjectionCurveFeatureData);
            Sketch swSketch = default(Sketch);
            object pointArray = null;
            double[] points = new double[11];
            bool status = false;
            int errors = 0;
            int warnings = 0;
            string fileName = null;

            //Open part document
            fileName = "C:\\Users\\Public\\Documents\\SOLIDWORKS\\SOLIDWORKS 2018\\samples\\tutorial\\api\\block20.sldprt";
            swModel = (ModelDoc2)swApp.OpenDoc6(fileName, (int)swDocumentTypes_e.swDocPART, (int)swOpenDocOptions_e.swOpenDocOptions_Silent, "", ref error
s, ref warnings);

            //Sketch a spline on the selected face
            swModelDocExt = (ModelDocExtension)swModel.Extension;
            status = swModelDocExt.SelectByID2("", "FACE", -0.0499223104334874, 0.0396239999998897, 0.00738137362270663, false, 0, null, 0);
            swSketchManager = (SketchManager)swModel.SketchManager;
            swSketchManager.InsertSketch(true);
            swModel.ClearSelection2(true);
            points[0] = -0.0624778997860176;
            points[1] = 0.00729572078180673;
            points[2] = 0;
            points[3] = -0.0364588790258153;
            points[4] = 0.0324586288177215;
            points[5] = 0;
            points[6] = 0.0104252377344665;
            points[7] = 0.0140473535914225;
            points[8] = 0;
            points[9] = 0.0646002912861796;
            points[10] = 0.0100590221094308;
            pointArray = points;
            swSketchSegment = (SketchSegment)swSketchManager.CreateSpline2((pointArray), false);
            swSketchManager.InsertSketch(true);
            swModel.ClearSelection2(true);

            //Insert projected curve
            status = swModelDocExt.SelectByID2("Sketch2", "SKETCH", 0, 0, 0, true, 0, null, 0);
            status = swModelDocExt.SelectByID2("", "FACE", -0.0497146993259321, 0, -0.0256283866693252, true, 0, null, 0);
            swFeature = (Feature)swModel.InsertProjectedSketch2(1);

            //Get projected curve data
            status = swModelDocExt.SelectByID2("Curve1", "REFCURVE", 0, 0, 0, false, 0, null, 0);
            swSelectionManager = (SelectionMgr)swModel.SelectionManager;
            swFeature = (Feature)swSelectionManager.GetSelectedObject6(1, -1);
            swProjectionCurveFeatureData = (ProjectionCurveFeatureData)swFeature.GetDefinition();
            swProjectionCurveFeatureData.AccessSelections(swModel, null);
            Debug.Print("Is reversed = " + swProjectionCurveFeatureData.Reverse);
            Debug.Print("Number of targeted faces = " + swProjectionCurveFeatureData.GetFaceArrayCount());
            swFeature = (Feature)swProjectionCurveFeatureData.Sketch;
            swSketch = (Sketch)swFeature.GetSpecificFeature2();
            Debug.Print("Name of sketch = " + swFeature.Name);
            swProjectionCurveFeatureData.ReleaseSelectionAccess();
        }
    }

    /// <summary>
    /// The SldWorks swApp variable is pre-assigned for you.
    /// </summary>
    public SldWorks swApp;
}
```

Get Projected Curve Feature Data Example (VBA)

This example shows how to get data for a projected curve feature.

```

'-----
' Preconditions:
' 1. Verify that the part document exists.
' 2. Open the Immediate window.
'
' Postconditions:
' 1. Opens the specified part document.
' 2. Selects a face and sketches a spline on that face.
' 3. Selects the sketch of the spline and a face.
' 4. Inserts a projected curve feature.
' 5. Gets some projected curve feature data and prints it
'    to the Immediate window.
' 6. Examines the Immediate window, FeatureManager design tree, and
'    the graphics area.
'
' NOTE: Because the part is used elsewhere, do not save changes.
'-----
Option Explicit

Dim swApp As SldWorks.SldWorks
Dim swModel As SldWorks.ModelDoc2
Dim swModelDocExt As SldWorks.ModelDocExtension
Dim swSketchManager As SldWorks.SketchManager
Dim swSketchSegment As SldWorks.SketchSegment
Dim swFeature As SldWorks.Feature
Dim swSelectionManager As SldWorks.SelectionMgr
Dim swProjectionCurveFeatureData As SldWorks.ProjectionCurveFeatureData
Dim swSketch As SldWorks.Sketch
Dim pointArray As Variant
Dim points(10) As Double
Dim status As Boolean
Dim errors As Long
Dim warnings As Long
Dim fileName As String

Sub main()

    Set swApp = Application.SldWorks

    'Open part document
    fileName = "C:\Users\Public\Documents\SOLIDWORKS\SOLIDWORKS 2018\samples\tutorial\api\block20.sldprt"
    Set swModel = swApp.OpenDoc6(fileName, swDocumentTypes_e.swDocPART, swOpenDocOptions_e.swOpenDocOptions_Silent, "", errors, warnings)

    'Sketch a spline on the selected face
    Set swModelDocExt = swModel.Extension
    status = swModelDocExt.SelectByID("", "FACE", -4.99223104334874E-02, 3.9623999998897E-02, 7.38137362270663E-03, False, 0, Nothing, 0)
    Set swSketchManager = swModel.SketchManager
    swSketchManager.InsertSketch True
    swModel.ClearSelection2 True
    points(0) = -6.24778997860176E-02
    points(1) = 7.29572078180673E-03
    points(2) = 0
    points(3) = -3.64588790258153E-02
    points(4) = 3.24586288177215E-02
    points(5) = 0
    points(6) = 1.04252377344665E-02
    points(7) = 1.40473535914225E-02
    points(8) = 0
    points(9) = 6.46002912861796E-02
    points(10) = 1.00590221094308E-02
    pointArray = points
    Set swSketchSegment = swSketchManager.CreateSpline2((pointArray), False)
    swSketchManager.InsertSketch True

    swModel.ClearSelection2 True

    'Insert projected curve
    status = swModelDocExt.SelectByID("Sketch2", "SKETCH", 0, 0, 0, True, 0, Nothing, 0)
    status = swModelDocExt.SelectByID("", "FACE", -4.97146993259321E-02, 0, -2.56283866693252E-02, True, 0, Nothing, 0)
    Set swFeature = swModel.InsertProjectedSketch2(1)

    'Get projected curve data
    status = swModelDocExt.SelectByID("Curve1", "REFCURVE", 0, 0, 0, False, 0, Nothing, 0)
    Set swSelectionManager = swModel.SelectionManager
    Set swFeature = swSelectionManager.GetSelectedObject6(1, -1)
    Set swProjectionCurveFeatureData = swFeature.GetDefinition
    swProjectionCurveFeatureData.AccessSelections swModel, Nothing
    Debug.Print "Is reversed = " & swProjectionCurveFeatureData.Reverse
    Debug.Print "Number of targeted faces = " & swProjectionCurveFeatureData.GetFaceArrayCount
    Set swFeature = swProjectionCurveFeatureData.Sketch
    Set swSketch = swFeature.GetSpecificFeature2
    Debug.Print "Name of sketch = " & swFeature.Name
    swProjectionCurveFeatureData.ReleaseSelectionAccess

End Sub

```

Get Projected Curve Feature Data Example (VB.NET)

This example shows how to get data for a projected curve feature.

```

'-----
' Preconditions:
' 1. Verify that the part document exists.
' 2. Open the Immediate window.
'
' Postconditions:
' 1. Opens the specified part document.
' 2. Selects a face and sketches a spline on that face.
' 3. Selects the sketch of the spline and a face.
' 4. Inserts a projected curve feature.
' 5. Gets some projected curve feature data and prints it
'    to the Immediate window.
' 6. Examine the Immediate window, FeatureManager design tree, and
'    the graphics area.
'
' NOTE: Because the part is used elsewhere, do not save changes.
'-----
Imports SolidWorks.Interop.sldworks
Imports SolidWorks.Interop.swconst
Imports System.Runtime.InteropServices
Imports System
Imports System.Diagnostics

Partial Class SolidWorksMacro

    Public Sub main()

        Dim swModel As ModelDoc2
        Dim swModelDocExt As ModelDocExtension
        Dim swSketchManager As SketchManager
        Dim swSketchSegment As SketchSegment
        Dim swFeature As Feature
        Dim swSelectionManager As SelectionMgr
        Dim swProjectionCurveFeatureData As ProjectionCurveFeatureData
        Dim swSketch As Sketch
        Dim pointArray As Object
        Dim points(10) As Double
        Dim status As Boolean
        Dim errors As Integer
        Dim warnings As Integer
        Dim fileName As String

        'Open part document
        fileName = "C:\Users\Public\Documents\SOLIDWORKS\SOLIDWORKS 2018\samples\tutorial\api\block20.sldprt"
        swModel = swApp.OpenDoc6(fileName, swDocumentTypes_e.swDocPART, swOpenDocOptions_e.swOpenDocOptions_Silent, "", errors, warnings)

        'Sketch a spline on the selected face
        swModelDocExt = swModel.Extension
        status = swModelDocExt.SelectByID("", "FACE", -0.0499223104334874, 0.039623999999897, 0.00738137362270663, False, 0, Nothing, 0)
        swSketchManager = swModel.SketchManager
        swSketchManager.InsertSketch(True)
        swModel.ClearSelection2(True)
        points(0) = -0.0624778997860176
        points(1) = 0.00729572078180673
        points(2) = 0
        points(3) = -0.0364588790258153
        points(4) = 0.0324586288177215
        points(5) = 0
        points(6) = 0.0104252377344665
        points(7) = 0.0140473535914225
        points(8) = 0
        points(9) = 0.0646002912861796
        points(10) = 0.0100590221094308
        pointArray = points
        swSketchSegment = swSketchManager.CreateSpline2((pointArray), False)
        swSketchManager.InsertSketch(True)
        swModel.ClearSelection2(True)

        'Insert projected curve
        status = swModelDocExt.SelectByID("Sketch2", "SKETCH", 0, 0, 0, True, 0, Nothing, 0)
        status = swModelDocExt.SelectByID("", "FACE", -0.0497146993259321, 0, -0.0256283866693252, True, 0, Nothing, 0)
        swFeature = swModel.InsertProjectedSketch2(1)

        'Get projected curve data
        status = swModelDocExt.SelectByID("Curve1", "REFCURVE", 0, 0, 0, False, 0, Nothing, 0)
        swSelectionManager = swModel.SelectionManager
        swFeature = swSelectionManager.GetSelectedObject6(1, -1)
        swProjectionCurveFeatureData = swFeature.GetDefinition
        swProjectionCurveFeatureData.AccessSelections(swModel, Nothing)
        Debug.Print("Is reversed = " & swProjectionCurveFeatureData.Reverse)
        Debug.Print("Number of targeted faces = " & swProjectionCurveFeatureData.GetFaceArrayCount)
        swFeature = swProjectionCurveFeatureData.Sketch
        swSketch = swFeature.GetSpecificFeature2
        Debug.Print("Name of sketch = " & swFeature.Name)
        swProjectionCurveFeatureData.ReleaseSelectionAccess()

    End Sub

    ''' <summary>
    ''' The SldWorks swApp variable is pre-assigned for you.
    ''' </summary>
    Public swApp As SldWorks

End Class

```

End Class

Get Projected Point Example (VBA)

This example shows how to get the point where an input point is projected on to a face and surface in the specified direction.

```

'-----
' Preconditions:
' 1. Open a part.
' 2. Select a face on the part.
'
' Postconditions:
' 1. Returns the point where the input point
'    is projected onto a face or surface, if
'    the input point is projected.
' 2. Examine the Immediate window.
'-----
Option Explicit

Sub main()

    Dim swApp As SldWorks.SldWorks
    Dim myModel As SldWorks.ModelDoc2
    Dim mathUtils As SldWorks.MathUtility

    Set swApp = Application.SldWorks
    Set myModel = swApp.ActiveDoc
    Set mathUtils = swApp.GetMathUtility()

    ' Preselect the face with which to work
    Dim mySelMgr As SldWorks.SelectionMgr
    Dim selObj As Object
    Dim faceToUse As SldWorks.Face2
    Dim surfaceToUse As SldWorks.Surface
    Dim selCount As Long
    Dim selType As Long

    Set mySelMgr = myModel.SelectionManager
    selCount = mySelMgr.GetSelectedObjectCount2(0)
    If (selCount > 0) Then
        selType = mySelMgr.GetSelectedObjectType3(1, 0)
        Set selObj = mySelMgr.GetSelectedObject6(1, 0)
        If (selType = SwConst.swSelFACES) Then
            Set faceToUse = selObj
        End If
    End If

    ' Do the ray intersection operation
    Dim basePoint(0 To 2) As Double, rayDir(0 To 2) As Double
    Dim vBasePoint As Variant, vVector As Variant
    Dim rayPoint As SldWorks.MathPoint, rayVector As SldWorks.MathVector
    Dim intersectPt As SldWorks.MathPoint
    Dim vPoint As Variant, vPoint2 As Variant
    Dim xPt As Double, yPt As Double, zPt As Double

    ' First try the face
    If Not faceToUse Is Nothing Then
        basePoint(0) = 0.04 '0.013
        basePoint(1) = 0.04 '0.056
        basePoint(2) = 1#
        vBasePoint = basePoint
        Set rayPoint = mathUtils.CreatePoint(vBasePoint)
        rayDir(0) = 0#
        rayDir(1) = 0#
        rayDir(2) = -1#
        vVector = rayDir
        Set rayVector = mathUtils.CreateVector(vVector)
        Set intersectPt = faceToUse.GetProjectedPointOn(rayPoint, rayVector)
        Debug.Print "Ray point = " & Format(basePoint(0), "##0.0#####") & " , " & Format(basePoint(1), "##0.0#####") & " , " & Format(basePoint(2), "##0.0#####")
        Debug.Print "Ray direction = " & Format(rayDir(0), "##0.0#####") & " , " & Format(rayDir(1), "##0.0#####") & " , " & Format(rayDir(2), "##0.0#####")
        If Not intersectPt Is Nothing Then
            vPoint = intersectPt.ArrayData
            xPt = vPoint(0)
            yPt = vPoint(1)
            zPt = vPoint(2)
            Debug.Print "    Face hit point = " & Format(xPt, "##0.0#####") & " , " & Format(yPt, "##0.0#####") & " , " & Format(zPt, "##0.0#####")
        Else
            Debug.Print "    No face hit point."
        End If

        ' Now try the surface, gotten from the face
        Set surfaceToUse = faceToUse.GetSurface()
        If Not surfaceToUse Is Nothing Then
            Set intersectPt = surfaceToUse.GetProjectedPointOn(rayPoint, rayVector)
            If Not intersectPt Is Nothing Then
                vPoint2 = intersectPt.ArrayData
                xPt = vPoint2(0)
                yPt = vPoint2(1)
                zPt = vPoint2(2)
                Debug.Print "    Surface hit point = " & Format(xPt, "##0.0#####") & " , " & Format(yPt, "##0.0#####") & " , " & Format(zPt, "##0.0#####")
            Else
                Debug.Print "    No surface hit point."
            End If
        End If
    End If
End Sub

```

End Sub

Get Properties of Sketch Pattern Feature Example (C#)

This example shows how to get the properties of a sketch pattern feature.

```
//-----
// Preconditions:
// 1. Verify that the specified document exists.
// 2. Open an Immediate window.
//
// Postconditions:
// 1. Creates Boss-Extrude2, Sketch3, and Sketch-Pattern1.
// 2. Inspect the Immediate window.
//
// NOTE: Because the model is used elsewhere, do not save changes.
//-----
using Microsoft.VisualBasic;
using System;
using System.Collections;
using System.Collections.Generic;
using System.Data;
using System.Diagnostics;
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System.Runtime.InteropServices;
namespace CreateSketchDrivenPattern_CSharp.csproj
{
    partial class SolidWorksMacro
    {
        ModelDoc2 Part;
        Feature myFeature;
        SketchPatternFeatureData swSketchPatt;
        double[] vBasePt;
        object skPoint;
        object vSkLines;
        Sketch swSketch;
        Feature swSketchFeat;
        MathTransform swPatternTransform;
        bool boolstatus;
        int i;
        int longstatus;
        int longwarnings;

        public void Main()
        {
            Part = (ModelDoc2)swApp.OpenDoc6("C:\\Users\\Public\\Documents\\SOLIDWORKS\\SOLIDWORKS
2019\\samples\\tutorial\\api\\block20.sldprt", 1, 0, "", ref longstatus, ref longwarnings);
            swApp.ActivateDoc2("block20", false, ref longstatus);

            boolstatus = Part.Extension.SelectByID("", "FACE", -0.0407921768468213, 0.0396239999998329, -0.0402814031592129, false,
            boolstatus = Part.Extension.SetUserPreferenceToggle((int)swUserPreferenceToggle_e.swSketchAddConstToRectEntity, (int)swl
            boolstatus = Part.Extension.SetUserPreferenceToggle((int)swUserPreferenceToggle_e.swSketchAddConstLineDiagonalType, (int)
            vSkLines = Part.SketchManager.CreateCornerRectangle(-0.0518589252521906, 0.0451811131877662, 0, -0.0357471289475484, 0.6
            Part.SketchManager.InsertSketch(true);

            boolstatus = Part.Extension.SelectByID("Sketch2", "SKETCH", 0, 0, 0, false, 4, null, 0);
            myFeature = Part.FeatureManager.FeatureExtrusion2(true, false, false, 0, 0, 0.00254, 0.00254, false, false, false,
            false, 0.0174532925199433, 0.0174532925199433, false, false, false, false, true, true, true,
            0, 0, false);

            Part.SketchManager.InsertSketch(true);
            boolstatus = Part.Extension.SelectByID("", "FACE", -0.00770328176440671, 0.0396239999998897, -0.00762437790422155, fals
            skPoint = Part.SketchManager.CreatePoint(-0.00527, 0.051345, 0.0);
            skPoint = Part.SketchManager.CreatePoint(-0.005854, 0.025783, 0.0);
            skPoint = Part.SketchManager.CreatePoint(-0.005888, -9E-06, 0.0);
            skPoint = Part.SketchManager.CreatePoint(0.019408, 0.051285, 0.0);
            skPoint = Part.SketchManager.CreatePoint(0.019093, 0.024628, 0.0);
            skPoint = Part.SketchManager.CreatePoint(0.019629, -0.000148, 0.0);
            skPoint = Part.SketchManager.CreatePoint(0.043756, 0.051962, 0.0);
            skPoint = Part.SketchManager.CreatePoint(0.043146, 0.025865, 0.0);
            skPoint = Part.SketchManager.CreatePoint(0.043401, 0.000225, 0.0);
            Part.ClearSelection2(true);
            Part.SketchManager.InsertSketch(true);

            boolstatus = Part.Extension.SelectByID("Boss-
Extrude2", "BODYFEATURE", -0.0477922378944982, 0.0421639999998433, 0.0233214950450815, false, 4, null, 0);
            boolstatus = Part.Extension.SelectByID("Sketch3", "SKETCH", 0, 0, 0, true, 64, null, 0);

            swSketchPatt =
            (SketchPatternFeatureData)Part.FeatureManager.CreateDefinition((int)swFeatureNameID_e.swFmSketchPattern);
            swSketchPatt.GeometryPattern = false;
            swSketchPatt.UseCentroid = true;
            swSketchFeat = (Feature)Part.FeatureManager.CreateFeature(swSketchPatt);
            swSketchPatt = (SketchPatternFeatureData)swSketchFeat.GetDefinition();

            swSketchPatt.AccessSelections(Part, null);

            swSketch = (Sketch)swSketchPatt.Sketch;
            i = swSketch.GetSketchPointsCount2();
        }
    }
}
```

```

swPatternTransform = swSketchPatt.GetTransform(i);

vBasePt = (double[])swSketchPatt.GetBasePoint();

Debug.Print(swSketchFeat.Name);
Debug.Print(" Create pattern using only geometry? " + swSketchPatt.GeometryPattern);
Debug.Print(" Pattern seed coordinates in mm: (" + vBasePt[0] * 1000.0 + ", " + vBasePt[1] * 1000.0 + ", " + vBasePt[2] * 1000.0 + ")");
Debug.Print(" Body count: " + swSketchPatt.GetPatternBodyCount());
Debug.Print(" Face count: " + swSketchPatt.GetPatternFaceCount());
Debug.Print(" Feature count: " + swSketchPatt.GetPatternFeatureCount());
Debug.Print(" Reference point type (-1 for centroid): " + swSketchPatt.GetReferencePointType());
Debug.Print(" Use centroid as the reference point? " + swSketchPatt.UseCentroid);
Debug.Print(" Propagate visual properties? " + swSketchPatt.PropagateVisualProperty);

swSketchPatt.ReleaseSelectionAccess();

}

/// <summary>
/// The SldWorks swApp variable is pre-assigned for you.
/// </summary>

public SldWorks swApp;

}

```

Get Properties of Sketch Pattern Feature Example (VBA)

This example shows how to create a sketch-driven pattern feature and gets some of its properties.

```

'-----
' Preconditions:
' 1. Verify that the specified document exists.
' 2. Open an Immediate window.
'
' Postconditions:
' 1. Creates Boss-Extrude2, Sketch3, and Sketch-Pattern1.
' 2. Inspect the Immediate window.
'
' NOTE: Because the model is used elsewhere, do not save changes.
'-----
Dim swApp                As SldWorks.SldWorks
Dim Part                 As SldWorks.ModelDoc2
Dim myFeature            As SldWorks.Feature
Dim swSketchPatt         As SldWorks.SketchPatternFeatureData
Dim vBasePt              As Variant
Dim skPoint              As Object
Dim vSkLines             As Variant
Dim swSketch             As SldWorks.Sketch
Dim swSketchFeat         As SldWorks.Feature
Dim swPatternTransform   As SldWorks.MathTransform
Dim boolstatus           As Boolean
Dim i                    As Long
Dim longstatus           As Long
Dim longwarnings         As Long

Option Explicit

Sub main()
    Set swApp = Application.SldWorks
    Set Part = swApp.OpenDoc6("C:\Users\Public\Documents\SOLIDWORKS\SOLIDWORKS 2019\samples\tutorial\api\block20.sldprt", 1, 0, "",
longstatus, longwarnings)
    swApp.ActivateDoc2 "block20", False, longstatus

    boolstatus = Part.Extension.SelectByID2("", "FACE", -4.07921768468213E-02, 3.96239999998329E-02, -4.02814031592129E-02, False,
0, Nothing, 0)
    boolstatus = Part.Extension.SetUserPreferenceToggle(swUserPreferenceToggle_e.swSketchAddConstToRectEntity,
swUserPreferenceOption_e.swDetailingNoOptionSpecified, False)
    boolstatus = Part.Extension.SetUserPreferenceToggle(swUserPreferenceToggle_e.swSketchAddConstLineDiagonalType,
swUserPreferenceOption_e.swDetailingNoOptionSpecified, True)
    vSkLines = Part.SketchManager.CreateCornerRectangle(-5.18589252521906E-02, 4.51811131877662E-02, 0, -3.57471289475484E-02,
2.86242963995278E-02, 0)
    Part.SketchManager.InsertSketch True

    boolstatus = Part.Extension.SelectByID2("Sketch2", "SKETCH", 0, 0, 0, False, 4, Nothing, 0)
    Set myFeature = Part.FeatureManager.FeatureExtrusion2(True, False, False, 0, 0, 0.00254, 0.00254, False, False, False, False,
1.74532925199433E-02, 1.74532925199433E-02, False, False, False, False, True, True, True, 0, 0, False)

    Part.SketchManager.InsertSketch True
    boolstatus = Part.Extension.SelectByID2("", "FACE", -7.70328176440671E-03, 3.96239999998897E-02, -7.62437790422155E-03, False,
0, Nothing, 0)
    Set skPoint = Part.SketchManager.CreatePoint(-0.00527, 0.051345, 0#)
    Set skPoint = Part.SketchManager.CreatePoint(-0.005854, 0.025783, 0#)
    Set skPoint = Part.SketchManager.CreatePoint(-0.005888, -0.000009, 0#)
    Set skPoint = Part.SketchManager.CreatePoint(0.019408, 0.051285, 0#)
    Set skPoint = Part.SketchManager.CreatePoint(0.019093, 0.024628, 0#)
    Set skPoint = Part.SketchManager.CreatePoint(0.019629, -0.000148, 0#)
    Set skPoint = Part.SketchManager.CreatePoint(0.043756, 0.051962, 0#)
    Set skPoint = Part.SketchManager.CreatePoint(0.043146, 0.025865, 0#)
    Set skPoint = Part.SketchManager.CreatePoint(0.043401, 0.000225, 0#)
    Part.ClearSelection2 True
    Part.SketchManager.InsertSketch True

    boolstatus = Part.Extension.SelectByID2("Boss-Extrude2", "BODYFEATURE", -4.77922378944982E-02, 4.21639999998433E-02,
2.33214950450815E-02, False, 4, Nothing, 0)
    boolstatus = Part.Extension.SelectByID2("Sketch3", "SKETCH", 0, 0, 0, True, 64, Nothing, 0)

    Set swSketchPatt = Part.FeatureManager.CreateDefinition(swFmSketchPattern)
    swSketchPatt.GeometryPattern = False
    swSketchPatt.UseCentroid = True
    Set swSketchFeat = Part.FeatureManager.CreateFeature(swSketchPatt)

    Set swSketchPatt = swSketchFeat.GetDefinition
    swSketchPatt.AccessSelections Part, Nothing

    Set swSketch = swSketchPatt.Sketch
    i = swSketch.GetSketchPointsCount2

    Set swPatternTransform = swSketchPatt.GetTransform(i)

    vBasePt = swSketchPatt.GetBasePoint

```

```

Debug.Print swSketchFeat.Name
Debug.Print " Create pattern using only geometry? " & swSketchPatt.GeometryPattern
Debug.Print " Pattern seed coordinates in mm: (" & vBasePt(0) * 1000# & ", " & vBasePt(1) * 1000# & ", " & vBasePt(2) * 1000#
& ")"
Debug.Print " Body count: " & swSketchPatt.GetPatternBodyCount
Debug.Print " Face count: " & swSketchPatt.GetPatternFaceCount
Debug.Print " Feature count: " & swSketchPatt.GetPatternFeatureCount
Debug.Print " Reference point type (-1 for centroid): " & swSketchPatt.GetReferencePointType
Debug.Print " Use centroid as the reference point? " & swSketchPatt.UseCentroid
Debug.Print " Propagate visual properties? " & swSketchPatt.PropagateVisualProperty

swSketchPatt.ReleaseSelectionAccess

End Sub

```

Get Properties of Sketch Pattern Feature Example (VB.NET)

This example shows how to get the properties of a sketch pattern feature.

```

'-----
' Preconditions:
' 1. Verify that the specified document exists.
' 2. Open an Immediate window.
'
' Postconditions:
' 1. Creates Boss-Extrude2, Sketch3, and Sketch-Pattern1.
' 2. Inspect the Immediate window.
'
' NOTE: Because the model is used elsewhere, do not save changes.
'-----
Imports SolidWorks.Interop.sldworks
Imports SolidWorks.Interop.swconst
Imports System.Runtime.InteropServices
Imports System
Imports System.Diagnostics

Partial Class SolidWorksMacro

    Dim Part As ModelDoc2
    Dim myFeature As Feature
    Dim swSketchPatt As SketchPatternFeatureData
    Dim vBasePt As Object
    Dim skPoint As Object
    Dim vSkLines As Object
    Dim swSketch As Sketch
    Dim swSketchFeat As Feature
    Dim swPatternTransform As MathTransform
    Dim boolstatus As Boolean
    Dim i As Integer
    Dim longstatus As Integer
    Dim longwarnings As Integer

    Sub main()

        Part = swApp.OpenDoc6("C:\Users\Public\Documents\SOLIDWORKS\SOLIDWORKS
2019\samples\tutorial\api\block20.sldprt", 1, 0, "", longstatus, longwarnings)
        swApp.ActivateDoc2("block20", False, longstatus)

        boolstatus = Part.Extension.SelectByID2("", "FACE", -0.0407921768468213, 0.0396239999998329, -0.0402814031592129, False, 0,
        boolstatus = Part.Extension.SetUserPreferenceToggle(swUserPreferenceToggle_e.swSketchAddConstToRectEntity, swUserPreference
        boolstatus = Part.Extension.SetUserPreferenceToggle(swUserPreferenceToggle_e.swSketchAddConstLineDiagonalType, swUserPrefer
        vSkLines = Part.SketchManager.CreateCornerRectangle(-0.0518589252521906, 0.0451811131877662, 0, -0.0357471289475484, 0.02862
        Part.SketchManager.InsertSketch(True)

        boolstatus = Part.Extension.SelectByID2("Sketch2", "SKETCH", 0, 0, 0, False, 4, Nothing, 0)
        myFeature = Part.FeatureManager.FeatureExtrusion2(True, False, False, 0, 0, 0.00254, 0.00254, False, False, False, False, 0.

        Part.SketchManager.InsertSketch(True)
        boolstatus = Part.Extension.SelectByID2("", "FACE", -0.00770328176440671, 0.0396239999998897, -0.00762437790422155, False, 0,
        skPoint = Part.SketchManager.CreatePoint(-0.00527, 0.051345, 0.0#)
        skPoint = Part.SketchManager.CreatePoint(-0.005854, 0.025783, 0.0#)
        skPoint = Part.SketchManager.CreatePoint(-0.005888, -0.000009, 0.0#)
        skPoint = Part.SketchManager.CreatePoint(0.019408, 0.051285, 0.0#)
        skPoint = Part.SketchManager.CreatePoint(0.019093, 0.024628, 0.0#)
        skPoint = Part.SketchManager.CreatePoint(0.019629, -0.000148, 0.0#)
        skPoint = Part.SketchManager.CreatePoint(0.043756, 0.051962, 0.0#)
        skPoint = Part.SketchManager.CreatePoint(0.043146, 0.025865, 0.0#)
        skPoint = Part.SketchManager.CreatePoint(0.043401, 0.000225, 0.0#)
        Part.ClearSelection2(True)
        Part.SketchManager.InsertSketch(True)

        boolstatus = Part.Extension.SelectByID2("Boss-
Extrude2", "BODYFEATURE", -0.0477922378944982, 0.0421639999998433, 0.0233214950450815, False, 4, Nothing, 0)
        boolstatus = Part.Extension.SelectByID2("Sketch3", "SKETCH", 0, 0, 0, True, 64, Nothing, 0)

        swSketchPatt = Part.FeatureManager.CreateDefinition(swFeatureNameID_e.swFmSketchPattern)
        swSketchPatt.GeometryPattern = False
        swSketchPatt.UseCentroid = True
        swSketchFeat = Part.FeatureManager.CreateFeature(swSketchPatt)
        swSketchPatt = swSketchFeat.GetDefinition

        swSketchPatt.AccessSelections(Part, Nothing)

        swSketch = swSketchPatt.Sketch
        i = swSketch.GetSketchPointsCount2

        swPatternTransform = swSketchPatt.GetTransform(i)

        vBasePt = swSketchPatt.GetBasePoint

        Debug.Print(swSketchFeat.Name)
        Debug.Print(" Create pattern using only geometry? " & swSketchPatt.GeometryPattern)
        Debug.Print(" Pattern seed coordinates in mm: (" & vBasePt(0) * 1000.0# & ", " & vBasePt(1) * 1000.0# & ", " & vBasePt(2)
        Debug.Print(" Body count: " & swSketchPatt.GetPatternBodyCount)
    
```

```
Debug.Print(" Face count: " & swSketchPatt.GetPatternFaceCount)  
Debug.Print(" Feature count: " & swSketchPatt.GetPatternFeatureCount)  
Debug.Print(" Reference point type (-1 for centroid): " & swSketchPatt.GetReferencePointType)  
Debug.Print(" Use centroid as the reference point? " & swSketchPatt.UseCentroid)  
Debug.Print(" Propagate visual properties? " & swSketchPatt.PropagateVisualProperty)
```

```
swSketchPatt.ReleaseSelectionAccess()
```

```
End Sub
```

```
''' <summary>  
''' The SldWorks swApp variable is pre-assigned for you.  
''' </summary>  
Public swApp As SldWorks
```

```
End Class
```

Get Properties of Sweep Feature Example (VBA)

This example shows how to get the properties of a sweep feature.

```

'-----
' Preconditions:
' 1. Open public_documents\samples\tutorial\api\cstick.sldprt.
' 2. Select Sweep1.
'
' Postconditions: Examine the Immediate window.
'
' NOTE: Because the part is used elsewhere, do not
' save changes.
'-----
Option Explicit

Sub main()

    Dim swApp As SldWorks.SldWorks
    Dim swModel As SldWorks.ModelDoc2
    Dim swSelMgr As SldWorks.SelectionMgr
    Dim swFeat As SldWorks.Feature
    Dim swSweep As SldWorks.SweepFeatureData
    Dim swProfFeat As SldWorks.Feature
    Dim swProfSketch As SldWorks.Sketch
    Dim swPathFeat As SldWorks.Feature
    Dim swPathSketch As SldWorks.Sketch
    Dim vGuideCurvesTypeArr As Variant
    Dim vGuideCurvesType As Variant
    Dim bRet As Boolean

    Set swApp = Application.SldWorks
    Set swModel = swApp.ActiveDoc
    Set swSelMgr = swModel.SelectionManager
    Set swFeat = swSelMgr.GetSelectedObject5(1)
    Set swSweep = swFeat.GetDefinition
    Set swProfFeat = swSweep.Profile
    Set swProfSketch = swProfFeat.GetSpecificFeature

    bRet = swSweep.AccessSelections(swModel, Nothing)
    Set swPathFeat = swSweep.Path
    Set swPathSketch = swPathFeat.GetSpecificFeature

    Debug.Print "File = " & swModel.GetPathName
    Debug.Print " " & swFeat.Name
    Debug.Print " Sweep path = " & swPathFeat.Name
    Debug.Print " Sweep path type = " & swSweep.GetPathType
    Debug.Print " Profile = " & swProfFeat.Name
    Debug.Print " Profile type as defined in swSelectType_e = " & swSweep.GetProfileType
    Debug.Print " Profile orientation/twist type as defined in swTwistControlType_e = " & swSweep.TwistControlType
    Debug.Print " Path alignment type as defined in swTangencyType_e = " & swSweep.PathAlignmentType
    Debug.Print " Align sweep with end faces? " & swSweep.AlignWithEndFaces
    Debug.Print " Automatically select all bodies in a multibody part? " & swSweep.AutoSelect
    Debug.Print " Type of tangency at the sweep's end as defined in swTangencyType_e = " & swSweep.EndTangencyType
    Debug.Print " Type of tangency at the sweep's start as defined in swTangencyType_e = " & swSweep.StartTangencyType

    Debug.Print " Is a boss sweep? " & swSweep.IsBossFeature
    Debug.Print " Is a thin-walled sweep feature? " & swSweep.IsThinFeature
    Debug.Print " Thin-walled type (0=1D, 1=1DReverse, 2=MidPlane, 3=2D) = " & swSweep.ThinWallType
    Debug.Print " Wall thickness in Direction 1 = " & swSweep.GetWallThickness(True) * 1000# & " mm"
    Debug.Print " Wall thickness in Direction 2 = " & swSweep.GetWallThickness(False) * 1000# & " mm"
    Debug.Print " Merge tangent faces? " & swSweep.MaintainTangency
    Debug.Print " Merge results for a multibody part? " & swSweep.Merge
    Debug.Print " Merge smooth faces if using guide curves? " & swSweep.MergeSmoothFaces

    vGuideCurvesTypeArr = swSweep.GetGuideCurvesType

    If Not IsEmpty(vGuideCurvesTypeArr) Then
        Debug.Print " Number of guide curves = " & UBound(vGuideCurvesTypeArr) + 1
        For Each vGuideCurvesType In vGuideCurvesTypeArr
            Debug.Print " Type as defined in swSelectType_e = " & vGuideCurvesType
        Next vGuideCurvesType
    End If

    swSweep.ReleaseSelectionAccess

End Sub

```

Create Radiate Surface Feature Example (VBA)

This example shows how to create a radiate surface feature.

```

'-----
' Preconditions:
' 1. Verify that the specified document template exists.
' 2. Open an Immediate window.
'
' Postconditions:
' 1. Creates a new model document with a feature extrusion.
' 2. Creates Boss-Extrude1 and Surface-Radiate1 in the graphics area and
'    FeatureManager design tree.
' 3. Inspect the Immediate window.
'-----

Dim swApp          As SldWorks.SldWorks
Dim Part           As SldWorks.ModelDoc2
Dim swSelMgr       As SldWorks.SelectionMgr
Dim swSelData      As SldWorks.SelectData
Dim swRadiate      As SldWorks.SurfaceRadiateFeatureData
Dim swFeat         As SldWorks.Feature
Dim swEnt          As SldWorks.Entity
Dim vRadEnt        As Variant
Dim swRadDirEnt    As SldWorks.Entity
Dim i              As Long
Dim boolStatus     As Boolean
Option Explicit

Sub main()

    Set swApp = Application.SldWorks
    Set Part = swApp.NewDocument("C:\ProgramData\SolidWorks\SOLIDWORKS 2015\templates\Part.prtdot", 0, 0, 0)
    Set Part = swApp.ActiveDoc

    boolStatus = Part.Extension.SelectByID2("Front Plane", "PLANE", -4.48901407839529E-02, 2.79954694016864E-02, 4.66820674117181E-03, False, 0, Nothing, 0)
    Part.SketchManager.InsertSketch True
    Part.ClearSelection2 True
    boolStatus = Part.Extension.SetUserPreferenceToggle(swUserPreferenceToggle_e.swSketchAddConstToRectEntity, swUserPreferenceOption_e.swDetailingNoOptionSpecified, False)
    boolStatus = Part.Extension.SetUserPreferenceToggle(swUserPreferenceToggle_e.swSketchAddConstLineDiagonalType, swUserPreferenceOption_e.swDetailingNoOptionSpecified, True)
    Dim vSkLines As Variant
    vSkLines = Part.SketchManager.CreateCornerRectangle(-5.55749908365768E-02, 3.29075527136081E-02, 0, 4.78203409524033E-02, -3.17145296545045E-02, 0)
    Part.ClearSelection2 True
    Part.SketchManager.InsertSketch True
    Part.ShowNamedView2 "Trimetric", 8
    Part.SketchManager.InsertSketch True
    Part.ClearSelection2 True
    boolStatus = Part.Extension.SelectByID2("Sketch1", "SKETCH", 0, 0, 0, False, 4, Nothing, 0)

    Dim myFeature As Object
    Set myFeature = Part.FeatureManager.FeatureExtrusion2(True, False, False, 0, 0, 0.00254, 0.00254, False, False, False, False, 1.74532925199433E-02, 1.74532925199433E-02, False, False, False, False, True, True, True, 0, 0, False)

    boolStatus = Part.Extension.SelectByID2("", "EDGE", -4.47337592343047E-02, 3.28467250718631E-02, 2.58132540182032E-03, False, 2, Nothing, 0)
    boolStatus = Part.Extension.SelectByID2("", "EDGE", -5.56265649287866E-02, 1.56695101210289E-02, 2.5672149453726E-03, True, 2, Nothing, 0)
    boolStatus = Part.Extension.SelectByID2("", "EDGE", -1.40113588298618E-02, -3.17157034173761E-02, 2.54079743683633E-03, True, 2, Nothing, 0)
    boolStatus = Part.Extension.SelectByID2("", "EDGE", 0.047780958393389, -5.42256709667299E-03, 2.56078163948814E-03, True, 2, Nothing, 0)
    boolStatus = Part.Extension.SelectByID2("", "FACE", 4.78203409524554E-02, -3.05747564971171E-03, 5.46558985774936E-04, True, 1, Nothing, 0)

    Part.InsertRadiateSurface 0.0254, False, False

    Set swSelMgr = Part.SelectionManager
    Set swSelData = swSelMgr.CreateSelectData

    boolStatus = Part.Extension.SelectByID2("Surface-Radiate1", "REFSURFACE", 0, 0, 0, False, 0, Nothing, 0)

    Set swFeat = swSelMgr.GetSelectedObject6(1, -1)
    Set swRadiate = swFeat.GetDefinition

    ' Get radiate surface data
    Debug.Print "File = " & Part.GetPathName
    Debug.Print " " & swFeat.Name
    Debug.Print " Distance: " & swRadiate.Distance * 1000# & " mm"
    Debug.Print " Flip? " & swRadiate.Flip
    Debug.Print " Propagate to tangent faces? " & swRadiate.PropagateToTangentFaces

    ' Roll back to get direction reference and radiated edges
    boolStatus = swRadiate.AccessSelections(Part, Nothing)

```

```
Set swRadDirEnt = swRadiate.DirectionReference  
Part.ClearSelection2 True  
vRadEnt = swRadiate.RadiatedEntities  
Debug.Print "Type as defined in swSelectType_e:"  
For i = 0 To swRadiate.GetRadiatedEntitiesCount - 1  
    Set swEnt = vRadEnt(i)  
    Debug.Print "  Radiated entity(" & i & ") = " & swEnt.GetType  
Next i  
swRadiate.ReleaseSelectionAccess
```

End Sub

Get Radius of Variable Radius Fillet Example (VBA)

This example shows how to get the radius of a variable radius fillet.

```
'-----  
'  
' Preconditions:  
'     (1) Model document is open.  
'     (2) Model contains a VarFillet2 feature.  
'  
' Postconditions: None  
'  
'-----  
Option Explicit  
  
Dim swApp As SldWorks.SldWorks  
Dim swModelDoc As SldWorks.ModelDoc2  
Dim swModelDocExt As SldWorks.ModelDocExtension  
Dim swSelMgr As SldWorks.SelectionMgr  
Dim boolstatus As Boolean  
Dim swFeat As SldWorks.Feature  
Dim swFeatData As SldWorks.VariableFilletFeatureData2  
Dim swEdge As SldWorks.Edge  
Dim Vert1 As SldWorks.Vertex  
Dim Vert2 As SldWorks.Vertex  
Dim radius1 As Double  
Dim radius2 As Double  
  
Sub main()  
Set swApp = Application.SldWorks  
Set swModelDoc = swApp.ActiveDoc  
Set swModelDocExt = swModelDoc.Extension  
Set swSelMgr = swModelDoc.SelectionManager  
  
boolstatus = swModelDocExt.SelectByID("VarFillet2", "BODYFEATURE", 0, 0, 0, False, 0, Nothing)  
Set swFeat = swSelMgr.GetSelectedObject5(1): Debug.Assert Not swFeat Is Nothing  
Set swFeatData = swFeat.GetDefinition()  
  
boolstatus = swFeatData.AccessSelections(swModelDoc, Nothing): Debug.Assert boolstatus  
Set swEdge = swFeatData.GetFilletEdgeAtIndex(0)  
Set Vert1 = swEdge.GetStartVertex(): Debug.Assert Not Vert1 Is Nothing  
radius1 = swFeatData.GetRadius2(Vert1, boolstatus)  
Set Vert2 = swEdge.GetEndVertex(): Debug.Assert Not Vert2 Is Nothing  
radius2 = swFeatData.GetRadius2(Vert2, boolstatus)  
Debug.Print "Radius1 = " & radius1 * 1000# & " mm"  
Debug.Print "Radius2 = " & radius2 * 1000# & " mm"  
swFeatData.ReleaseSelectionAccess  
  
End Sub
```

Get Reference Axis Data Example (C#)

This example shows how to get reference axis data.

```
//-----
// Preconditions:
// 1. Open a part, assembly, or drawing document.
// 2. Select a reference axis.
//
// Postconditions: Inspect the Immediate window.
//-----
using Microsoft.VisualBasic;
using System;
using System.Collections;
using System.Collections.Generic;
using System.Data;
using System.Diagnostics;
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System.Runtime.InteropServices;

namespace ReferenceAxisFeatDataType_CSharp.csproj
{
    partial class SolidWorksMacro
    {
        public void Main()
        {
            ModelDoc2 swModel = default(ModelDoc2);
            SelectionMgr swSelMgr = default(SelectionMgr);
            Feature swFeat = default(Feature);
            RefAxis swRefAxis = default(RefAxis);
            RefAxisFeatureData swRefAxisData = default(RefAxisFeatureData);
            double[] swMathAxis = null;

            swModel = (ModelDoc2)swApp.ActiveDoc;
            swSelMgr = (SelectionMgr)swModel.SelectionManager;
            swFeat = (Feature)swSelMgr.GetSelectedObject6(1, -1);
            swRefAxis = (RefAxis)swFeat.GetSpecificFeature2();
            swMathAxis = (double[])swRefAxis.GetRefAxisParams();

            Debug.Print("File = " + swModel.GetPathName());
            Debug.Print(" " + swFeat.Name);
            Debug.Print("    Start point = (" + swMathAxis[0] * 1000.0 + ", " + swMathAxis[1] * 1000.0 + ", " + swMathAxis[2] *
1000.0 + ") mm");
            Debug.Print("    End point = (" + swMathAxis[3] * 1000.0 + ", " + swMathAxis[4] * 1000.0 + ", " + swMathAxis[5] *
1000.0 + ") mm");

            swRefAxisData = (RefAxisFeatureData)swFeat.GetDefinition();
            Debug.Print("    Type = " + swRefAxisData.Type);

        }

        public SldWorks swApp;
    }
}
```

Get Reference Axis Data Example (VBA)

This example shows how to get reference axis data.

```
'-----
' Preconditions:
' 1. Open a part, assembly, or drawing document.
' 2. Select a reference axis.
```

```
' Postconditions: Inspect the Immediate window.
'-----
```

Option Explicit

Sub main()

```
    Dim swApp                As SldWorks.SldWorks
    Dim swModel              As SldWorks.ModelDoc2
    Dim swSelMgr             As SldWorks.SelectionMgr
    Dim swFeat              As SldWorks.Feature
    Dim swRefAxis            As SldWorks.RefAxis
    Dim swRefAxisData        As SldWorks.RefAxisFeatureData
    Dim swMathAxis           As Variant
```

```
    Set swApp = Application.SldWorks
    Set swModel = swApp.ActiveDoc
    Set swSelMgr = swModel.SelectionManager
    Set swFeat = swSelMgr.GetSelectedObject6(1, -1)
    Set swRefAxis = swFeat.GetSpecificFeature2
    swMathAxis = swRefAxis.GetRefAxisParams
```

```
    Debug.Print "File = " & swModel.GetPathName
```

```
    Debug.Print " " & swFeat.Name
```

```
    Debug.Print "      Start point = (" & swMathAxis(0) * 1000# & ", " & swMathAxis(1) * 1000# & ", " & swMathAxis(2) * 1000# & ")
```

```
mm"    Debug.Print "      End point = (" & swMathAxis(3) * 1000# & ", " & swMathAxis(4) * 1000# & ", " & swMathAxis(5) * 1000# & ") mm"
```

```
    Set swRefAxisData = swFeat.GetDefinition
```

```
    Debug.Print "      Type = " & swRefAxisData.Type
```

End Sub

Get Reference Axis Data Example (VB.NET)

This example shows how to get reference axis data.

```
'-----
' Preconditions:
' 1. Open a part, assembly, or drawing document.
' 2. Select a reference axis.
```

```
'-----
' Postconditions: Inspect the Immediate window.
```

```
Imports SolidWorks.Interop.sldworks
Imports SolidWorks.Interop.swconst
Imports System.Runtime.InteropServices
Imports System
Imports System.Diagnostics
```

```
Partial Class SolidWorksMacro
```

```
    Sub main()
```

```
        Dim swModel As ModelDoc2
        Dim swSelMgr As SelectionMgr
        Dim swFeat As Feature
        Dim swRefAxis As RefAxis
        Dim swRefAxisData As RefAxisFeatureData
        Dim swMathAxis As Object
```

```
        swModel = swApp.ActiveDoc
        swSelMgr = swModel.SelectionManager
        swFeat = swSelMgr.GetSelectedObject6(1, -1)
        swRefAxis = swFeat.GetSpecificFeature2
        swMathAxis = swRefAxis.GetRefAxisParams
```

```
        Debug.Print("File = " & swModel.GetPathName)
        Debug.Print(" " & swFeat.Name)
        Debug.Print("    Start point = (" & swMathAxis(0) * 1000.0# & ", " & swMathAxis(1) * 1000.0# & ", " & swMathAxis(2) *
1000.0# & ") mm")
        Debug.Print("    End point = (" & swMathAxis(3) * 1000.0# & ", " & swMathAxis(4) * 1000.0# & ", " & swMathAxis(5) * 1000.0#
& ") mm")
```

```
        swRefAxisData = swFeat.GetDefinition
        Debug.Print("    Type = " & swRefAxisData.Type)
```

```
    End Sub
```

```
    Public swApp As SldWorks
```

```
End Class
```

Get Reference Curve Information Example (VBA)

This example shows how to get information about a reference curve (for example, a projected curve) through a set of points.

```
'-----
' Preconditions:
' 1. Verify that the part document exists.
' 2. Open the Immediate window.
'
' Postconditions:
' 1. Opens the specified part document.
' 2. Selects a face and sketches a spline on that face.
' 3. Selects the sketch of the spline and a face.
' 4. Inserts a projected curve feature and selects it.
' 5. Gets reference curve information.
' 6. Examine the Immediate window, FeatureManager design tree, and
'    the graphics area.
'
' NOTE: Because the part is used elsewhere, do not save changes.
'-----
Option Explicit

Dim swApp As SldWorks.SldWorks
Dim swModel As SldWorks.ModelDoc2
Dim swModelDocExt As SldWorks.ModelDocExtension
Dim swSketchManager As SldWorks.SketchManager
Dim swSketchSegment As SldWorks.SketchSegment
Dim swFeature As SldWorks.Feature
Dim swSelectionManager As SldWorks.SelectionMgr
Dim swRefCurve As SldWorks.ReferenceCurve
Dim swCurve As SldWorks.Curve
Dim swEdge As SldWorks.Edge
Dim swSketch As SldWorks.Sketch
Dim pointArray As Variant
Dim points(10) As Double
Dim status As Boolean
Dim errors As Long
Dim warnings As Long
Dim fileName As String
Dim vCurveParam As Variant
Dim nStartPt(2) As Double
Dim nEndPt(2) As Double
Dim vStartPt As Variant
Dim vEndPt As Variant
Dim vTessPts As Variant
Dim i As Long

Function TessLength(vTessPts As Variant) As Double
    Dim i As Long
    For i = 0 To ((UBound(vTessPts) + 1) / 3 - 2)
        TessLength = TessLength + Sqr((vTessPts(3 * i + 0) - vTessPts(3 * i + 3)) ^ 2# + (vTessPts(3 * i + 1) - vTessPts(3 * i + 4)) ^ 2# + (vTessPts(3 * i + 2) - vTessPts(3 * i + 5)) ^ 2#)
    Next i
End Function

Sub main()
    Set swApp = Application.SldWorks

    'Open part document
    fileName = "C:\Users\Public\Documents\SOLIDWORKS\SOLIDWORKS 2018\samples\tutorial\api\block20.sldprt"
    Set swModel = swApp.OpenDoc6(fileName, swDocumentTypes_e.swDocPART, swOpenDocOptions_e.swOpenDocOptions_Silent, "", errors, warnings)

    'Sketch a spline on the selected face
    Set swModelDocExt = swModel.Extension
    status = swModelDocExt.SelectByID("", "FACE", -4.99223104334874E-02, 3.96239999998897E-02, 7.38137362270663E-03, False, 0, Nothing, 0)
    Set swSketchManager = swModel.SketchManager
    swSketchManager.InsertSketch True
    swModel.ClearSelection2 True
    points(0) = -6.24778997860176E-02
    points(1) = 7.29572078180673E-03
    points(2) = 0
    points(3) = -3.64588790258153E-02
    points(4) = 3.24586288177215E-02
    points(5) = 0
    points(6) = 1.04252377344665E-02
    points(7) = 1.40473535914225E-02
    points(8) = 0
    points(9) = 6.46002912861796E-02
    points(10) = 1.00590221094308E-02
    pointArray = points
    Set swSketchSegment = swSketchManager.CreateSpline2((pointArray), False)
    swSketchManager.InsertSketch True
    swModel.ClearSelection2 True

    'Insert projected curve
    status = swModelDocExt.SelectByID("Sketch2", "SKETCH", 0, 0, 0, True, 0, Nothing, 0)
    status = swModelDocExt.SelectByID("", "FACE", -4.97146993259321E-02, 0, -2.56283866693252E-02, True, 0, Nothing, 0)
    Set swFeature = swModel.InsertProjectedSketch2(1)

    'Select reference curve and get reference curve information
    status = swModelDocExt.SelectByID("Curve1", "REFCURVE", 0, 0, 0, False, 0, Nothing, 0)
    Set swSelectionManager = swModel.SelectionManager
    Set swFeature = swSelectionManager.GetSelectedObject6(1, -1)
    Set swRefCurve = swFeature.GetSpecificFeature
    Set swEdge = swRefCurve.GetFirstSegment
    Debug.Print "Feature = " & swFeature.Name
```

```

While Not swEdge Is Nothing
    Set swCurve = swEdge.GetCurve
    vCurveParam = swEdge.GetCurveParams2
    nStartPt(0) = vCurveParam(0)
    nStartPt(1) = vCurveParam(1)
    nStartPt(2) = vCurveParam(2)
    nEndPt(0) = vCurveParam(3)
    nEndPt(1) = vCurveParam(4)
    nEndPt(2) = vCurveParam(5)

    vStartPt = nStartPt
    vEndPt = nEndPt

    vTessPts = swCurve.GetTessPts(0.000001, 0.000001, (vStartPt), (vEndPt))

    Debug.Print "   Curve(" & i & ")"
    Debug.Print "       Start = (" & vCurveParam(0) * 1000# & ", " & vCurveParam(1) * 1000# & ", " & vCurveParam(2) * 1000# & ") mm"
    Debug.Print "       End   = (" & vCurveParam(3) * 1000# & ", " & vCurveParam(4) * 1000# & ", " & vCurveParam(5) * 1000# & ") mm"
    Debug.Print "       Length = " & swCurve.GetLength2(vCurveParam(6), vCurveParam(7)) * 1000# & " mm"
    Debug.Print "       TessLength(" & UBound(vTessPts) & ") = " & TessLength(vTessPts) * 1000# & " mm"

    i = i + 1

    Set swEdge = swRefCurve.GetNextSegment
Wend

End Sub

```

Get Reference Point Data Example (C#)

This example shows how to get reference point data.

```
//-----
// Preconditions:
// 1. Open a part, assembly, or drawing document that has a
//    reference point.
// 2. Select the reference point.
//
// Postconditions:
// 1. Selects the feature used to create the reference point.
// 2. Press F5 after each feature selection to continue.
// 3. Inspect the Immediate window.
//-----
using Microsoft.VisualBasic;
using System;
using System.Collections;
using System.Collections.Generic;
using System.Data;
using System.Diagnostics;
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System.Runtime.InteropServices;
namespace ReferencePointFeatDataType_CSharp.csproj
{
    partial class SolidWorksMacro
    {
        public void Main()
        {
            ModelDoc2 swModel = default(ModelDoc2);
            SelectionMgr swSelMgr = default(SelectionMgr);
            Feature swFeat = default(Feature);
            RefPoint swRefPt = default(RefPoint);
            RefPointFeatureData swRefPtData = default(RefPointFeatureData);
            MathPoint swMathPt = default(MathPoint);
            object[] vEntArr = null;
            object vEnt = null;
            Entity swEnt = default(Entity);
            SketchPoint swSkPt = default(SketchPoint);
            SketchSegment swSkSeg = default(SketchSegment);
            bool bRet = false;

            swModel = (ModelDoc2)swApp.ActiveDoc;
            swSelMgr = (SelectionMgr)swModel.SelectionManager;
            swFeat = (Feature)swSelMgr.GetSelectedObject6(1, -1);
            swRefPt = (RefPoint)swFeat.GetSpecificFeature2();
            swRefPtData = (RefPointFeatureData)swFeat.GetDefinition();
            swMathPt = swRefPt.GetRefPoint();

            Debug.Print("File = " + swModel.GetPathName());
            Debug.Print(" " + swFeat.Name);
            Debug.Print(" Pt = (" + ((double[])swMathPt.ArrayData)[0] * 1000.0 + ", " + ((double[])swMathPt.ArrayData)[1] *
1000.0 + ", " + ((double[])swMathPt.ArrayData)[2] * 1000.0 + ") mm");
            Debug.Print(" AlongCurveOption = " + swRefPtData.AlongCurveOption);
            Debug.Print(" Distance = " + swRefPtData.Distance * 1000.0 + " mm");
            Debug.Print(" Type = " + swRefPtData.Type);

            bRet = swRefPtData.AccessSelections(swModel, null);
            vEntArr = (object[])swRefPtData.Selections;

            foreach (object vEnt_loopVariable in vEntArr)
            {
                vEnt = vEnt_loopVariable;
                if ((null != vEnt))
                {
                    if (vEnt is Entity)
                    {
                        {
                            swEnt = (Entity)vEnt;
                            bRet = swEnt.Select4(true, null);
                            Debug.Print("Entity used to create reference point selected.");
                            System.Diagnostics.Debugger.Break();
                        }
                    }
                    else if (vEnt is SketchPoint)
                    {
                        {
                            swSkPt = (SketchPoint)vEnt;
                            bRet = swSkPt.Select4(true, null);
                            Debug.Print("SketchPoint used to create reference point selected.");
                            System.Diagnostics.Debugger.Break();
                        }
                    }
                    else if (vEnt is SketchSegment)
                    {
                        {
                            swSkSeg = (SketchSegment)vEnt;
                            bRet = swSkSeg.Select4(true, null);
                            Debug.Print("SketchSegment used to create reference point selected.");
                            System.Diagnostics.Debugger.Break();
                        }
                    }
                    else
                }
            }
        }
    }
}
```

```
        {  
        }  
    }  
}  
  
    swRefPtData.ReleaseSelectionAccess();  
  
}  
  
public SldWorks swApp;  
  
}  
}
```

Get Reference Point Data Example (VBA)

This example shows how to get reference point data.

```
'-----
' Preconditions:
' 1. Open a part, assembly, or drawing document that has a
'   reference point.
' 2. Select the reference point.
'
' Postconditions:
' 1. Selects the feature used to create the reference point.
' 2. Press F5 after each feature selection to continue.
' 3. Inspect the Immediate window.
'-----
```

Option Explicit

```
Sub main()
    Dim swApp As SldWorks.SldWorks
    Dim swModel As SldWorks.ModelDoc2
    Dim swSelMgr As SldWorks.SelectionMgr
    Dim swFeat As SldWorks.Feature
    Dim swRefPt As SldWorks.RefPoint
    Dim swRefPtData As SldWorks.RefPointFeatureData
    Dim swMathPt As SldWorks.MathPoint
    Dim vEntArr As Variant
    Dim vEnt As Variant
    Dim swEnt As SldWorks.Entity
    Dim swSkPt As SldWorks.SketchPoint
    Dim swSkSeg As SldWorks.SketchSegment
    Dim bRet As Boolean

    Set swApp = Application.SldWorks
    Set swModel = swApp.ActiveDoc
    Set swSelMgr = swModel.SelectionManager
    Set swFeat = swSelMgr.GetSelectedObject6(1, -1)
    Set swRefPt = swFeat.GetSpecificFeature2
    Set swRefPtData = swRefPt.GetDefinition
    Set swMathPt = swRefPt.GetRefPoint

    Debug.Print "File = " & swModel.GetPathName
    Debug.Print " " & swFeat.Name
    Debug.Print " Pt = (" & swMathPt.ArrayData(0) * 1000# & ", " & swMathPt.ArrayData(1) * 1000# & ", " & swMathPt.ArrayData(2)
* 1000# & ") mm"
    Debug.Print " AlongCurveOption = " & swRefPtData.AlongCurveOption
    Debug.Print " Distance = " & swRefPtData.Distance * 1000# & " mm"
    Debug.Print " Type = " & swRefPtData.Type

    bRet = swRefPtData.AccessSelections(swModel, Nothing)
    vEntArr = swRefPtData.Selections

    For Each vEnt In vEntArr
        If Not Nothing Is vEnt Then
            If TypeOf vEnt Is SldWorks.Entity Then
                Set swEnt = vEnt
                bRet = swEnt.Select4(True, Nothing)
                Debug.Print "Entity used to create reference point selected."
                Stop
            ElseIf TypeOf vEnt Is SldWorks.SketchPoint Then
                Set swSkPt = vEnt
                bRet = swSkPt.Select4(True, Nothing)
                Debug.Print "SketchPoint used to create reference point selected."
                Stop
            ElseIf TypeOf vEnt Is SldWorks.SketchSegment Then
                Set swSkSeg = vEnt
                bRet = swSkSeg.Select4(True, Nothing)
                Debug.Print "SketchSegment used to create reference point selected."
                Stop
            Else
                End If
            End If
        End If
    Next

    swRefPtData.ReleaseSelectionAccess
End Sub
```

Get Reference Point Data Example (VB.NET)

This example shows how to get reference point data.

```
'-----
' Preconditions:
' 1. Open a part, assembly, or drawing document that has a
'   reference point.
' 2. Select the reference point.
'
' Postconditions:
' 1. Selects the feature used to create the reference point.
' 2. Press F5 after each feature selection to continue.
' 3. Inspect the Immediate window.
'-----
```

```
Imports SolidWorks.Interop.sldworks
Imports SolidWorks.Interop.swconst
Imports System.Runtime.InteropServices
Imports System
Imports System.Diagnostics
```

```
Partial Class SolidWorksMacro
```

```
    Sub main()
```

```
        Dim swModel As ModelDoc2
        Dim swSelMgr As SelectionMgr
        Dim swFeat As Feature
        Dim swRefPt As RefPoint
        Dim swRefPtData As RefPointFeatureData
        Dim swMathPt As MathPoint
        Dim vEntArr As Object
        Dim vEnt As Object
        Dim swEnt As Entity
        Dim swSkPt As SketchPoint
        Dim swSkSeg As SketchSegment
        Dim bRet As Boolean
```

```
        swModel = swApp.ActiveDoc
        swSelMgr = swModel.SelectionManager
        swFeat = swSelMgr.GetSelectedObject6(1, -1)
        swRefPt = swFeat.GetSpecificFeature2
        swRefPtData = swFeat.GetDefinition
        swMathPt = swRefPt.GetRefPoint
```

```
        Debug.Print("File = " & swModel.GetPathName)
        Debug.Print(" " & swFeat.Name)
        Debug.Print(" Pt = (" & swMathPt.ArrayData(0) * 1000.0# & ", " & swMathPt.ArrayData(1) * 1000.0# & ", " &
swMathPt.ArrayData(2) * 1000.0# & ") mm")
        Debug.Print(" AlongCurveOption = " & swRefPtData.AlongCurveOption)
        Debug.Print(" Distance = " & swRefPtData.Distance * 1000.0# & " mm")
        Debug.Print(" Type = " & swRefPtData.Type)
```

```
        bRet = swRefPtData.AccessSelections(swModel, Nothing)
        vEntArr = swRefPtData.Selections
```

```
        For Each vEnt In vEntArr
            If Not Nothing Is vEnt Then
                If TypeOf vEnt Is Entity Then
                    swEnt = vEnt
                    bRet = swEnt.Select4(True, Nothing)
                    Debug.Print("Entity used to create reference point selected.")
                    Stop
                ElseIf TypeOf vEnt Is SketchPoint Then
                    swSkPt = vEnt
                    bRet = swSkPt.Select4(True, Nothing)
                    Debug.Print("SketchPoint used to create reference point selected.")
                    Stop
                ElseIf TypeOf vEnt Is SketchSegment Then
                    swSkSeg = vEnt
                    bRet = swSkSeg.Select4(True, Nothing)
                    Debug.Print("SketchSegment used to create reference point selected.")
                    Stop
                Else
                    End If
            End If
        Next
```

```
        swRefPtData.ReleaseSelectionAccess()
```

```
    End Sub
```

```
    Public swApp As SldWorks
```

```
End Class
```


Get Referenced Display State Example (C#)

This example shows how to get the active display state of a component.

```
//-----
// Preconditions:
// 1. Open an assembly that contains two instances of the same
//    component in different display states.
// 2. Open the Immediate window.
//
// Postconditions: Inspect the Immediate window for the display states.
//-----

using Microsoft.VisualBasic;
using System;
using System.Collections;
using System.Collections.Generic;
using System.Data;
using System.Diagnostics;
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
namespace ReferencedDisplayState_CSharp.csproj
{
    partial class SolidWorksMacro
    {
        public void Main()
        {
            AssemblyDoc assem = default(AssemblyDoc);
            assem = (AssemblyDoc)swApp.ActiveDoc;

            ModelDoc2 model = default(ModelDoc2);
            model = (ModelDoc2)assem;

            Configuration assemConfig = default(Configuration);
            assemConfig = model.ConfigurationManager.ActiveConfiguration;

            Component2 root = default(Component2);
            root = assemConfig.GetRootComponent3(false);

            object[] comps = null;
            comps = (object[])root.GetChildren();

            object vComp = null;
            foreach (object vComp_loopVariable in comps)
            {
                vComp = vComp_loopVariable;
                Component2 comp = default(Component2);
                comp = (Component2)vComp;

                string refConfigName = null;
                refConfigName = comp.ReferencedConfiguration;

                ModelDoc2 compModel = default(ModelDoc2);
                compModel = (ModelDoc2)comp.GetModelDoc2();

                compModel.Visible = true;

                Configuration cmActiveConfig = default(Configuration);
                cmActiveConfig = compModel.ConfigurationManager.ActiveConfiguration;

                Debug.Print(comp.Name2);
                Debug.Print(" " + cmActiveConfig.Name + " <" + comp.ReferencedDisplayState + ">");

                compModel.Visible = false;
            }
        }

        public SldWorks swApp;
    }
}
```

Get Referenced Display State Example (VBA)

This example shows how to get the active display state of a component.

```
'-----  
' Preconditions:  
' 1. Open an assembly that contains two instances of the same  
'    component in different display states.  
' 2. Open the Immediate window.  
'-----  
' Postconditions: Inspect the Immediate window for the display states.  
'-----
```

Option Explicit

```
Sub main()  
    Dim swApp As SldWorks.SldWorks  
    Set swApp = Application.SldWorks  
  
    Dim assem As AssemblyDoc  
    Set assem = swApp.ActiveDoc  
  
    Dim model As ModelDoc2  
    Set model = assem  
  
    Dim assemConfig As Configuration  
    Set assemConfig = model.ConfigurationManager.ActiveConfiguration  
  
    Dim root As Component2  
    Set root = assemConfig.GetRootComponent3(False)  
  
    Dim comps  
    comps = root.GetChildren  
  
    Dim vComp  
    For Each vComp In comps  
        Dim comp As Component2  
        Set comp = vComp  
  
        Dim refConfigName As String  
        refConfigName = comp.ReferencedConfiguration  
  
        Dim compModel As ModelDoc2  
        Set compModel = comp.GetModelDoc2  
  
        compModel.Visible = True  
  
        Dim cmActiveConfig As Configuration  
        Set cmActiveConfig = compModel.ConfigurationManager.ActiveConfiguration  
  
        Debug.Print comp.Name2  
        Debug.Print " " & cmActiveConfig.Name & " <" & comp.ReferencedDisplayState & ">"  
  
        compModel.Visible = False  
    Next  
End Sub
```

Get Referenced Display State Example (VB.NET)

This example shows how to get the active display state of a component.

```
'-----
' Preconditions:
' 1. Open an assembly that contains two instances of the same
'    component in different display states.
' 2. Open the Immediate window.
'
' Postconditions: Inspect the Immediate window for the display states.
'-----
```

```
Imports SolidWorks.Interop.sldworks
Imports SolidWorks.Interop.swconst
Imports System
Imports System.Diagnostics

Partial Class SolidWorksMacro

    Sub main()

        Dim assem As AssemblyDoc
        assem = swApp.ActiveDoc

        Dim model As ModelDoc2
        model = assem

        Dim assemConfig As Configuration
        assemConfig = model.ConfigurationManager.ActiveConfiguration

        Dim root As Component2
        root = assemConfig.GetRootComponent3(False)

        Dim comps As Object
        comps = root.GetChildren

        Dim vComp As Object
        For Each vComp In comps
            Dim comp As Component2
            comp = vComp

            Dim refConfigName As String
            refConfigName = comp.ReferencedConfiguration

            Dim compModel As ModelDoc2
            compModel = comp.GetModelDoc2

            compModel.Visible = True

            Dim cmActiveConfig As Configuration
            cmActiveConfig = compModel.ConfigurationManager.ActiveConfiguration

            Debug.Print(comp.Name2)
            Debug.Print(" " & cmActiveConfig.Name & " <" & comp.ReferencedDisplayState & ">")

            compModel.Visible = False
        Next
    End Sub

    Public swApp As SldWorks
End Class
```

Get Render References (C#)

This example shows how to get the render stock (SOLIDWORKS-supplied) references for a model.

```
//-----
// Preconditions:
// 1. Verify that the specified part to open and
//    kitchen background scene files exist.
// 2. Open the Immediate window.
//
// Postconditions:
// 1. Inserts the kitchen background scene in the part.
// 2. Gets the names of the render references.
// 3. Examine the Immediate window.
//
// NOTE: Because the part is used elsewhere, do
// not save changes.
//-----
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System;
using System.Diagnostics;

namespace GetRenderStockReferencesCSharp.csproj
{
    partial class SolidWorksMacro
    {
        public void Main()
        {
            ModelDoc2 swModel;
            ModelDocExtension swModelDocExt;
            string modelName = null;
            object[] renderReferences = null;
            bool status = false;
            int errors = 0;
            int warnings = 0;
            int i = 0;

            modelName = "C:\\Users\\Public\\Documents\\SOLIDWORKS\\SOLIDWORKS 2018\\samples\\tutorial\\api\\toaster.sldprt";
            swModel = (ModelDoc2)swApp.OpenDoc6(modelName, (int)swDocumentTypes_e.swDocPART,
            (int)swOpenDocOptions_e.swOpenDocOptions_Silent, "", ref errors, ref warnings);
            swModelDocExt = (ModelDocExtension)swModel.Extension;

            // Insert kitchen background scene
            // and rebuild the model to see it
            status = swModelDocExt.InsertScene("C:\\Program Files\\SOLIDWORKS Corp\\SOLIDWORKS\\data\\graphics\\scenes\\03
presentation scenes\\00 kitchen_background.p2s");
            status = swModel.ForceRebuild3(true);

            // Get the render stock references for the
            // kitchen background scene and print
            // them to the Immediate window
            renderReferences = (object[])swModelDocExt.GetRenderStockReferences();
            for (i = 0; i <= renderReferences.GetUpperBound(0); i++)
            {
                Debug.Print("Render reference: " + renderReferences[i]);
            }
        }

        /// <summary>
        /// The SldWorks swApp variable is pre-assigned for you.
        /// </summary>
        public SldWorks swApp;
    }
}
```

Get Render References (VBA)

This example shows how to get the render stock (SOLIDWORKS-supplied) references for a model.

```
'-----
' Preconditions:
' 1. Verify that the specified part to open and
'    kitchen background scene files exist.
' 2. Open the Immediate window.
'
' Postconditions:
' 1. Inserts the kitchen background scene in the part.
' 2. Gets the names of the render references.
' 3. Examine the Immediate window.
'
' NOTE: Because the part is used elsewhere, do
' not save changes.
'-----
Option Explicit

Dim swApp As SldWorks.SldWorks
Dim swModel As SldWorks.ModelDoc2
Dim swModelDocExt As SldWorks.ModelDocExtension
Dim modelName As String
Dim renderReferences As Variant
Dim status As Boolean
Dim errors As Long, warnings As Long
Dim i As Long

Sub main()

Set swApp = Application.SldWorks
modelName = "C:\Users\Public\Documents\SOLIDWORKS\SOLIDWORKS 2018\samples\tutorial\api\toaster.sldprt"
Set swModel = swApp.OpenDoc6(modelName, swDocPART, swOpenDocOptions_Silent, "", errors, warnings)
Set swModelDocExt = swModel.Extension

' Insert kitchen background scene
' and rebuild the model to see it
status = swModelDocExt.InsertScene("C:\Program Files\SOLIDWORKS Corp\SOLIDWORKS\data\graphics\scenes\03 presentation scenes\00 kitchen_background.p2s")
status = swModel.ForceRebuild3(True)

' Get the render stock references for the
' kitchen background scene and print
' them to the Immediate window
renderReferences = swModelDocExt.GetRenderStockReferences
For i = 0 To UBound(renderReferences)
    Debug.Print "Render reference: " & renderReferences(i)
Next i

End Sub
```

Get Render References (VB.NET)

This example shows how to get the render stock (SOLIDWORKS-supplied) references for a model.

```

'-----
' Preconditions:
' 1. Verify that the specified part to open and
'    kitchen background scene files exist.
' 2. Open the Immediate window.
'
' Postconditions:
' 1. Inserts the kitchen background scene in the part.
' 2. Gets the names of the render references.
' 3. Examine the Immediate window.
'
' NOTE: Because the part is used elsewhere, do
' not save changes.
'-----
Imports SolidWorks.Interop.sldworks
Imports SolidWorks.Interop.swconst
Imports System
Imports System.Diagnostics

Partial Class SolidWorksMacro

    Public Sub Main()

        Dim swModel As ModelDoc2
        Dim swModelDocExt As ModelDocExtension
        Dim modelName As String
        Dim renderReferences As Object
        Dim status As Boolean
        Dim errors As Long, warnings As Long
        Dim i As Long

        modelName = "C:\Users\Public\Documents\SOLIDWORKS\SOLIDWORKS 2018\samples\tutorial\api\toaster.sldprt"
        swModel = swApp.OpenDoc6(modelName, swDocumentTypes_e.swDocPART, swOpenDocOptions_e.swOpenDocOptions_Silent, "", errors,
warnings)
        swModelDocExt = swModel.Extension

        ' Insert kitchen background scene
        ' and rebuild the model to see it
        status = swModelDocExt.InsertScene("C:\Program Files\SOLIDWORKS Corp\SOLIDWORKS\data\graphics\scenes\03 presentation scenes\00
kitchen_background.p2s")
        status = swModel.ForceRebuild3(True)

        ' Get the render stock references for the
        ' kitchen background scene and print
        ' them to the Immediate window
        renderReferences = swModelDocExt.GetRenderStockReferences
        For i = 0 To UBound(renderReferences)
            Debug.Print("Render reference: " & renderReferences(i))
        Next i

    End Sub

    ''' <summary>
    ''' The SldWorks swApp variable is pre-assigned for you.
    ''' </summary>
    Public swApp As SldWorks

End Class

```

Get Revision Table Example (VBA)

This example shows how to get information from a revision table annotation.

```
'-----  
' Preconditions:  
' 1. Open a drawing whose current sheet  
'    contains a revision table annotation.  
' 2. Open the Immediate window.  
'  
' Postconditions:  
' 1. Gets the number of columns and rows  
'    in the revision table annotation and  
'    the text in each row.  
' 2. Examine the Immediate window.  
'-----  
Option Explicit  
  
Sub ProcessTableAnn(swApp As SldWorks.SldWorks, swModel As SldWorks.ModelDoc2, swTable As SldWorks.TableAnnotation)  
  
    Dim sRowStr As String  
    Dim i As Long  
    Dim j As Long  
  
    Debug.Print "Number of columns: " & swTable.ColumnCount  
    Debug.Print "Number of rows: " & swTable.RowCount  
    For i = 0 To swTable.RowCount - 1  
        sRowStr = ""  
        For j = 0 To swTable.ColumnCount - 1  
            sRowStr = sRowStr & "|" & swTable.Text2(i, j, True)  
        Next j  
        Debug.Print " " & sRowStr & "|"  
    Next i  
  
End Sub  
  
Sub main()  
  
    Dim swApp As SldWorks.SldWorks  
    Dim swModel As SldWorks.ModelDoc2  
    Dim swDraw As SldWorks.DrawingDoc  
    Dim swSheet As SldWorks.Sheet  
    Dim swRevTable As SldWorks.RevisionTableAnnotation  
    Dim swTable As SldWorks.TableAnnotation  
    Dim bRet As Boolean  
  
    Set swApp = Application.SldWorks  
    Set swModel = swApp.ActiveDoc  
    Set swDraw = swModel  
    Set swSheet = swDraw.GetCurrentSheet  
    Set swRevTable = swSheet.RevisionTable  
    Set swTable = swRevTable  
    Debug.Print "File = " & swModel.GetPathName  
    ProcessTableAnn swApp, swModel, swTable  
  
End Sub
```

Get Rib Feature Data Example (C#)

This example shows how to get rib feature data.

```
//-----
// Preconditions:
// 1. Verify that the specified model document exists.
// 2. Open an Immediate window.
//
// Postconditions:
// 1. Opens the part document.
// 2. Creates Shell1, Plane1, and Rib1.
// 3. Inspect the FeatureManager design tree, the graphics area, and the
//    Immediate window.
//
// NOTE: Because the model is used elsewhere, do not save changes.
//-----
using Microsoft.VisualBasic;
using System;
using System.Collections;
using System.Collections.Generic;
using System.Data;
using System.Diagnostics;
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System.Runtime.InteropServices;
namespace InsertRib_CSharp.csproj
{
    partial class SolidWorksMacro
    {
        ModelDoc2 Part;
        RefPlane myRefPlane;
        SketchSegment skSegment;
        SelectionMgr swSelMgr;
        Feature swFeat;
        RibFeatureData2 swRibFeat;
        bool boolstatus;
        int longstatus;
        int longwarnings;

        public void Main()
        {
            Part = swApp.OpenDoc6("C:\\Users\\Public\\Documents\\SOLIDWORKS\\SOLIDWORKS
2018\\samples\\tutorial\\api\\block20.sldprt", 1, 0, "", ref longstatus, ref longwarnings);
            swApp.ActivateDoc2("block20", false, ref longstatus);
            Part = (ModelDoc2)swApp.ActiveDoc;

            boolstatus = Part.Extension.SelectByID("", "FACE", -0.00878816842651986, 0.0396239999998897, -0.0292468281514857, false, 0, 0, 0);
            Part.InsertFeatureShell(0.00254, false);

            boolstatus = Part.Extension.SelectByID("", "FACE", 0.00264031138414111, 0.028407059059532, -0.0613970439424634, true, 0, 0, 0);
            boolstatus = Part.Extension.SelectByID("", "FACE", -0.059937899786064, 0.0277866864457792, -0.00877977980189826, true, 0, 0, 0);

            myRefPlane = (RefPlane)Part.FeatureManager.InsertRefPlane(128, 0, 128, 0, 0, 0);
            Part.ClearSelection2(true);

            boolstatus = Part.Extension.SelectByID("Plane1", "PLANE", 0.00664896553058725, 0.109417877974863, 0.0524178648701081, true, 0, 0, 0);
            Part.SketchManager.InsertSketch(true);

            skSegment = Part.SketchManager.CreateLine(-0.085797, 0.021082, 0.0, -0.03423, 0.035134, 0.0);
            skSegment = Part.SketchManager.CreateLine(-0.03423, 0.035134, 0.0, 0.007726, 0.025357, 0.0);
            skSegment = Part.SketchManager.CreateLine(0.007726, 0.025357, 0.0, 0.11514, 0.039624, 0.0);
            Part.ClearSelection2(true);

            Part.SketchManager.InsertSketch(true);
            Part.ClearSelection2(true);
            boolstatus = Part.Extension.SelectByID("Sketch2", "SKETCH", 0, 0, 0, false, 0, null, 0);
            Part.FeatureManager.InsertRib(true, false, 0.00254, 0, false, false, true, 0.0174532925199433, false, false);

            swSelMgr = (SelectionMgr)Part.SelectionManager;
            swFeat = (Feature)swSelMgr.GetSelectedObject6(1, -1);
            swRibFeat = (RibFeatureData2)swFeat.GetDefinition();

            Debug.Print("Rib feature type as defined in swRibType_e: " + swRibFeat.Type);
            Debug.Print("Thickness: " + swRibFeat.Thickness);
            Debug.Print("Extrusion direction as defined in swRibExtrusionDirection_e: " + swRibFeat.ExtrusionDirection);
            Debug.Print("Rib has a draft? " + swRibFeat.EnableDraft);
            if (swRibFeat.EnableDraft)
            {
                Debug.Print("    Draft angle: " + swRibFeat.DraftAngle);
                Debug.Print("    Draft outward? " + swRibFeat.DraftOutward);
            }
            Debug.Print("Add material to reverse side of the rib? " + swRibFeat.FlipSide);
            Debug.Print("Rib is extruded on two sides of the midplane? " + swRibFeat.IsTwoSided);
            if (!swRibFeat.IsTwoSided)
            {
                Debug.Print("Single-sided rib is extruded on the reverse side? " + swRibFeat.ReverseThicknessDir);
            }
        }
    }
}
```

```
}

/// <summary>
/// The SldWorks swApp variable is pre-assigned for you.
/// </summary>

public SldWorks swApp;

}
```

Get Rib Feature Data Example (VBA)

This example shows how to get rib feature data.

```

'-----
' Preconditions:
' 1. Verify that the specified model document exists.
' 2. Open an Immediate window.
'
' Postconditions:
' 1. Opens the part document.
' 2. Creates Shell1, Plane1, and Rib1.
' 3. Inspect the FeatureManager design tree, the graphics area, and the
'    Immediate window.
'
' NOTE: Because the model is used elsewhere, do not save changes.
'-----
Dim swApp As SldWorks.SldWorks
Dim Part As SldWorks.ModelDoc2
Dim myRefPlane As SldWorks.RefPlane
Dim skSegment As SldWorks.SketchSegment
Dim swSelMgr As SldWorks.SelectionMgr
Dim swFeat As SldWorks.Feature
Dim swRibFeat As SldWorks.RibFeatureData2
Dim boolstatus As Boolean
Dim longstatus As Long, longwarnings As Long
Option Explicit
Sub main()

    Set swApp = Application.SldWorks

    Set Part = swApp.OpenDoc6("C:\Users\Public\Documents\SOLIDWORKS\SOLIDWORKS 2018\samples\tutorial\api\block20.sldprt", 1, 0, "",
longstatus, longwarnings)
    swApp.ActivateDoc2 "block20", False, longstatus
    Set Part = swApp.ActiveDoc

    boolstatus = Part.Extension.SelectByID2("", "FACE", -8.78816842651986E-03, 3.96239999998897E-02, -2.92468281514857E-02, False,
1, Nothing, 0)
    Part.InsertFeatureShell 0.00254, False

    boolstatus = Part.Extension.SelectByID2("", "FACE", 2.64031138414111E-03, 0.028407059059532, -6.13970439424634E-02, True, 0,
Nothing, 0)
    boolstatus = Part.Extension.SelectByID2("", "FACE", -0.059937899786064, 2.77866864457792E-02, -8.77977980189826E-03, True, 1,
Nothing, 0)

    Set myRefPlane = Part.FeatureManager.InsertRefPlane(128, 0, 128, 0, 0, 0)
    Part.ClearSelection2 True

    boolstatus = Part.Extension.SelectByID2("Plane1", "PLANE", 6.64896553058725E-03, 0.109417877974863, 5.24178648701081E-02,
False, 0, Nothing, 0)
    Part.SketchManager.InsertSketch True

    Set skSegment = Part.SketchManager.CreateLine(-0.085797, 0.021082, 0#, -0.03423, 0.035134, 0#)
    Set skSegment = Part.SketchManager.CreateLine(-0.03423, 0.035134, 0#, 0.007726, 0.025357, 0#)
    Set skSegment = Part.SketchManager.CreateLine(0.007726, 0.025357, 0#, 0.111514, 0.039624, 0#)
    Part.ClearSelection2 True

    Part.SketchManager.InsertSketch True
    Part.ClearSelection2 True
    boolstatus = Part.Extension.SelectByID2("Sketch2", "SKETCH", 0, 0, 0, False, 0, Nothing, 0)
    Part.FeatureManager.InsertRib True, False, 0.00254, 0, False, False, True, 1.74532925199433E-02, False, False

    Set swSelMgr = Part.SelectionManager
    Set swFeat = swSelMgr.GetSelectedObject6(1, -1)
    Set swRibFeat = swFeat.GetDefinition

    Debug.Print "Rib feature type as defined in swRibType_e: " & swRibFeat.Type
    Debug.Print "Thickness: " & swRibFeat.Thickness
    Debug.Print "Extrusion direction as defined in swRibExtrusionDirection_e: " & swRibFeat.ExtrusionDirection
    Debug.Print "Rib has a draft? " & swRibFeat.EnableDraft
    If swRibFeat.EnableDraft Then
        Debug.Print "    Draft angle: " & swRibFeat.DraftAngle
        Debug.Print "    Draft outward? " & swRibFeat.DraftOutward
    End If
    Debug.Print "Add material to reverse side of the rib? " & swRibFeat.FlipSide
    Debug.Print "Rib is extruded on two sides of the midplane? " & swRibFeat.IsTwoSided
    If Not swRibFeat.IsTwoSided Then
        Debug.Print "Single-sided rib is extruded on the reverse side? " & swRibFeat.ReverseThicknessDir
    End If

End Sub

```

Get Rib Feature Data Example (VB.NET)

This example shows how to get rib feature data.

```

'-----
' Preconditions:
' 1. Verify that the specified model document exists.
' 2. Open an Immediate window.
'
' Postconditions:
' 1. Opens the part document.
' 2. Creates Shell1, Plane1, and Rib1.
' 3. Inspect the FeatureManager design tree, the graphics area, and the
'    Immediate window.
'
' NOTE: Because the model is used elsewhere, do not save changes.
'-----
Imports SolidWorks.Interop.sldworks
Imports SolidWorks.Interop.swconst
Imports System.Runtime.InteropServices
Imports System
Imports System.Diagnostics

Partial Class SolidWorksMacro

    Dim Part As ModelDoc2
    Dim myRefPlane As RefPlane
    Dim skSegment As SketchSegment
    Dim swSelMgr As SelectionMgr
    Dim swFeat As Feature
    Dim swRibFeat As RibFeatureData2
    Dim boolstatus As Boolean
    Dim longstatus As Integer, longwarnings As Integer

    Sub main()

        Part = swApp.OpenDoc6("C:\Users\Public\Documents\SOLIDWORKS\SOLIDWORKS
2018\samples\tutorial\api\block20.sldprt", 1, 0, "", longstatus, longwarnings)
        swApp.ActivateDoc2("block20", False, longstatus)
        Part = swApp.ActiveDoc

        boolstatus = Part.Extension.SelectByID2("", "FACE", -0.00878816842651986, 0.0396239999998897, -0.0292468281514857, False, 1,
        Part.InsertFeatureShell(0.00254, False)

        boolstatus = Part.Extension.SelectByID2("", "FACE", 0.00264031138414111, 0.028407059059532, -0.0613970439424634, True, 0, No
        boolstatus = Part.Extension.SelectByID2("", "FACE", -0.059937899786064, 0.0277866864457792, -0.00877977980189826, True, 1, P

        myRefPlane = Part.FeatureManager.InsertRefPlane(128, 0, 128, 0, 0, 0)
        Part.ClearSelection2(True)

        boolstatus = Part.Extension.SelectByID2("Plane1", "PLANE", 0.00664896553058725, 0.109417877974863, 0.0524178648701081, False
        Part.SketchManager.InsertSketch(True)

        skSegment = Part.SketchManager.CreateLine(-0.085797, 0.021082, 0.0#, -0.03423, 0.035134, 0.0#)
        skSegment = Part.SketchManager.CreateLine(-0.03423, 0.035134, 0.0#, 0.007726, 0.025357, 0.0#)
        skSegment = Part.SketchManager.CreateLine(0.007726, 0.025357, 0.0#, 0.111514, 0.039624, 0.0#)
        Part.ClearSelection2(True)

        Part.SketchManager.InsertSketch(True)
        Part.ClearSelection2(True)
        boolstatus = Part.Extension.SelectByID2("Sketch2", "SKETCH", 0, 0, 0, False, 0, Nothing, 0)
        Part.FeatureManager.InsertRib(True, False, 0.00254, 0, False, False, True, 0.0174532925199433, False, False)

        swSelMgr = Part.SelectionManager
        swFeat = swSelMgr.GetSelectedObject6(1, -1)
        swRibFeat = swFeat.GetDefinition

        Debug.Print("Rib feature type as defined in swRibType_e: " & swRibFeat.Type)
        Debug.Print("Thickness: " & swRibFeat.Thickness)
        Debug.Print("Extrusion direction as defined in swRibExtrusionDirection_e: " & swRibFeat.ExtrusionDirection)
        Debug.Print("Rib has a draft? " & swRibFeat.EnableDraft)
        If swRibFeat.EnableDraft Then
            Debug.Print("    Draft angle: " & swRibFeat.DraftAngle)
            Debug.Print("    Draft outward? " & swRibFeat.DraftOutward)
        End If
        Debug.Print("Add material to reverse side of the rib? " & swRibFeat.FlipSide)
        Debug.Print("Rib is extruded on two sides of the midplane? " & swRibFeat.IsTwoSided)
        If Not swRibFeat.IsTwoSided Then
            Debug.Print("Single-sided rib is extruded on the reverse side? " & swRibFeat.ReverseThicknessDir)
        End If

    End Sub

''' <summary>
''' The SldWorks swApp variable is pre-assigned for you.
''' </summary>
Public swApp As SldWorks

```

End Class

Get Ruled-Surface Feature Data Example (C#)

This example shows how to get ruled-surface feature data.

```
//-----
// Preconditions:
// 1. Open a part document that contains a ruled-surface feature.
// 2. Select the ruled-surface feature.
// 3. Open the Immediate window.
//
// Postconditions:
// 1. Gets the type of ruled-surface feature.
// 2. Examine the Immediate window.
//-----
using Microsoft.VisualBasic;
using System;
using System.Collections;
using System.Collections.Generic;
using System.Data;
using System.Diagnostics;
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System.Runtime.InteropServices;

namespace RuledSurfaceFeatDataType_CSharp.csproj
{
    partial class SolidWorksMacro
    {
        ModelDoc2 swModel;
        Feature swFeat;
        SelectionMgr swSelMgr;

        RuledSurfaceFeatureData swRuledSurfFeat;

        public void Main()
        {
            swModel = (ModelDoc2)swApp.ActiveDoc;

            swSelMgr = (SelectionMgr)swModel.SelectionManager;
            swFeat = (Feature)swSelMgr.GetSelectedObject6(1, -1);
            swRuledSurfFeat = (RuledSurfaceFeatureData)swFeat.GetDefinition();

            // Ruled-surface feature type as defined in swRuledSurfaceType_e
            Debug.Print("Ruled-surface feature type: " + swRuledSurfFeat.Type);
        }

        public SldWorks swApp;
    }
}
```

Get Ruled-Surface Feature Data Example (VBA)

This example shows how to get ruled-surface feature data.

```
'-----  
' Preconditions:  
' 1. Open a part document that contains a ruled-surface feature.  
' 2. Select the ruled-surface feature.  
' 3. Open the Immediate window.  
'  
' Postconditions:  
' 1. Gets the type of ruled-surface feature.  
' 2. Examine the Immediate window.  
'-----  
Option Explicit  
  
Dim swApp As SldWorks.SldWorks  
Dim swModel As SldWorks.ModelDoc2  
Dim swFeat As SldWorks.Feature  
Dim swSelMgr As SldWorks.SelectionMgr  
Dim swRuledSurfFeat As SldWorks.RuledSurfaceFeatureData  
  
Sub main()  
    Set swApp = Application.SldWorks  
    Set swModel = swApp.ActiveDoc  
  
    Set swSelMgr = swModel.SelectionManager  
    Set swFeat = swSelMgr.GetSelectedObject6(1, -1)  
    Set swRuledSurfFeat = swFeat.GetDefinition  
  
    ' Ruled-surface feature type as defined in swRuledSurfaceType_e  
    Debug.Print "Ruled-surface feature type: " & swRuledSurfFeat.Type  
  
End Sub
```

Get Ruled-Surface Feature Data Example (VB.NET)

This example shows how to get ruled-surface feature data.

```
'-----  
' Preconditions:  
' 1. Open a part document that contains a ruled-surface feature.  
' 2. Select the ruled-surface feature.  
' 3. Open the Immediate window.  
'-----  
' Postconditions:  
' 1. Gets the type of ruled-surface feature.  
' 2. Examine the Immediate window.  
'-----
```

```
Imports SolidWorks.Interop.sldworks
```

```
Imports SolidWorks.Interop.swconst
```

```
Imports System.Runtime.InteropServices
```

```
Imports System
```

```
Imports System.Diagnostics
```

```
Partial Class SolidWorksMacro
```

```
    Dim swModel As ModelDoc2
```

```
    Dim swFeat As Feature
```

```
    Dim swSelMgr As SelectionMgr
```

```
    Dim swRuledSurfFeat As RuledSurfaceFeatureData
```

```
    Sub main()
```

```
        swModel = swApp.ActiveDoc
```

```
        swSelMgr = swModel.SelectionManager
```

```
        swFeat = swSelMgr.GetSelectedObject6(1, -1)
```

```
        swRuledSurfFeat = swFeat.GetDefinition
```

```
        ' Ruled-surface feature type as  
defined in swRuledSurfaceType_e
```

```
        Debug.Print("Ruled-surface  
feature type: " & swRuledSurfFeat.Type)
```

```
    End Sub
```

```
    Public swApp As SldWorks
```

```
End Class
```

Get Scale Factor of Each Model View Example (C++)

This example shows how to get the scale factor of each model view in a part document.

```
//-----
// Preconditions:
// 1. Start Microsoft Visual Studio 2019 or later.
//    a. Click File > New > Project > (C++ Windows) Console App.
//    b. Click Next.
//    c. Type the name of your project in Project Name.
//    d. Click Create.
//    e. Right-click project, select Add, New Item, Select ATL > ATL Support, Click Add.
//    f. Click Project > projectname Properties > Configuration Properties > C/C++ and type the path to sldworks.tlb and swconst.tlb, typically C:\Program Files\SOLIDWORKS Corp\SOLIDWORKS, in Additional Include Directories.
//    g. Click Apply or OK.
// 2. Replace the code in the code window with This code.
// 3. Click Debug > Start Debugging.
// 4. Start up SOLIDWORKS and open a part document.
// 5. Click Window > Viewport > Four View.
// 6. Click a model view and spin the middle-mouse button forward or back.
// 7. Click Debug > Start Debugging.
//
// Postconditions:
// 1. Gets the number of model views in the part document.
// 2. Gets the scale factor of each model view.
// 3. Examine the Immediate Window.
//-----

//This code
#include "framework.h"
#include <iostream>
#include <sstream>
#import "sldworks.tlb" raw_interfaces_only, raw_native_types, no_namespace, named_guids // SOLIDWORKS type library
#import "swconst.tlb" raw_interfaces_only, raw_native_types, no_namespace, named_guids // SOLIDWORKS constants type library

int _tmain(int argc, _TCHAR* argv[])
{
    //Initialize COM
    //Do this before using ATL smart pointers so that
    //COM is available
    CoInitialize(NULL);

    //Use a block so that the smart pointers are destructed when
    //scope of this block is exited
    {
        //Use ATL smart pointers
        CComPtr<ISldWorks> swApp;
        if(swApp.CoCreateInstance(L"SldWorks.Application", NULL, CLSCTX_LOCAL_SERVER) != S_OK){
            return(0);
        }

        HRESULT hres;

        CComPtr<IModelDoc2> swDoc;
        swApp->get_IActiveDoc2(&swDoc);

        //Get number of model views
        CComPtr<IModelDocExtension> swDocExt;
        hres = swDoc->get_Extension(&swDocExt);
        long nbrModelViews;
        hres = swDocExt->GetModelViewCount(&nbrModelViews);
        _bstr_t s;
        s = (long)nbrModelViews;
        OutputDebugString("Number of model views: " + s + "\n");

        //Get an ISafeArrayUtility object
        CComPtr<IDispatch> dispatchSafeArray = NULL;
        CComPtr<ISafeArrayUtility> swSafeArray = NULL;
        hres = swApp->GetSafeArrayUtility(&dispatchSafeArray);
        hres = dispatchSafeArray->QueryInterface<ISafeArrayUtility>(&swSafeArray);

        //Pack a Variant SafeArray of Dispatch-based objects
        IDispatch** dispArray = new IDispatch*[nbrModelViews];
        VARIANT vPacked;
        ::VariantInit(&vPacked);
        int arrType = swDispatchArray;
        hres = swSafeArray->PackVariant(&vPacked, nbrModelViews, arrType, (long*)dispArray);

        //Get the model views in the model document
        hres = swDocExt->GetModelViews(&vPacked);

        //Get scale factor of each model view
        for (int i = 0; i < nbrModelViews; i++)
        {
            CComPtr<IModelView> swModelView;
            IDispatch *pDisp;
            hres = swSafeArray->GetDispatch(vPacked, i, &pDisp);
            pDisp->QueryInterface(IID_IModelView, (void*)&swModelView);
            _bstr_t t;
            double scale;
            hres = swModelView->get_Scale2(&scale);
            t = (double)scale;
            OutputDebugString("Scale of this model view is: " + t + "\n");
        }
    }

    return 0;
}
```

Get Scale Factor of Each Model View Example (C#)

This example shows how to get the scale factor of each model view in a part document.

```
//-----
// Preconditions:
// 1. Open a part document.
// 2. Click Window > Viewport > Four View.
// 3. Click a model view and spin the middle-mouse
//    button forward or back.
// 4. Open the Immediate Window.
//
// Postconditions:
// 1. Gets the number of model views in the part document.
// 2. Gets the scale factor of each model view.
// 3. Examine the Immediate Window.
//-----

using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System.Runtime.InteropServices;
using System;
using System.Diagnostics;

namespace GetModelViewScalesCSharp.csproj
{
    public partial class SolidWorksMacro
    {
        public void Main()
        {
            ModelDoc2 SwModel = default(ModelDoc2);
            ModelDocExtension swModelDocExt = default(ModelDocExtension);
            ModelView swModView = default(ModelView);
            object[] modViews = null;
            long Count = 0;

            SwModel = (ModelDoc2)swApp.ActiveDoc;
            swModelDocExt = (ModelDocExtension)SwModel.Extension;

            // Get model views
            modViews = (object[])swModelDocExt.GetModelViews();

            // Get number of model views
            Count = swModelDocExt.GetModelViewCount();
            Debug.Print("Number of model views: " + Count);

            // Get the scale factor of each model view
            for (long i = 0; i < Count; i++)
            {
                swModView = (ModelView)modViews[i];
                Debug.Print("Scale factor of this model view is: " + swModView.Scale2);
            }
        }

        /// <summary>
        /// The SldWorks swApp variable is pre-assigned for you.
        /// </summary>
        public SldWorks swApp;
    }
}
```

Get Scale Factor of Each Model View Example (VBA)

This example shows how to get the scale factor of each model view in a part document.

```
'-----  
' Preconditions:  
' 1. Open a part document.  
' 2. Click Window > Viewport > Four View.  
' 3. Click a model view and spin the middle-mouse  
'    button forward or back.  
' 4. Open the Immediate Window.  
'  
' Postconditions:  
' 1. Gets the number of model views in the part document.  
' 2. Gets the scale factor of each model view.  
' 3. Examine the Immediate Window.  
'-----  
Option Explicit  
  
Dim swApp As SldWorks.SldWorks  
Dim SwModel As SldWorks.ModelDoc2  
Dim swModelDocExt As SldWorks.ModelDocExtension  
Dim swModView As SldWorks.ModelView  
Dim swModViews As Variant  
Dim index As Long  
Dim Count As Long  
  
Sub main()  
  
    Set swApp = Application.SldWorks  
    Set SwModel = swApp.ActiveDoc  
    Set swModelDocExt = SwModel.Extension  
  
    ' Get model views  
    swModViews = swModelDocExt.GetModelViews  
  
    ' Get number of model views  
    Count = swModelDocExt.GetModelViewCount  
    Debug.Print "Number of model views: " & Count  
  
    ' Get the scale factor of each model view  
    For index = LBound(swModViews) To UBound(swModViews)  
        Set swModView = swModViews(index)  
        Debug.Print "Scale of this model view is: " & swModView.Scale2  
    Next index  
  
End Sub
```

Get Scale Factor of Each Model View Example (VB.NET)

This example shows how to get the scale factor of each model view in a part document.

```
'-----
' Preconditions:
' 1. Open a part document.
' 2. Click Window > Viewport > Four View.
' 3. Click a model view and spin the middle-mouse
'    button forward or back.
' 4. Open the Immediate Window.
'
' Postconditions:
' 1. Gets the number of model views in the part document.
' 2. Gets the scale factor of each model view.
' 3. Examine the Immediate Window.
'-----
Imports SolidWorks.Interop.sldworks
Imports SolidWorks.Interop.swconst
Imports System.Runtime.InteropServices
Imports System
Imports System.Diagnostics

Partial Class SolidWorksMacro

    Public Sub main()

        Dim SwModel As ModelDoc2
        Dim swModelDocExt As ModelDocExtension
        Dim swModView As ModelView
        Dim swModViews As Object
        Dim index As Integer
        Dim Count As Integer

        SwModel = swApp.ActiveDoc
        swModelDocExt = SwModel.Extension

        ' Get model views
        swModViews = swModelDocExt.GetModelViews

        ' Get number of model views
        Count = swModelDocExt.GetModelViewCount
        Debug.Print("Number of model views: " & Count)

        ' Get the scale factor of each model view
        For index = LBound(swModViews) To UBound(swModViews)
            swModView = swModViews(index)
            Debug.Print("Scale of this model view is: " & swModView.Scale2)
        Next index

    End Sub

    ''' <summary>
    ''' The SldWorks swApp variable is pre-assigned for you.
    ''' </summary>
    Public swApp As SldWorks

End Class
```

Get Section Properties for Planar Faces Example (VBA)

This example shows how to get section properties for planar faces.

```

'-----
' Preconditions:
' 1. Open a part that has at least three planar faces.
' 2. Select three planar faces.
' 3. Open the Immediate window.
'
' Postconditions:
' 1. Gets the section properties of the three selected
'    faces.
' 2. Examine the Immediate window.
'-----
Option Explicit

Dim swApp As SldWorks.SldWorks
Dim swModelDoc As SldWorks.ModelDoc2
Dim swModelDocExt As SldWorks.ModelDocExtension
Dim swSelMgr As SldWorks.SelectionMgr
Dim boolstatus As Boolean

Sub main()

    Set swApp = Application.SldWorks
    Set swModelDoc = swApp.ActiveDoc
    Set swModelDocExt = swModelDoc.Extension
    Set swSelMgr = swModelDoc.SelectionManager

    boolstatus = swModelDocExt.SelectByID("", "FACE", 0, 0, 0, False, 0, Nothing, swSelectOptionDefault)
    Dim face1(0 To 0) As Object
    Set face1(0) = swSelMgr.GetSelectedObject5(1)
    swModelDoc.ClearSelection2 True
    Dim vFace1 As Variant
    vFace1 = face1
    Dim v1 As Variant

    Debug.Print "Section properties:"

    ' Get the section properties for the selected face
    v1 = swModelDocExt.GetSectionProperties2((vFace1))
    Dim i As Integer
    Debug.Print "  First selected face:"
    For i = 0 To UBound(v1)
        Debug.Print "    " & v1(i); ""
    Next i

    boolstatus = swModelDocExt.SelectByID("", "FACE", 0, 0, 0, False, 0, Nothing, swSelectOptionDefault)
    Dim face2(0 To 0) As Object
    Set face2(0) = swSelMgr.GetSelectedObject5(1)
    swModelDoc.ClearSelection2 True
    Dim vFace2 As Variant
    vFace2 = face2
    Dim v2 As Variant

    ' Get the section properties for the selected face
    v2 = swModelDocExt.GetSectionProperties2((vFace2))
    Debug.Print "  Second selected face:"
    For i = 0 To UBound(v2)
        Debug.Print "    " & v2(i)
    Next i

    boolstatus = swModelDocExt.SelectByID("", "FACE", 0, 0, 0, False, 0, Nothing, swSelectOptionDefault)
    Dim face3(0 To 0) As Object
    Set face3(0) = swSelMgr.GetSelectedObject5(1)
    swModelDoc.ClearSelection2 True
    Dim vFace3 As Variant
    vFace3 = face3
    Dim v3 As Variant

    ' Get the section properties for the selected face
    v3 = swModelDocExt.GetSectionProperties2((vFace3))
    Debug.Print "  Third selected face:"
    For i = 0 To UBound(v3)
        Debug.Print "    " & v3(i)
    Next i

End Sub

```

Get Section Properties for Faces from Section Planes Example (VBA)

This example shows how to get the section properties of the selected items.

NOTE: Through the user interface, it is possible to extract the section properties from various combinations of selected entities. This is accessible through the SOLIDWORKS API with `IModelDocExtension::GetSectionProperties`. This method has only one input parameter, an array of entities to consider with the current selection set. The entity array must only contain these types of objects: Face2 and Sketch. For planar faces and reference surfaces, this is the Face2 object. A sketch corresponds to a Sketch object. However, for a face on a section plane, there is no API object. This sample code shows how to use this method to allow for faces obtained from section planes.

```
'-----
' Preconditions:
' 1. Open a part, fully resolved assembly, or drawing.
' 2. Select one of the following:
'   * Sketch
'   * Planar model face
'   * Face on a section plane
'   * Crosshatch section face in a section view
'     in a drawing or a sketch
' 3. Open the Immediate window.
'
' Postconditions:
' 1. Maintains the selection set. (See NOTES)
' 2. Examine the Immediate window.
'
' NOTES:
' * Array of objects passed into this method are
'   added to the selection set.
' * Outputted values are the same as those obtained
'   through the user interface.
' * Reasons for failures are the same as those
'   in the user interface.
'-----
Option Explicit

Sub main()
    Const PI As Double = 3.14159

    Dim swApp As SldWorks.SldWorks
    Dim swModel As SldWorks.ModelDoc2
    Dim swModelExt As SldWorks.ModelDocExtension
    Dim swSelMgr As SldWorks.SelectionMgr
    Dim swFace As SldWorks.Face2
    Dim swFeat() As SldWorks.Feature
    Dim swFaceEnt() As SldWorks.Entity
    Dim swSketch As SldWorks.Sketch
    Dim nSelType As Long
    Dim swSelObj() As Object
    Dim vSelObj As Variant
    Dim vSectionProp As Variant
    Dim nSelCount As Long
    Dim nNumObj As Long
    Dim i As Long

    Set swApp = Application.SldWorks
    Set swModel = swApp.ActiveDoc
    Set swModelExt = swModel.Extension
    Set swSelMgr = swModel.SelectionManager

    Debug.Print "File = " & swModel.GetPathName

    nSelCount = swSelMgr.GetSelectedObjectCount
    For i = 1 To nSelCount
        nSelType = swSelMgr.GetSelectedObjectType2(i)
        Debug.Print " SelType(" & i & ") = " & nSelType
        Select Case nSelType
            Case swSelFACES, swSelREFSURFACES
                nNumObj = nNumObj + 1
                ReDim Preserve swSelObj(nNumObj - 1)
                ReDim Preserve swFaceEnt(nNumObj - 1)
                Set swFace = swSelMgr.GetSelectedObject5(i)
                Set swFaceEnt(nNumObj - 1) = swFace
                Set swSelObj(nNumObj - 1) = swFace
            Case swSelSKETCHES
                nNumObj = nNumObj + 1
                ReDim Preserve swSelObj(nNumObj - 1)
                ReDim Preserve swFeat(nNumObj - 1)
                Set swFeat(nNumObj - 1) = swSelMgr.GetSelectedObject5(i)
                Set swSketch = swFeat(nNumObj - 1).GetSpecificFeature
                Set swSelObj(nNumObj - 1) = swSketch
                Debug.Print " " & swFeat(nNumObj - 1).Name

            Case swSelMANIPULATORS
                ' Section face in a part or assembly when in a section view
                ' There is no corresponding API object for this,
                ' so leave it selected
        End Select
    Next i
    Debug.Print ""

    ' Deselect faces and sketches; otherwise, user-interface selections
    ' are added to array parameter; leave section faces selected
    If Not IsEmpty(swFaceEnt) Then
        For i = 0 To UBound(swFaceEnt)
            swFaceEnt(i).Deselect
        Next i
    End If
```

```

If Not IsEmpty(swFeat) Then
    For i = 0 To UBound(swFeat)
        swFeat(i).Deselect
    Next i
End If

vSelObj = swSelObj

' This adds the array of faces or sketches to the selection set
' Because the faces or sketches have been deselected, this
' preserves the selection set
vSectionProp = swModelExt.GetSectionProperties((vSelObj))

' Return code from:
'   IModelDocExtension::GetSectionProperties
'   0 = success
'   1 = invalid input
'   2 = selected faces are not in the same or parallel planes
'   3 = unable to compute section properties
Debug.Print "   Return code" = " & vSectionProp(0)
Debug.Print ""
Debug.Print "   Area" = " & vSectionProp(1) * 1000000# & " mm^2"
Debug.Print "   Centroid" = (" & vSectionProp(2) * 1000# & ", " & vSectionProp(3) * 1000# & ", " & vSectionProp(4) * 1000# & ") mm"
Debug.Print "   Ixx" = " & vSectionProp(5) * 1000000000000# & " mm^4"
Debug.Print "   Iyy" = " & vSectionProp(6) * 1000000000000# & " mm^4"
Debug.Print "   Izz" = " & vSectionProp(7) * 1000000000000# & " mm^4"
Debug.Print "   Ixy" = " & vSectionProp(8) * 1000000000000# & " mm^4"
Debug.Print "   Izx" = " & vSectionProp(9) * 1000000000000# & " mm^4"
Debug.Print "   Iyz" = " & vSectionProp(10) * 1000000000000# & " mm^4"
Debug.Print "   Polar MOI" = " & vSectionProp(11) * 1000000000000# & " mm^4"
Debug.Print "   Angle princ & part axes" = " & vSectionProp(12) * 180# / PI & " deg"
Debug.Print "   Ix" = " & vSectionProp(13) * 1000000000000# & " mm^4"
Debug.Print "   Iy" = " & vSectionProp(14) * 1000000000000# & " mm^4"

End Sub

```

Get Section View Data Example (C#)

This example shows how to get the data for a section view in a part.

```
//-----
// Preconditions:
// 1. Open public_documents\samples\tutorial\cosmosxpress\aw_rubber_duct.sldprt.
// 2. Create a section view with three sections:
//    a. Right-click Front in the FeatureManager design tree
//       and click Section View.
//    b. Set X Rotation to 45.00deg in Section 1.
//    c. Select Section 2, click Top, and
//       set X Rotation to 45.00deg.
//    d. Select Section 3, click Right, and
//       set X Rotation to 45.00deg.
//    e. Click Save twice.
// 3. Click View > Display > Section View twice.
// 4. Open the Immediate window.
//
// Postconditions:
// 1. Gets and prints data about each section in
//    the section view to the Immediate window.
// 2. Examine the Immediate window.
//
// NOTE: Because the part is used elsewhere, do not save changes.
//-----
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System.Runtime.InteropServices;
using System;
using System.Diagnostics;

namespace Macro1CSharp.csproj
{
    public partial class SolidWorksMacro
    {
        public void Main()
        {
            ModelDoc2 swModel = default(ModelDoc2);
            ModelViewManager swModelViewMgr = default(ModelViewManager);
            SectionViewData swSectionViewData = default(SectionViewData);
            SelectionMgr swSelMgr = default(SelectionMgr);
            bool status = false;

            swModel = (ModelDoc2)swApp.ActiveDoc;
            swSelMgr = (SelectionMgr)swModel.SelectionManager;

            //Get section view
            swModelViewMgr = (ModelViewManager)swModel.ModelViewManager;
            swSectionViewData = (SectionViewData)swModelViewMgr.GetSectionViewData("");

            DisplayDebugInformation(swSectionViewData, swSelMgr);

            swModel.ClearSelection2(true);

        }
        //Select the planes and print each section's data
        public void DisplayDebugInformation(SectionViewData data, SelectionMgr selMgr)
        {
            Feature p1 = default(Feature);
            Feature p2 = default(Feature);
            Feature p3 = default(Feature);
            MathTransform swMathTransform = default(MathTransform);
            double[] transform = null;

            p1 = (Feature)data.FirstPlane;
            if ((p1 != null))
                p1.Select2(true, 1);
            p2 = (Feature)data.SecondPlane;
            if ((p2 != null))
                p2.Select2(true, 2);
            p3 = (Feature)data.ThirdPlane;
            if ((p3 != null))
                p3.Select2(true, 4);

            Debug.Print("-----Section 1-----");
            Debug.Print("Offset: " + data.FirstOffset);
            Debug.Print("Rotation X: " + data.FirstRotationX);
            Debug.Print("Rotation Y: " + data.FirstRotationY);
            Debug.Print("Color: " + data.FirstColor);
            Debug.Print("Reverse direction: " + data.FirstReverseDirection);
            Debug.Print("-----Section 2-----");
            Debug.Print("Offset: " + data.SecondOffset);
            Debug.Print("Rotation X: " + data.SecondRotationX);
            Debug.Print("Rotation Y: " + data.SecondRotationY);
            Debug.Print("Color: " + data.SecondColor);
            Debug.Print("Reverse direction: " + data.SecondReverseDirection);
            Debug.Print("-----Section 3-----");
            Debug.Print("Offset: " + data.ThirdOffset);
            Debug.Print("Rotation X: " + data.ThirdRotationX);
            Debug.Print("Rotation Y: " + data.ThirdRotationY);
            Debug.Print("Color: " + data.ThirdColor);
            Debug.Print("Reverse direction: " + data.ThirdReverseDirection);
            swMathTransform = (MathTransform)data.GetDynamicCenterTransform(3);
            Debug.Print("Dynamic center transform:");
            transform = (double[])swMathTransform.ArrayData;
            if ((transform != null))
            {
                Debug.Print(" Rotate: " + transform[0].ToString("###0.0#####") + " " + transform[1].ToString("###0.0#####") + " " + transform[2].ToString("###0.0#####");
                Debug.Print(" " + transform[3].ToString("###0.0#####") + " " + transform[4].ToString("###0.0#####") + " " + transform[5].ToString("###0.0#####");
                Debug.Print(" " + transform[6].ToString("###0.0#####") + " " + transform[7].ToString("###0.0#####") + " " + transform[8].ToString("###0.0#####");
                Debug.Print(" Translate: " + transform[9].ToString("###0.0#####") + " " + transform[10].ToString("###0.0#####") + " " + transform[11].ToString("###0.0#####");
            }
            Debug.Print("-----Section Cap-----");
            Debug.Print("Show section cap: " + data.ShowSectionCap);
        }
    }
}
```

```
    /// <summary>  
    /// The Sldworks swApp variable is pre-assigned for you.  
    /// </summary>  
    public SldWorks swApp;  
}
```

Get Section View Data Example (VBA)

This example shows how to get the data for a section view in a part.

```

'-----
' Preconditions:
' 1. Open public_documents\samples\tutorial\cosmosxpress\aw_rubber_duct.sldprt.
' 2. Create a section view with three sections:
'   a. Right-click Front in the FeatureManager design tree
'      and click Section View.
'   b. Set X Rotation to 45.00deg in Section 1.
'   c. Select Section 2, click Top, and
'      set X Rotation to 45.00deg.
'   d. Select Section 3, click Right, and
'      set X Rotation to 45.00deg.
'   e. Click Save twice.
' 3. Click View > Display > Section View twice.
' 4. Open the Immediate window.
'
' Postconditions:
' 1. Gets and prints data about each section in
'    the section view to the Immediate window.
' 2. Examine the Immediate window.
'
' NOTE: Because the part is used elsewhere, do not save changes.
'-----
Option Explicit

Sub main()

    Dim swApp As SldWorks.SldWorks
    Dim swModel As SldWorks.ModelDoc2
    Dim swModelViewMgr As SldWorks.ModelViewManager
    Dim swSectionViewData As SldWorks.SectionViewData
    Dim swSelMgr As SldWorks.SelectionMgr
    Dim status As Boolean

    Set swApp = Application.SldWorks
    Set swModel = swApp.ActiveDoc
    Set swSelMgr = swModel.SelectionManager

    'Get section view
    Set swModelViewMgr = swModel.ModelViewManager
    Set swSectionViewData = swModelViewMgr.GetSectionViewData("")

    DisplayDebugInformation swSectionViewData, swSelMgr

    swModel.ClearSelection2 True

End Sub

'Select the planes and print each section's data
Sub DisplayDebugInformation(data As SectionViewData, selMgr As SelectionMgr)
    Dim p1 As SldWorks.Feature
    Dim p2 As SldWorks.Feature
    Dim p3 As SldWorks.Feature
    Dim swMathTransform As SldWorks.MathTransform
    Dim transform As Variant

    Set p1 = data.FirstPlane
    If Not p1 Is Nothing Then p1.Select2 True, 1

    Set p2 = data.SecondPlane
    If Not p2 Is Nothing Then p2.Select2 True, 2

    Set p3 = data.ThirdPlane
    If Not p3 Is Nothing Then p3.Select2 True, 4

    Debug.Print "-----Section 1-----"
    Debug.Print "Offset: " & data.FirstOffset
    Debug.Print "Rotation X: " & data.FirstRotationX
    Debug.Print "Rotation Y: " & data.FirstRotationY
    Debug.Print "Color: " & data.FirstColor
    Debug.Print "Reverse direction: " & data.FirstReverseDirection

    Debug.Print "-----Section 2-----"
    Debug.Print "Offset: " & data.SecondOffset
    Debug.Print "Rotation X: " & data.SecondRotationX
    Debug.Print "Rotation Y: " & data.SecondRotationY
    Debug.Print "Color: " & data.SecondColor
    Debug.Print "Reverse direction: " & data.SecondReverseDirection

    Debug.Print "-----Section 3-----"
    Debug.Print "Offset: " & data.ThirdOffset
    Debug.Print "Rotation X: " & data.ThirdRotationX
    Debug.Print "Rotation Y: " & data.ThirdRotationY
    Debug.Print "Color: " & data.ThirdColor
    Debug.Print "Reverse direction: " & data.ThirdReverseDirection
    Set swMathTransform = data.GetDynamicCenterTransform(3)
    Debug.Print "Dynamic center transform:"
    transform = swMathTransform.ArrayData()
    If Not IsEmpty(transform) Then
        Debug.Print " Rotate: " & Format(transform(0), "###0.0#####") & " " & Format(transform(1), "###0.0#####") & " " & Format(transform(2),
"###0.0#####")
        Debug.Print " " & Format(transform(3), "###0.0#####") & " " & Format(transform(4), "###0.0#####") & " " & Format(transform(5),
"###0.0#####")
        Debug.Print " " & Format(transform(6), "###0.0#####") & " " & Format(transform(7), "###0.0#####") & " " & Format(transform(8),

```

```
###0.0####")
    Debug.Print "  Translate: " & Format(transform(9), "###0.0####") & " " & Format(transform(10), "###0.0####") & " " & Format(transform(11),
"###0.0####")
    End If
    Debug.Print "-----Section Cap-----"
    Debug.Print "Show section cap: " & data.ShowSectionCap
End Sub
```

Get Section View Data Example (VB.NET)

This example shows how to get the data for a section view in a part.

```

Preconditions:
1. Open public_documents\samples\tutorial\cosmosxpress\aw_rubber_duct.sldprt.
2. Create a section view with three sections:
   a. Right-click Front in the FeatureManager design tree
      and click Section View.
   b. Set X Rotation to 45.00deg in Section 1.
   c. Select Section 2, click Top, and
      set X Rotation to 45.00deg.
   d. Select Section 3, click Right, and
      set X Rotation to 45.00deg.
   e. Click Save twice.
3. Click View > Display > Section View twice.
4. Open the Immediate window.

Postconditions:
1. Gets and prints data about each section in
   the section view to the Immediate window.
2. Examine the Immediate window.

NOTE: Because the part is used elsewhere, do not save changes.

```

```

Imports SolidWorks.Interop.sldworks
Imports SolidWorks.Interop.swconst
Imports System.Runtime.InteropServices
Imports System
Imports System.Diagnostics

Partial Class SolidWorksMacro

    Public Sub main()

        Dim swModel As ModelDoc2
        Dim swModelViewMgr As ModelViewManager
        Dim swSectionViewData As SectionViewData
        Dim swSelMgr As SelectionMgr
        Dim status As Boolean

        swModel = swApp.ActiveDoc
        swSelMgr = swModel.SelectionManager

        'Get section view
        swModelViewMgr = swModel.ModelViewManager
        swSectionViewData = swModelViewMgr.GetSectionViewData("")

        DisplayDebugInformation(swSectionViewData, swSelMgr)

        swModel.ClearSelection2(True)

    End Sub

    'Select the planes and print each section's data
    Sub DisplayDebugInformation(ByVal data As SectionViewData, ByVal selMgr As SelectionMgr)
        Dim p1 As Feature
        Dim p2 As Feature
        Dim p3 As Feature
        Dim swMathTransform As MathTransform
        Dim transform As Object

        p1 = data.FirstPlane
        If Not p1 Is Nothing Then p1.Select2(True, 1)
        p2 = data.SecondPlane
        If Not p2 Is Nothing Then p2.Select2(True, 2)
        p3 = data.ThirdPlane
        If Not p3 Is Nothing Then p3.Select2(True, 4)

        Debug.Print("-----Section 1-----")
        Debug.Print("Offset: " & data.FirstOffset)
        Debug.Print("Rotation X: " & data.FirstRotationX)
        Debug.Print("Rotation Y: " & data.FirstRotationY)
        Debug.Print("Color: " & data.FirstColor)
        Debug.Print("Reverse direction: " & data.FirstReverseDirection)
        Debug.Print("-----Section 2-----")
        Debug.Print("Offset: " & data.SecondOffset)
        Debug.Print("Rotation X: " & data.SecondRotationX)
        Debug.Print("Rotation Y: " & data.SecondRotationY)
        Debug.Print("Color: " & data.SecondColor)
        Debug.Print("Reverse direction: " & data.SecondReverseDirection)
        Debug.Print("-----Section 3-----")
        Debug.Print("Offset: " & data.ThirdOffset)
        Debug.Print("Rotation X: " & data.ThirdRotationX)
        Debug.Print("Rotation Y: " & data.ThirdRotationY)
        Debug.Print("Color: " & data.ThirdColor)
        Debug.Print("Reverse direction: " & data.ThirdReverseDirection)
        swMathTransform = data.GetDynamicCenterTransform(3)
        Debug.Print("Dynamic center transform:")
        transform = swMathTransform.ArrayData()
        If Not IsNothing(transform) Then
            Debug.Print(" Rotate: " & Format(transform(0), "###0.0#####") & " " & Format(transform(1), "###0.0#####") & " " & Format(transform(2), "###0.0#####") & " " & Format(transform(3), "###0.0#####") & " " & Format(transform(4), "###0.0#####") & " " & Format(transform(5), "###0.0#####") & " " & Format(transform(6), "###0.0#####") & " " & Format(transform(7), "###0.0#####") & " " & Format(transform(8), "###0.0#####") & " " & Format(transform(9), "###0.0#####") & " " & Format(transform(10), "###0.0#####") & " " & Format(transform(11), "###0.0#####"))
        End If
        Debug.Print("-----Section Cap-----")
        Debug.Print("Show section cap: " & data.ShowSectionCap)

    End Sub

    ''' <summary>
    ''' The SldWorks swApp variable is pre-assigned for you.
    ''' </summary>

```

Public swApp As SldWorks

End Class

Get Sectioned Bodies Example (VBA)

This example shows how to get sectioned bodies and create a new part using the sectioned bodies.

```
'-----
' Preconditions:
' 1. Verify that the specified assembly document to open exists.
' 2. Verify that the specified part document template exists.
' 3. Open the Immediate window.
' 4. Step through the macro by pressing F8.
'
' Postconditions:
' 1. Opens the specified assembly document.
' 2. Creates a section view.
' 3. Selects a component in the section view.
' 4. Creates a new part.
' 5. Creates a feature using the body in the component
'    in the section view selected in step 3.
' 6. Examine the FeatureManager design tree, graphics
'    area, and Immediate window.
'
' NOTE: Because the assembly is used elsewhere, do not
' save changes.
'-----
Option Explicit

Dim swApp As SldWorks.SldWorks
Dim swModel As SldWorks.ModelDoc2
Dim swModelDocExt As SldWorks.ModelDocExtension
Dim swModelViewMgr As SldWorks.ModelViewManager
Dim swSelMgr As SldWorks.SelectionMgr
Dim swComp As SldWorks.Component2
Dim swModelView As SldWorks.ModelView
Dim swSectionViewData As SldWorks.SectionViewData
Dim vBodyArr As Variant
Dim vBody As Variant
Dim swBody As SldWorks.Body2
Dim swNewPart As SldWorks.PartDoc
Dim swFeat As SldWorks.Feature
Dim status As Boolean
Dim errors As Long
Dim warnings As Long
Dim fileName As String

Sub main()

    Set swApp = Application.SldWorks

    'Open assembly, create section view, and select
    'a component in the section view
    fileName = "C:\Users\Public\Documents\SOLIDWORKS\SOLIDWORKS 2018\samples\tutorial\api\landing_gear.sldasm"
    Set swModel = swApp.OpenDoc6(fileName, swDocumentTypes_e.swDocASSEMBLY, swOpenDocOptions_e.swOpenDocOptions_Silent, "", errors, warnings)
    Set swModelDocExt = swModel.Extension
    status = swModelDocExt.SelectByID2("Plane1", "PLANE", 0, 0, 0, True, 1, Nothing, 0)
    Set swModelViewMgr = swModel.ModelViewManager
    Set swSectionViewData = swModelViewMgr.CreateSectionViewData()
    swSectionViewData.FirstPlane = Nothing
    swSectionViewData.FirstReverseDirection = False
    swSectionViewData.FirstOffset = 0
    swSectionViewData.FirstRotationX = 0
    swSectionViewData.FirstRotationY = 0
    swSectionViewData.FirstColor = 16711680
    swSectionViewData.ShowSectionCap = True
    swSectionViewData.KeepCapColor = False
    swSectionViewData.GraphicsOnlySection = False
    status = swModelViewMgr.CreateSectionView(swSectionViewData)
    swModel.ClearSelection2 True
    status = swModelDocExt.SelectByID2("oleostrut-1@landing_gear", "COMPONENT", 0, 0, 0, False, 0, Nothing, 0)
    Set swSelMgr = swModel.SelectionManager
    Set swComp = swSelMgr.GetSelectedObjectsComponent4(1, -1)

    Set swModelView = swModel.ActiveView
    status = swModelView.GetDisplayState(swViewDisplayType_e.swIsViewSectioned)
    If status Then
        'Create new part using selected component
        Set swModel = swApp.NewDocument("C:\ProgramData\SolidWorks\SolidWorks 2014\templates\part.prt", swDwgPaperAsize, 0, 0)
        Set swNewPart = swModel
        vBodyArr = swComp.GetSectionedBodies(swModelView)
        For Each vBody In vBodyArr
            Set swBody = vBody
            Set swFeat = swNewPart.CreateFeatureFromBody3(swBody, False, swCreateFeatureBodyCheck)
        Next vBody
    Else
        Debug.Print "Model view does not contain a section view."
    End If
End Sub
```

Get Selected Faces on Processed Body Example (C#)

This example shows how to get the selected faces on a processed body.

```
//-----
// Preconditions:
// 1. Verify that the specified part document template
//    and part exist.
// 2. Open the Immediate window.
//
// Postconditions:
// 1. Opens the specified part.
// 2. Creates a new part with an imported body
//    from the processed body of the original part.
// 3. Gets the body in the new part and selects
//    multiple faces on that body.
// 4. Examine the Immediate window.
//
// NOTE: Because the part is used elsewhere, do not
// save changes.
//-----
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System.Runtime.InteropServices;
using System;
using System.Diagnostics;

namespace Macro1CSharp.csproj
{
    public partial class SolidWorksMacro
    {
        public void Main()
        {
            // 1 radian = 180°/p = 57.295779513° or approximately 57.3°
            const double RadPerDeg = 1.0 / 57.3;
            const double MaxUAngle = 1.0 * RadPerDeg;
            const double MaxVAngle = 1.0 * RadPerDeg;

            ModelDoc2 swModel = default(ModelDoc2);
            ModelDocExtension swModelDocExt = default(ModelDocExtension);
            Body2 swBody = default(Body2);
            Body2 swProcBody = default(Body2);
            PartDoc swPart = default(PartDoc);
            PartDoc swNewPart = default(PartDoc);
            Feature swFeat = default(Feature);
            object[] vBodies = new object[1];
            int errors = 0;
            int warnings = 0;
            string fileName = null;
            bool status = false;
            object[] swBodyVar = new object[1];
            object[] swSelFaceVar = new object[4];
            int swSelFaceCount = 0;

            //Open part
            fileName = "C:\\Users\\Public\\Documents\\SOLIDWORKS\\SOLIDWORKS 2018\\samples\\tutorial\\api\\block.sldprt";
            swModel = (ModelDoc2)swApp.OpenDoc6(fileName, (int)swDocumentTypes_e.swDocPART, (int)swOpenDocOptions_e.swOpenDocOptions_Silent, "", ref error
s, ref warnings);
            swPart = (PartDoc)swModel;

            //Get and process body in part
            vBodies = (object[])swPart.GetBodies2((int)swBodyType_e.swSolidBody, false);
            swBody = (Body2)vBodies[0];
            swProcBody = (Body2)swBody.GetProcessedBody2(MaxUAngle, MaxVAngle);

            //Create new part containing processed body
            swNewPart = (PartDoc)swApp.NewDocument("C:\\ProgramData\\SOLIDWORKS\\SOLIDWORKS 2014\\templates\\part.prtdot", 0, 0, 0);
            swFeat = (Feature)swNewPart.CreateFeatureFromBody3(swProcBody, false, (int)swCreateFeatureBodyOpts_e.swCreateFeatureBodyCheck);
            Debug.Print("Original part: " + swModel.GetPathName());
            Debug.Print(" Title: " + swModel.GetTitle());
            Debug.Print(" Body faces: " + swBody.GetFaceCount());
            Debug.Print(" Processed body faces: " + swProcBody.GetFaceCount());

            //Select multiple faces in new part
            swModel = (ModelDoc2)swNewPart;
            Debug.Print("New part title: " + swModel.GetTitle());
            Debug.Print(" Body faces: " + swBody.GetFaceCount());
            Debug.Print(" Processed body faces: " + swProcBody.GetFaceCount());
            swModelDocExt = (ModelDocExtension)swModel.Extension;
            status = swModelDocExt.SelectByID2("", "FACE", -0.0258707587273648, -0.00453920675113295, -0.00750000000022055, false, 0, null, 0);
            status = swModelDocExt.SelectByID2("", "FACE", -0.016247803762667, 0, -0.0112417538793466, true, 0, null, 0);
            status = swModelDocExt.SelectByID2("", "FACE", -0.0149546544521968, -0.026689891165347, 0, true, 0, null, 0);
            status = swModelDocExt.SelectByID2("", "FACE", -0.0208314165242882, -0.0200000000001523, -0.00322480979224338, true, 0, null, 0);

            //Get selected faces in body
            swBodyVar = (object[])swNewPart.GetBodies2((int)swBodyType_e.swAllBodies, true);
            if ((swBodyVar == null))
            {
                Debug.Print(" Did not get any bodies.");
            }
            else
            {
                swBody = (Body2)swBodyVar[0];
                Debug.Print(" Name of processed body: " + swBody.Name);
            }
            swProcBody = (Body2)swBody.GetProcessedBodyWithSelFace();
            swSelFaceVar = (object[])swProcBody.GetSelectedFaces();
            if ((swSelFaceVar != null))
            {
                swSelFaceCount = swProcBody.GetSelectedFaceCount();
                Debug.Print(" Number of faces selected in processed body: " + swSelFaceCount);
            }
            else
            {
                Debug.Print(" No faces selected in processed body.");
            }
        }
    }
}
```

```
    /// <summary>  
    /// The Sldworks swApp variable is pre-assigned for you.  
    /// </summary>  
    public SldWorks swApp;  
}
```

Get Selected Faces on Processed Body Example (VBA)

This example shows how to get the selected faces on a processed body.

```

'-----
' Preconditions:
' 1. Verify that the specified part document template
'    and part exist.
' 2. Open the Immediate window.
'
' Postconditions:
' 1. Opens the specified part.
' 2. Creates a new part with an imported body
'    from the processed body of the original part.
' 3. Gets the body in the new part and selects
'    multiple faces on that body.
' 4. Examine the Immediate window.
'
' NOTE: Because the part is used elsewhere, do not
' save changes.
'-----
Option Explicit

Sub main()

    ' 1 radian = 180°/p = 57.295779513° or approximately 57.3°
    Const RadPerDeg As Double = 1# / 57.3
    Const MaxUAngle As Double = 1# * RadPerDeg
    Const MaxVAngle As Double = 1# * RadPerDeg

    Dim swApp As SldWorks.SldWorks
    Dim swModel As SldWorks.ModelDoc2
    Dim swModelDocExt As SldWorks.ModelDocExtension
    Dim swBody As SldWorks.Body2
    Dim swProcBody As SldWorks.Body2
    Dim swPart As SldWorks.PartDoc
    Dim swNewPart As SldWorks.PartDoc
    Dim swFeat As SldWorks.Feature
    Dim vBodies As Variant
    Dim errors As Long
    Dim warnings As Long
    Dim fileName As String
    Dim status As Boolean
    Dim swBodyVar As Variant
    Dim swSelFaceVar As Variant
    Dim swSelFaceCount As Long

    Set swApp = Application.SldWorks

    'Open part
    fileName = "C:\Users\Public\Documents\SOLIDWORKS\SOLIDWORKS 2018\samples\tutorial\api\block.sldprt"
    Set swModel = swApp.OpenDoc6(fileName, swDocumentTypes_e.swDocPART, swOpenDocOptions_e.swOpenDocOptions_Silent, "", errors, warnings)
    Set swPart = swModel

    'Get and process body in part
    vBodies = swPart.GetBodies2(swSolidBody, False)
    Set swBody = vBodies(0)
    Set swProcBody = swBody.GetProcessedBody2(MaxUAngle, MaxVAngle)

    'Create new part containing processed body
    Set swNewPart = swApp.NewDocument("C:\ProgramData\SOLIDWORKS\SOLIDWORKS 2015\templates\part.prtdot", 0, 0, 0)
    Set swFeat = swNewPart.CreateFeatureFromBody3(swProcBody, False, swCreateFeatureBodyCheck)
    Debug.Print "Original part: " & swModel.GetPathName
    Debug.Print "    Title: " & swModel.GetTitle
    Debug.Print "    Body faces: " & swBody.GetFaceCount
    Debug.Print "    Processed body faces: " & swProcBody.GetFaceCount

    'Select multiple faces in new part
    Set swModel = swNewPart
    Debug.Print "New part title: " & swModel.GetTitle
    Debug.Print "    Body faces: " & swBody.GetFaceCount
    Debug.Print "    Processed body faces: " & swProcBody.GetFaceCount
    Set swModelDocExt = swModel.Extension
    status = swModelDocExt.SelectByID("", "FACE", -2.58707587273648E-02, -4.53920675113295E-03, -7.50000000022055E-03, False, 0, Nothing, 0)
    status = swModelDocExt.SelectByID("", "FACE", -0.016247803762667, 0, -1.12417538793466E-02, True, 0, Nothing, 0)
    status = swModelDocExt.SelectByID("", "FACE", -1.49546544521968E-02, -0.026689891165347, 0, True, 0, Nothing, 0)
    status = swModelDocExt.SelectByID("", "FACE", -2.08314165242882E-02, -2.00000000001523E-02, -3.22480979224338E-03, True, 0, Nothing, 0)

    'Get selected faces in processed body
    swBodyVar = swNewPart.GetBodies2(swAllBodies, True)
    If IsEmpty(swBodyVar) Then
        Debug.Print "    Did not get any bodies."
    Else
        Set swBody = swBodyVar(0)
        Debug.Print "    Name of processed body: " & swBody.Name
    End If
    Set swProcBody = swBody.GetProcessedBodyWithSelFace
    swSelFaceVar = swProcBody.GetSelectedFaces
    If Not IsEmpty(swSelFaceVar) Then
        swSelFaceCount = swProcBody.GetSelectedFaceCount
        Debug.Print "    Number of faces selected in processed body: " & swSelFaceCount
    Else
        Debug.Print "    No faces selected in processed body."
    End If

End Sub

```

Get Selected Faces on Processed Body Example (VB.NET)

This example shows how to get the selected faces on a processed body.

```

-----
Preconditions:
1. Verify that the specified part document template
   and part exist.
2. Open the Immediate window.
Postconditions:
1. Opens the specified part.
2. Creates a new part with an imported body
   from the processed body of the original part.
3. Gets the body in the new part and selects
   multiple faces on that body.
4. Examine the Immediate window.
NOTE: Because the part is used elsewhere, do not
save changes.
-----
Imports SolidWorks.Interop.sldworks
Imports SolidWorks.Interop.swconst
Imports System.Runtime.InteropServices
Imports System
Imports System.Diagnostics

Partial Class SolidWorksMacro

    Public Sub main()

        ' 1 radian = 180°/p = 57.295779513° or approximately 57.3°
        Const RadPerDeg As Double = 1.0# / 57.3
        Const MaxUAngle As Double = 1.0# * RadPerDeg
        Const MaxVAngle As Double = 1.0# * RadPerDeg

        Dim swModel As ModelDoc2
        Dim swModelDocExt As ModelDocExtension
        Dim swBody As Body2
        Dim swProcBody As Body2
        Dim swPart As PartDoc
        Dim swNewPart As PartDoc
        Dim swFeat As Feature
        Dim vBodies(0) As Object
        Dim errors As Integer
        Dim warnings As Integer
        Dim fileName As String
        Dim status As Boolean
        Dim swBodyVar(0) As Object
        Dim swSelFaceVar(3) As Object
        Dim swSelFaceCount As Integer

        'Open part
        fileName = "C:\Users\Public\Documents\SOLIDWORKS\SOLIDWORKS 2018\samples\tutorial\api\block.sldprt"
        swModel = swApp.OpenDoc6(fileName, swDocumentTypes_e.swDocPART, swOpenDocOptions_e.swOpenDocOptions_Silent, "", errors, warnings)
        swPart = swModel

        'Get and process body in part
        vBodies = swPart.GetBodies2(swBodyType_e.swSolidBody, False)
        swBody = vBodies(0)
        swProcBody = swBody.GetProcessedBody2(MaxUAngle, MaxVAngle)

        'Create new part containing processed body
        swNewPart = swApp.NewDocument("C:\ProgramData\SOLIDWORKS\SOLIDWORKS 2015\templates\part.prtdot", 0, 0, 0)
        swFeat = swNewPart.CreateFeatureFromBody3(swProcBody, False, swCreateFeatureBodyOpts_e.swCreateFeatureBodyCheck)
        Debug.Print("Original part: " & swModel.GetPathName)
        Debug.Print(" Title: " & swModel.GetTitle)
        Debug.Print(" Body faces: " & swBody.GetFaceCount)
        Debug.Print(" Processed body faces: " & swProcBody.GetFaceCount)

        'Select multiple faces in new part
        swModel = swNewPart
        Debug.Print("New part title: " & swModel.GetTitle)
        Debug.Print(" Body faces: " & swBody.GetFaceCount)
        Debug.Print(" Processed body faces: " & swProcBody.GetFaceCount)
        swModelDocExt = swModel.Extension
        status = swModelDocExt.SelectByID("", "FACE", -0.0258707587273648, -0.00453920675113295, -0.00750000000022055, False, 0, Nothing, 0)
        status = swModelDocExt.SelectByID("", "FACE", -0.016247803762667, 0, -0.0112417538793466, True, 0, Nothing, 0)
        status = swModelDocExt.SelectByID("", "FACE", -0.0149546544521968, -0.026689891165347, 0, True, 0, Nothing, 0)
        status = swModelDocExt.SelectByID("", "FACE", -0.0208314165242882, -0.0200000000001523, -0.00322480979224338, True, 0, Nothing, 0)

        'Get selected faces in body
        swBodyVar = swNewPart.GetBodies2(swBodyType_e.swAllBodies, True)
        If IsNothing(swBodyVar) Then
            Debug.Print(" Did not get any bodies.")
        Else
            swBody = swBodyVar(0)
            Debug.Print(" Name of processed body: " & swBody.Name)
        End If
        swProcBody = swBody.GetProcessedBodyWithSelFace
        swSelFaceVar = swProcBody.GetSelectedFaces
        If Not IsNothing(swSelFaceVar) Then
            swSelFaceCount = swProcBody.GetSelectedFaceCount
            Debug.Print(" Number of faces selected in processed body: " & swSelFaceCount)
        Else
            Debug.Print(" No faces selected in processed body.")
        End If

    End Sub

    ''' <summary>
    ''' The SldWorks swApp variable is pre-assigned for you.
    ''' </summary>
    Public swApp As SldWorks

End Class

```

Get Selected Feature Example

This example shows how to get the selected feature and its type, and return the object to the calling method.

```
LPFEATURE CGetSelectedFeatureApp::GetSelectedFeature1( LPMODELDOC2 pModelDoc, CString* Message )
```

```
{
    long lres = 0;
    LPSELECTIONMGR pSelectMgr = NULL;
    HRESULT hres =NULL;
    LPUNKNOWN pUnk = NULL;
    LPFEATURE pSelectedFeature = NULL;
    BSTR bFeatureTypeName;
    BSTR bFeatureName;

    // Get the Selection Manager for this doc
    if(S_OK != pModelDoc->get_ISelectionManager( &pSelectMgr ) ||
    pSelectMgr == NULL)
    {
        *Message = "Unable to get the Selection Manager.\r\n";
    }
    else if(S_OK != pSelectMgr->GetSelectedObjectType( 1, &lres )
    || lres == 0)
    {
        // Get the Selected Object type
        *Message = "No Feature is currently selected.\r\n";
    }
    else if(S_OK != pSelectMgr->IGetSelectedObject2( 1, &pUnk )
    || pUnk == NULL)
    {
        // Get the Selected Object of Unknown Type
        *Message = "Unable to get the selected Object.\r\n";
    }
    else if(S_OK != pUnk->QueryInterface( IID_IFeature,
    (LPVOID *)&pSelectedFeature) || pSelectedFeature == NULL)
    {
        // Get the Selected Feature
        *Message = "Unable to get the selected Feature.\r\n";
    }
    else if(S_OK != pSelectedFeature->GetTypeName(&bFeatureTypeName))
    bFeatureTypeName == NULL)
    {
        // Get the Selected Feature Type Name

        *Message = "Unable to get the selected Feature Type Name.\r\n";
    }
    else if(S_OK != pSelectedFeature->get_Name(&bFeatureName))
    bFeatureName == NULL)
    {
        // Get the Selected Feature Name
        *Message = "Unable to get the selected Feature Name.\r\n";
    }
    else
    {
        // Everything is okay. Clean up and report back with object.

        // Release the Selection Manager
        pSelectMgr->Release();
        // Release the unknown object
        pUnk->Release();

        // Form the return message
        CString FeatureTypeName(bFeatureTypeName);
        CString FeatureName(bFeatureName);
        *Message = "Found a Feature called " + FeatureName + " of type "
        + FeatureTypeName + ".\r\n";

        SysFreeString(bFeatureTypeName);
    }
}
```

```
SysFreeString(bFeatureName);  
// Return with Feature object  
return pSelectedFeature;  
}  
// Something went wrong. Clean up and report back without object.  
// Release the Selection Manager  
if(pSelectMgr)pSelectMgr->Release();  
// Release the unknown object  
if(pUnk)pUnk->Release();  
// Return with no object  
return NULL;  
}
```

Get Selected Object Example (VBA)

This example shows how to select and get that object in a SOLIDWORKS Design document.

```
'-----  
' Preconditions.  
' 1. Open public_documents\samples\tutorial\api\box.sldprt.  
' 2. Open the Immediate window.  
'  
' Postconditions:  
' 1. Selects the Boss-Extrude1 feature.  
' 2. Examine the Immediate window and graphics area.  
'-----  
Option Explicit  
  
Dim swApp As SldWorks.SldWorks  
Dim Model As ModelDoc2  
Dim feature As feature  
Dim boolstatus As Variant  
  
Sub main()  
  
    Set swApp = Application.SldWorks  
    Set Model = swApp.ActiveDoc  
  
    boolstatus = Model.Extension.SelectByID2("Boss-Extrude1", "BODYFEATURE", 0, 0, 0, False, 0, Nothing, swSelectOptionDefault)  
  
    If boolstatus = True Then  
        Dim SelMgr As SelectionMgr  
        Set SelMgr = Model.SelectionManager  
        Set feature = SelMgr.GetSelectedObject6(1, 0)  
        Debug.Print (feature.Name & " selected")  
    Else  
        Debug.Print "Error"  
    End If  
  
End Sub
```

Get Selected Objects and Types in an Assembly Example (VBA)

This example shows how to determine what is currently selected in an assembly document.

NOTE: You can select many features and entities in SOLIDWORKS Design. In most cases, it is obvious what is selected, but sometimes it is not clear or it is ambiguous. This example is a diagnostic tool to determine what is currently selected. It shows several techniques and methods to get a reference to what is selected.

```

'-----
' Preconditions:
' 1. Open an assembly.
' 2. Select a feature or entity in the assembly.
' 3. Open the Immediate window.
'
' Postconditions: Examine the Immediate window.
'-----
Option Explicit

Sub main()

    Dim swApp As SldWorks.SldWorks
    Dim swModel As SldWorks.ModelDoc2
    Dim swSelMgr As SldWorks.SelectionMgr
    Dim swSelObj As Object
    Dim swFeat As SldWorks.Feature
    Dim swEnt As SldWorks.Entity
    Dim swBody As SldWorks.Body2
    Dim swSelComp As SldWorks.Component2
    Dim swSelModel As SldWorks.ModelDoc2
    Dim nSelType As Long
    Dim sFeatName As String
    Dim bRet As Boolean

    ' Disables Visual Basic's implicit error on QueryInterface
    On Error Resume Next

    Set swApp = CreateObject("SldWorks.Application")
    Set swModel = swApp.ActiveDoc
    Set swSelMgr = swModel.SelectionManager

    ' Could either have a feature or entity selected
    ' Do not try to get entity directly
    ' from feature because feature might be Nothing
    ' Instead, use ISelectionMgr
    Set swFeat = swSelMgr.GetSelectedObject6(1, -1)
    Set swEnt = swSelMgr.GetSelectedObject6(1, -1)
    Set swBody = swSelMgr.GetSelectedObject6(1, -1)
    Set swSelObj = swSelMgr.GetSelectedObject6(1, -1)
    Set swSelComp = swSelMgr.GetSelectedObjectsComponent3(1, -1)
    Debug.Print "Selected Type      = " & swSelMgr.GetSelectedObjectType3(1, -1)
    If Not swFeat Is Nothing Then
        Debug.Print "Feature          = " & swFeat.Name & " [" & swFeat.GetTypeName & "]"
    End If

    If Not swBody Is Nothing Then
        Debug.Print "    Body selected"
    End If

    If swFeat Is Nothing And swEnt Is Nothing And swBody Is Nothing And Not swSelObj Is Nothing Then
        Debug.Print "    Unknown object"
    End If

    ' Could not get component from ISelectionMgr,
    ' so try and get component through IEntity
    If swSelComp Is Nothing Then
        Set swSelComp = swEnt.GetComponent
    End If

    If Not swSelComp Is Nothing Then
        Set swSelModel = swSelComp.GetModelDoc2
        Debug.Print "Component Name      = " & swSelComp.Name2
        Debug.Print "Model Path          = " & swSelModel.GetPathName
    End If
End Sub

```

Get Selection Specification Example (C#)

This example shows how to get the selection specification of a selected object.

```
//-----
// Preconditions:
// 1. Open a model document that contains at least two features.
// 2. Select two features in the FeatureManager design tree.
// 3. Open an Immediate window.
//
// Postconditions:
// 1. Gets the selection specifications of the first and second selections.
// 2. Inspect the Immediate window.
//-----
using Microsoft.VisualBasic;
using System;
using System.Collections;
using System.Collections.Generic;
using System.Data;
using System.Diagnostics;
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System.Runtime.InteropServices;
namespace GetSelectionSpecification_CSharp.csproj
{
    partial class SolidWorksMacro
    {
        ModelDoc2 Part;
        SelectionMgr SelMgr;
        Feature selObject;
        string selectByString;
        string objectTypeStr;
        int objectTypeInt;
        double X;
        double Y;
        double Z;

        public void Main()
        {
            Part = (ModelDoc2)swApp.ActiveDoc;
            SelMgr = (SelectionMgr)Part.SelectionManager;

            // Get the selection specification of the first selected object
            selObject = (Feature)SelMgr.GetSelectedObject6(1, -1);
            SelMgr.GetSelectByIdSpecification(selObject, out selectByString, out objectTypeStr, out objectTypeInt);
            Debug.Print("Name of selected feature: " + selectByString);
            Debug.Print("Type of object: " + objectTypeStr);
            Debug.Print("Type of object as defined in swSelectType_e: " + objectTypeInt);

            Debug.Print("");

            // Get the selection specification of the selected object in position 2 of the selection list
            SelMgr.GetSelectionSpecification(2, out selectByString, out objectTypeStr, out objectTypeInt, out X, out Y, out Z);
            Debug.Print("Name of selected feature: " + selectByString);
            Debug.Print("Type of object: " + objectTypeStr);
            Debug.Print("Type of object as defined in swSelectType_e: " + objectTypeInt);
            Debug.Print("X Coordinate: " + X);
            Debug.Print("Y Coordinate: " + Y);
            Debug.Print("Z Coordinate: " + Z);

        }

        public SldWorks swApp;
    }
}
```

Get Selection Specification Example (VBA)

This example shows how to get the selection specification of a selected object.

```

'-----
' Preconditions:
' 1. Open a model document that contains at least two features.
' 2. Select two features in the FeatureManager design tree.
' 3. Open an Immediate window.
'
' Postconditions:
' 1. Gets the selection specifications of the first and second selections.
' 2. Inspect the Immediate window.
'-----

Dim swApp As SldWorks.SldWorks
Dim Part As SldWorks.ModelDoc2
Dim SelMgr As SldWorks.SelectionMgr
Dim selObject As SldWorks.Feature
Dim selectByString As String
Dim objectTypeStr As String
Dim objectTypeInt As Long
Dim X As Double
Dim Y As Double
Dim Z As Double

Option Explicit

Sub main()
    Set swApp = Application.SldWorks
    Set Part = swApp.ActiveDoc
    Set SelMgr = Part.SelectionManager

    ' Get the selection specification of the first selected object
    Set selObject = SelMgr.GetSelectedObject6(1, -1)
    SelMgr.GetSelectByIdSpecification selObject, selectByString, objectTypeStr, objectTypeInt
    Debug.Print "Name of selected feature: " & selectByString
    Debug.Print "Type of object: " & objectTypeStr
    Debug.Print "Type of object as defined in swSelectType_e: " & objectTypeInt

    Debug.Print ""

    ' Get the selection specification of the selected object in position 2 of the selection list
    SelMgr.GetSelectionSpecification 2, selectByString, objectTypeStr, objectTypeInt, X, Y, Z
    Debug.Print "Name of selected feature: " & selectByString
    Debug.Print "Type of object: " & objectTypeStr
    Debug.Print "Type of object as defined in swSelectType_e: " & objectTypeInt
    Debug.Print "X Coordinate: " & X
    Debug.Print "Y Coordinate: " & Y
    Debug.Print "Z Coordinate: " & Z

End Sub

```

Get Selection Specification Example (VB.NET)

This example shows how to get the selection specification of a selected object.

```

'-----
' Preconditions:
' 1. Open a model document that contains at least two features.
' 2. Select two features in the FeatureManager design tree.
' 3. Open an Immediate window.
'
' Postconditions:
' 1. Gets the selection specifications of the first and second selections.
' 2. Inspect the Immediate window.
'-----
Imports SolidWorks.Interop.sldworks
Imports SolidWorks.Interop.swconst
Imports System.Runtime.InteropServices
Imports System
Imports System.Diagnostics

Partial Class SolidWorksMacro

    Dim Part As ModelDoc2
    Dim SelMgr As SelectionMgr
    Dim selObject As Feature
    Dim selectByString As String
    Dim objectTypeStr As String
    Dim objectTypeInt As Integer
    Dim X As Double
    Dim Y As Double
    Dim Z As Double

    Sub main()

        Part = swApp.ActiveDoc
        SelMgr = Part.SelectionManager

        ' Get the selection specification of the first selected object
        selObject = SelMgr.GetSelectedObject6(1, -1)
        SelMgr.GetSelectByIdSpecification(selObject, selectByString, objectTypeStr, objectTypeInt)
        Debug.Print("Name of selected feature: " & selectByString)
        Debug.Print("Type of object: " & objectTypeStr)
        Debug.Print("Type of object as defined in swSelectType_e: " & objectTypeInt)

        Debug.Print("")

        ' Get the selection specification of the selected object in position 2 of the selection list
        SelMgr.GetSelectionSpecification(2, selectByString, objectTypeStr, objectTypeInt, X, Y, Z)
        Debug.Print("Name of selected feature: " & selectByString)
        Debug.Print("Type of object: " & objectTypeStr)
        Debug.Print("Type of object as defined in swSelectType_e: " & objectTypeInt)
        Debug.Print("X Coordinate: " & X)
        Debug.Print("Y Coordinate: " & Y)
        Debug.Print("Z Coordinate: " & Z)

    End Sub

    Public swApp As SldWorks

End Class

```

Get Selections for Reference Axis Feature Example (C#)

This example shows how to get the selections for a reference axis feature.

```
//-----
// Preconditions:
// 1. Verify that the document to open exists.
// 2. Open the Immediate window.
//
// Postconditions:
// 1. Opens the specified document.
// 2. Gets the Axis1 feature.
// 3. Gets the entities that define Axis1.
// 4. Gets the type and name of the entities that define
//    Axis1.
// 5. Examine the Immediate window.
//-----
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System.Runtime.InteropServices;
using System;
using System.Diagnostics;

namespace IRefAxisFeatureDataGetSelectionCSharp.csproj
{
    public partial class SolidWorksMacro
    {

        public void Main()
        {
            ModelDoc2 swModel = default(ModelDoc2);
            PartDoc swPart = default(PartDoc);
            Feature swFeat = default(Feature);
            RefAxisFeatureData swRefAxisFeatureData = default(RefAxisFeatureData);
            Entity swEntity = default(Entity);
            int warnings = 0;
            int errors = 0;
            string fileName = null;
            object[] obj = null;
            object types = null;
            string aType = null;
            string name = null;
            int i = 0;
            int[] entTypes;

            fileName = "C:\\Users\\Public\\Documents\\SOLIDWORKS\\SOLIDWORKS 2018\\samples\\tutorial\\api\\button.sldprt";
            swModel = (ModelDoc2)swApp.OpenDoc6(fileName, (int)swDocumentTypes_e.swDocPART, (int)swOpenDocOptions_e.swOpenDocOptions_Silent, "", ref error
s, ref warnings);
            swPart = (PartDoc)swModel;

            swFeat = (Feature)swPart.FeatureByName("Axis1");
            swRefAxisFeatureData = (RefAxisFeatureData)swFeat.GetDefinition();
            swRefAxisFeatureData.AccessSelections(swModel, null);
            obj = (Object[])swRefAxisFeatureData.GetSelections(out types);
            swRefAxisFeatureData.ReleaseSelectionAccess();

            entTypes = (int[])types;

            Debug.Print("Entity:");
            Debug.Print("");
            for (i = 0; i < obj.GetLength(0); i++)
            {
                swEntity = (Entity)obj[i];
                swFeat = (Feature)swEntity;
                swEntity.Select(false);
                name = swFeat.GetNameForSelection(out aType);
                Debug.Print("  Type: " + entTypes[i]);
                Debug.Print("  Name: " + name);
                Debug.Print("");
            }
        }

        /// <summary>
        /// The SldWorks swApp variable is pre-assigned for you.
        /// </summary>
        public SldWorks swApp;
    }
}
```

Get Selections for Reference Axis Feature Example (VBA)

This example shows how to get the selections for a reference axis feature.

```
'-----
' Preconditions:
' 1. Verify that the document to open exists.
' 2. Open the Immediate window.
'
' Postconditions:
' 1. Opens the specified document.
' 2. Gets the Axis1 feature.
' 3. Gets the entities that define Axis1.
' 4. Gets the type and name of the entities that define
'    Axis1.
' 5. Examine the Immediate window.
'-----
Option Explicit

Dim swApp As SldWorks.SldWorks
Dim swModel As SldWorks.ModelDoc2
Dim swPart As SldWorks.PartDoc
Dim swFeat As SldWorks.Feature
Dim swRefAxisFeatureData As SldWorks.RefAxisFeatureData
Dim swEntity As SldWorks.Entity
Dim warnings As Long
Dim errors As Long
Dim fileName As String
Dim obj As Variant
Dim types As Variant
Dim aType As String
Dim name As String
Dim i As Long

Sub main()

    Set swApp = Application.SldWorks
    fileName = "C:\Users\Public\Documents\SOLIDWORKS\SOLIDWORKS 2018\samples\tutorial\api\button.sldprt"
    Set swModel = swApp.OpenDoc6(fileName, swDocumentTypes_e.swDocPART, swOpenDocOptions_e.swOpenDocOptions_Silent, "", errors, warnings)
    Set swPart = swModel

    Set swFeat = swPart.FeatureByName("Axis1")
    Set swRefAxisFeatureData = swFeat.GetDefinition
    swRefAxisFeatureData.AccessSelections swModel, Nothing
    obj = swRefAxisFeatureData.GetSelections(types)
    swRefAxisFeatureData.ReleaseSelectionAccess

    Debug.Print "Entity:"
    Debug.Print ""
    For i = 0 To UBound(obj)
        Set swEntity = obj(i)
        Set swFeat = swEntity
        swEntity.Select False
        name = swFeat.GetNameForSelection(aType)
        Debug.Print "  Type: " & types(i)
        Debug.Print "  Name: " & name
        Debug.Print ""
    Next i

End Sub
```

Get Selections for Reference Axis Feature Example (VB.NET)

This example shows how to get the selections for a reference axis feature.

```

-----
Preconditions:
1. Verify that the document to open exists.
2. Open the Immediate window.
Postconditions:
1. Opens the specified document.
2. Gets the Axis1 feature.
3. Gets the entities that define Axis1.
4. Gets the type and name of the entities that define Axis1.
5. Examine the Immediate window.
-----
Imports SolidWorks.Interop.sldworks
Imports SolidWorks.Interop.swconst
Imports System.Runtime.InteropServices
Imports System
Imports System.Diagnostics

Partial Class SolidWorksMacro

    Public Sub main()

        Dim swModel As ModelDoc2
        Dim swPart As PartDoc
        Dim swFeat As Feature
        Dim swRefAxisFeatureData As RefAxisFeatureData
        Dim swEntity As Entity
        Dim warnings As Integer
        Dim errors As Integer
        Dim fileName As String
        Dim obj() As Object = Nothing
        Dim types() As Integer = Nothing
        Dim aType As String
        Dim name As String
        Dim i As Integer

        fileName = "C:\Users\Public\Documents\SOLIDWORKS\SOLIDWORKS 2018\samples\tutorial\api\button.sldprt"
        swModel = swApp.OpenDoc6(fileName, swDocumentTypes_e.swDocPART, swOpenDocOptions_e.swOpenDocOptions_Silent, "", errors, warnings)
        swPart = swModel

        swFeat = swPart.FeatureByName("Axis1")
        swRefAxisFeatureData = swFeat.GetDefinition
        swRefAxisFeatureData.AccessSelections(swModel, Nothing)
        obj = swRefAxisFeatureData.GetSelections(types)
        swRefAxisFeatureData.ReleaseSelectionAccess()

        Debug.Print("Entity:")
        Debug.Print("")
        For i = 0 To UBound(obj)
            swEntity = obj(i)
            swFeat = swEntity
            swEntity.Select(False)
            name = swFeat.GetNameForSelection(aType)
            Debug.Print("  Type: " & types(i))
            Debug.Print("  Name: " & name)
            Debug.Print("")
        Next i

    End Sub

    ''' <summary>
    ''' The SldWorks swApp variable is pre-assigned for you.
    ''' </summary>
    Public swApp As SldWorks

```

End Class

Get Sense for Each Coedge in a Loop (VBA)

This example gets the coedges in a loop and their senses.

```
'-----
' Preconditions:
' 1. Open a model document.
' 2. Select a face on the model.
' 3. Open the Immediate window.
'
' Postconditions:
' 1. Traverses the loops on the selected face.
' 2. Gets the number of coedges on the selected face
'    and the sense (i.e., true if the coedge has the same
'    direction as the underlying edge, false if not)
'    for each coedge.
' 3. Examine the Immediate window.
'-----
```

Option Explicit

```
Dim swApp As SldWorks.SldWorks
Dim swModel As SldWorks.ModelDoc2
Dim swSelMgr As SldWorks.SelectionMgr
Dim myEdge As SldWorks.CoEdge
Dim myLoop As SldWorks.Loop2
Dim myFace As SldWorks.Face2
Dim myEdges As Variant
Dim index As Long
Dim count As Long

Sub main()
    Set swApp = Application.SldWorks
    Set swModel = swApp.ActiveDoc
    Set swSelMgr = swModel.SelectionManager
    Set myFace = swSelMgr.GetSelectedObject6(1, -1)

    ' Get the first loop on the selected face
    Set myLoop = myFace.GetFirstLoop
    ' Get the number of coedges in the loop
    count = myLoop.GetCoEdgeCount
    Debug.Print count

    ' Get the coedges in the loop
    myEdges = myLoop.GetCoEdges
    ' For each coedge, get its sense
        For index = LBound(myEdges) To UBound(myEdges)
            Set myEdge = myEdges(index)
            Debug.Print myEdge.GetSense
        Next index
End Sub
```

Get, Set, And Remove Tracking IDs On Body (VBA)

This example shows how to get, set, and remove tracking IDs on a body. Tracking IDs reliably associate application-specific data across modeling operations.

```
'-----
' Preconditions: Part document with a single part is open.
'               The selected body has not been assigned any
'               tracking IDs.
'
' Directions:   Run the macro once, and the
'               the selected body is assigned
'               a tracking ID of 12. Run
'               the macro again, and the selected body
'               is assigned tracking ID of 16. The
'               the tracking ID is then deleted from the
'               body.
'
' Postconditions: None
'-----
```

Option Explicit

Sub main()

```
Dim swApp           As SldWorks.SldWorks
Dim swModel         As SldWorks.ModelDoc2
Dim swModelExtn     As ModelDocExtension
Dim swSelMgr        As SldWorks.SelectionMgr
Dim swBody          As SldWorks.Body2
Dim Cookies         As Long
Dim vTrackingIDs    As Variant
Dim TrackingID      As Long
Dim bRebuild        As Boolean
```

```
Set swApp = CreateObject("SldWorks.Application")
```

```
Cookies = swApp.RegisterTrackingDefinition("Tracking_API")
```

```
If (Cookies = -1) Then
    End ' No valid cookies found
End If
```

```
Stop
```

```
Set swModel = swApp.ActiveDoc
Set swSelMgr = swModel.SelectionManager
```

```
Stop ' Select body in Bodies folder
```

```
Set swBody = swSelMgr.GetSelectedObject6(1, -1)
```

```
Debug.Print "-----"
```

```
Dim NbrTrackingIds As Long
NbrTrackingIds = swBody.GetTrackingIDsCount(Cookies)
Debug.Print "Body Tracking IDs:"
If (NbrTrackingIds = 0) Then
```

```
    Debug.Print " No tracking IDs"
```

```
    Debug.Print " Setting tracking ID..."
```

```
    Debug.Print " SetTrackingID Result (if 0, then call successful): " & swBody.SetTrackingID(Cookies, 12)
```

```
    Debug.Print " GetTrackingIDs Result (if 0, then call successful): " & swBody.GetTrackingIDs(Cookies, vTrackingIDs)
```

```
    Debug.Print " Tracking ID after SetTrackingID call : " & vTrackingIDs(0)
```

End

Else

```
Debug.Print " Tracking IDs exist..."
Debug.Print " Getting tracking ID..."
Debug.Print " GetTrackingIDs Result (if 0, then call successful): " & swBody.GetTrackingIDs(Cookies, vTrackingIDs)
Debug.Print " Tracking ID before SetTracingID call : " & vTrackingIDs(0)

Debug.Print " "
Debug.Print " SetTrackingID Result (if 0, then call successful): " & swBody.SetTrackingID(Cookies, 16)
Debug.Print " GetTrackingIDs Result (if 0, then call successful): " & swBody.GetTrackingIDs(Cookies, vTrackingIDs)
Debug.Print " Tracking ID after SetTrackingID call: " & vTrackingIDs(0)
Debug.Print " "
Debug.Print " Remove Tracking ID..."
Debug.Print " RemoveTrackingID Result (if 0, then call successful): " & swBody.RemoveTrackingID(Cookies)
Debug.Print " Number of tracking IDs on this body: " & swBody.GetTrackingIDsCount(Cookies)

Debug.Print "-----"
```

End If

End Sub

Get, Set, And Remove Tracking IDs On Edge (VBA)

This example shows how to get, set, and remove tracking IDs on an edge. Tracking IDs reliably associate application-specific data across modeling operations.

```
' -----
'
' Preconditions: Part document with a single part is open.
'               The selected edge on the part has not
'               been assigned any tracking IDs.
'
' Directions:   Run the macro once, and the
'               the selected edge is assigned
'               a tracking ID of 12. Run
'               the macro again, and the selected edge
'               is assigned a tracking ID of 16. The
'               the tracking ID is then deleted from the
'               edge.
'
' Postconditions: None
'
' -----
```

Option Explicit

```
Sub main()

    Dim swApp           As SldWorks.SldWorks
    Dim swModel         As SldWorks.ModelDoc2
    Dim swModelExtn     As ModelDocExtension
    Dim swSelMgr        As SldWorks.SelectionMgr
    Dim swEdge          As SldWorks.Edge
    Dim Cookies         As Long
    Dim vTrackingIDs    As Variant
    Dim TrackingID      As Long
    Dim bRebuild        As Boolean

    Set swApp = CreateObject("SldWorks.Application")

    Cookies = swApp.RegisterTrackingDefinition("Tracking_API")
    If (Cookies = -1) Then
        End ' No valid cookies found
    End If

    Stop

    Set swModel = swApp.ActiveDoc
    Set swSelMgr = swModel.SelectionManager

    Stop ' Select and edge on the part

    Set swEdge = swSelMgr.GetSelectedObject6(1, -1)

    Debug.Print "-----"
    Dim NbrTrackingIds As Long
    NbrTrackingIds = swEdge.GetTrackingIDsCount(Cookies)
    Debug.Print "Edge Tracking IDs:"
    If (NbrTrackingIds = 0) Then

        Debug.Print " No tracking IDs"
        Debug.Print " Setting tracking ID..."
    End If
End Sub
```

```
Debug.Print "      SetTrackingID Result (if 0, then call successful): " & swEdge.SetTrackingID(Cookies, 12)
Debug.Print "      GetTrackingIDs Result (if 0, then call successful): " & swEdge.GetTrackingIDs(Cookies, vTrackingIDs)
Debug.Print "      Tracking ID after SetTrackingID call : " & vTrackingIDs(0)
End
```

Else

```
Debug.Print "      Tracking IDs exist..."
Debug.Print "      Getting tracking ID..."
Debug.Print "      GetTrackingIDs Result (if 0, then call successful): " & swEdge.GetTrackingIDs(Cookies, vTrackingIDs)
Debug.Print "      Tracking ID before SetTracingID call : " & vTrackingIDs(0)
```

```
Debug.Print " "
Debug.Print "      SetTrackingID Result (if 0, then call successful): " & swEdge.SetTrackingID(Cookies, 16)
Debug.Print "      GetTrackingIDs Result (if 0, then call successful): " & swEdge.GetTrackingIDs(Cookies, vTrackingIDs)
Debug.Print "      Tracking ID after SetTrackingID call: " & vTrackingIDs(0)
Debug.Print " "
Debug.Print "      Remove Tracking ID..."
Debug.Print "      RemoveTrackingID Result (if 0, then call successful): " & swEdge.RemoveTrackingID(Cookies)
Debug.Print "      Number of tracking IDs on this body: " & swEdge.GetTrackingIDsCount(Cookies)
```

```
Debug.Print "-----"
```

End If

End Sub

Get, Set, And Remove Tracking IDs On Face (VBA)

This example shows how to get, set, and remove tracking IDs on a face. Tracking IDs reliably associate application-specific data across modeling operations.

```
'-----
'
' Preconditions: Part document with a single part is open.
'               The selected face on the part has not been
'               assigned any tracking IDs.
'
' Directions:   Run the macro once, and the
'               the selected face is assigned
'               a tracking ID of 12. Run
'               the macro again, and the selected face
'               is assigned a tracking ID of 16. The
'               the tracking ID is then deleted from the
'               face.
'
' Postconditions: None
'-----
```

Option Explicit

Sub main()

```
    Dim swApp           As SldWorks.SldWorks
    Dim swModel         As SldWorks.ModelDoc2
    Dim swModelExtn     As ModelDocExtension
    Dim swSelMgr        As SldWorks.SelectionMgr
    Dim swFace          As SldWorks.Face2
    Dim Cookies         As Long
    Dim vTrackingIDs    As Variant
    Dim TrackingID      As Long
    Dim bRebuild        As Boolean

    Set swApp = CreateObject("SldWorks.Application")

    Cookies = swApp.RegisterTrackingDefinition("Tracking_API")

    If (Cookies = -1) Then
        End ' No valid cookies found
    End If

    Stop

    Set swModel = swApp.ActiveDoc
    Set swSelMgr = swModel.SelectionManager

    Stop ' Select face on the part

    Set swFace = swSelMgr.GetSelectedObject6(1, -1)

    Debug.Print "-----"
    Dim NbrTrackingIds As Long
    NbrTrackingIds = swFace.GetTrackingIDsCount(Cookies)
    Debug.Print "Face Tracking IDs:"
    If (NbrTrackingIds = 0) Then

        Debug.Print " No tracking IDs"
        Debug.Print " Setting tracking ID..."
```

```
Debug.Print "      SetTrackingID Result (if 0, then call successful): " & swFace.SetTrackingID(Cookies, 12)
Debug.Print "      GetTrackingIDs Result (if 0, then call successful): " & swFace.GetTrackingIDs(Cookies, vTrackingIDs)
Debug.Print "      Tracking ID after SetTrackingID call : " & vTrackingIDs(0)
End
```

Else

```
Debug.Print "      Tracking IDs exist..."
Debug.Print "      Getting tracking ID..."
Debug.Print "      GetTrackingIDs Result (if 0, then call successful): " & swFace.GetTrackingIDs(Cookies, vTrackingIDs)
Debug.Print "      Tracking ID before SetTracingID call : " & vTrackingIDs(0)

Debug.Print " "
Debug.Print "      SetTrackingID Result (if 0, then call successful): " & swFace.SetTrackingID(Cookies, 16)
Debug.Print "      GetTrackingIDs Result (if 0, then call successful): " & swFace.GetTrackingIDs(Cookies, vTrackingIDs)
Debug.Print "      Tracking ID after SetTrackingID call: " & vTrackingIDs(0)
Debug.Print " "
Debug.Print "      Remove Tracking ID..."
Debug.Print "      RemoveTrackingID Result (if 0, then call successful): " & swFace.RemoveTrackingID(Cookies)
Debug.Print "      Number of tracking IDs on this body: " & swFace.GetTrackingIDsCount(Cookies)

Debug.Print "-----"
```

End If

End Sub

Get, Set, And Remove Tracking IDs On Vertex (VBA)

This example shows how to get, set, and remove tracking IDs on a vertex. Tracking IDs reliably associate application-specific data across modeling operations.

```
' -----
' Preconditions: Part document with a single part is open.
'               The selected vertex has not been
'               assigned any tracking IDs.
'
' Directions:   Run the macro once, and the
'               the selected vertex is assigned
'               a tracking ID of 12. Run
'               the macro again, and the selected vertex
'               is assigned tracking ID of 16. The
'               the tracking ID is then deleted from the
'               vertex.
'
' Postconditions: None
' -----
```

Option Explicit

Sub main()

```
Dim swApp          As SldWorks.SldWorks
Dim swModel        As SldWorks.ModelDoc2
Dim swModelExtn    As ModelDocExtension
Dim swSelMgr       As SldWorks.SelectionMgr
Dim swVertex       As SldWorks.Vertex
Dim Cookies        As Long
Dim vTrackingIDs   As Variant
Dim TrackingID     As Long
Dim bRebuild       As Boolean
```

```
Set swApp = CreateObject("SldWorks.Application")
```

```
Cookies = swApp.RegisterTrackingDefinition("Tracking_API")
If (Cookies = -1) Then
    End ' No valid cookies found
End If
```

```
Stop
```

```
Set swModel = swApp.ActiveDoc
Set swSelMgr = swModel.SelectionManager
```

```
Stop ' Select vertex on the part
```

```
Set swVertex = swSelMgr.GetSelectedObject6(1, -1)
```

```
Debug.Print "-----"
```

```
Dim NbrTrackingIds As Long
NbrTrackingIds = swVertex.GetTrackingIDsCount(Cookies)
Debug.Print "Vertex Tracking IDs:"
If (NbrTrackingIds = 0) Then
```

```
    Debug.Print "  No tracking IDs"
    Debug.Print "  Setting tracking ID..."
    Debug.Print "    SetTrackingID Result (if 0, then call successful): " & swVertex.SetTrackingID(Cookies, 12)
    Debug.Print "    GetTrackingIDs Result (if 0, then call successful): " & swVertex.GetTrackingIDs(Cookies, vTrackingIDs)
    Debug.Print "    Tracking ID after SetTrackingID call : " & vTrackingIDs(0)
```

End

Else

```
Debug.Print " Tracking IDs exist..."
Debug.Print " Getting tracking ID..."
Debug.Print " GetTrackingIDs Result (if 0, then call successful): " & swVertex.GetTrackingIDs(Cookies, vTrackingIDs)
Debug.Print " Tracking ID before SetTracingID call : " & vTrackingIDs(0)

Debug.Print " "
Debug.Print " SetTrackingID Result (if 0, then call successful): " & swVertex.SetTrackingID(Cookies, 16)
Debug.Print " GetTrackingIDs Result (if 0, then call successful): " & swVertex.GetTrackingIDs(Cookies, vTrackingIDs)
Debug.Print " Tracking ID after SetTrackingID call: " & vTrackingIDs(0)
Debug.Print " "
Debug.Print " Remove Tracking ID..."
Debug.Print " RemoveTrackingID Result (if 0, then call successful): " & swVertex.RemoveTrackingID(Cookies)
Debug.Print " Number of tracking IDs on this body: " & swVertex.GetTrackingIDsCount(Cookies)

Debug.Print "-----"
```

End If

End Sub

Get Sheet in Multi-sheet Drawing Example (C#)

This example shows how to get each sheet in a multi-sheet drawing document regardless whether the sheet is loaded.

```
//-----
// Preconditions:
// 1. Click File > Open.
// 2. Open public_documents\samples\tutorial\advdrawings\foodprocessor.sldrw.
// 3. Click Select sheets to open > Selected > Sheet1* (load) > OK > Open.
// 4. Open the Immediate window.
//
// Postconditions:
// 1. Loads Sheet1 only.
// 2. Mouse over Sheet2, Sheet3, and Sheet4 tabs and examine the
//    Immediate window.
//
// NOTE: Because this drawing is used elsewhere, do not save changes.
//-----

using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System;
using System.Diagnostics;
namespace SheetDrawingDocCSharp.csproj
{
    partial class SolidWorksMacro
    {
        public void Main()
        {
            ModelDoc2 swModel = default(ModelDoc2);
            DrawingDoc swDraw = default(DrawingDoc);
            string[] vSheetName = null;
            int i = 0;
            bool bRet = false;
            string sheetName;

            swModel = (ModelDoc2)swApp.ActiveDoc;
            swDraw = (DrawingDoc)swModel;

            // Get the sheets in the drawing document
            vSheetName = (string[])swDraw.GetSheetNames();
            // Traverse the sheets and determine whether
            // they're loaded
            for (i = 0; i < vSheetName.Length; i++)
            {
                sheetName = (string)vSheetName[i];
                bRet = swDraw.ActivateSheet(sheetName);
                Sheet swSheet = default(Sheet);
                swSheet = (Sheet)swDraw.get_Sheet(vSheetName[i]);
                if ((swSheet.IsLoaded()))
                {
                    Debug.Print(vSheetName[i] + " is loaded.");
                }
                else
                {
                    Debug.Print(vSheetName[i] + " is not loaded.");
                }
            }
        }
    }
    /// <summary>
    /// The SldWorks swApp variable is pre-assigned for you.
    /// </summary>
    public SldWorks swApp;
}
}
```

Get Sheet in Multi-sheet Drawing Example (VBA)

This example shows how to get each sheet in a multi-sheet drawing regardless whether the sheet is loaded.

```
'-----  
' Preconditions:  
' 1. Click File > Open.  
' 2. Open public_documents\samples\tutorial\advdrawings\foodprocessor.sldrw.  
' 3. Click Select sheets to open > Selected > Sheet1* (load) > OK >Open.  
' 4. Open the Immediate window.  
'  
' Postconditions:  
' 1. Loads Sheet1 only.  
' 2. Mouse over Sheet2, Sheet3, and Sheet4 tabs and examine the  
'    Immediate window.  
'  
' NOTE: Because this drawing is used elsewhere, do not save changes.  
'-----
```

Option Explicit

```
Dim swApp As SldWorks.SldWorks  
Dim swModel As SldWorks.ModelDoc2  
Dim swDraw As SldWorks.DrawingDoc  
Dim vSheetName As Variant  
Dim i As Integer  
Dim bRet As Boolean  
  
Sub main()  
  
    Set swApp = Application.SldWorks  
    Set swModel = swApp.ActiveDoc  
    Set swDraw = swModel  
  
    ' Get the sheets in the drawing document  
    vSheetName = swDraw.GetSheetNames  
  
    ' Traverse the drawing sheets and determine whether  
    ' they're loaded  
    For i = 0 To UBound(vSheetName)  
        bRet = swDraw.ActivateSheet(vSheetName(i))  
        Dim swSheet As Sheet  
        Set swSheet = swDraw.Sheet(vSheetName(i))  
        'swSheet = swDraw.GetCurrentSheet  
        If (swSheet.IsLoaded) Then  
            Debug.Print (vSheetName(i) & " is loaded.")  
        Else  
            Debug.Print (vSheetName(i) & " is not loaded.")  
        End If  
    Next i  
  
End Sub
```

Get Sheet in Multi-Sheet Drawing Example (VB.NET)

This example shows how to get each sheet in a multi-sheet drawing regardless whether the sheet is loaded.

```
'-----
' Preconditions:
' 1. Click File > Open.
' 2. Open public_documents\samples\tutorial\advdrawings\foodprocessor.sldrw.
' 3. Click Select sheets to open > Selected > Sheet1* (load) > OK >Open.
' 4. Open the Immediate window.
'
' Postconditions:
' 1. Loads Sheet1 only.
' 2. Mouse over Sheet2, Sheet3, and Sheet4 tabs and examine the
'    Immediate window.
'
' NOTE: Because this drawing is used elsewhere, do not save changes.
'-----
```

```
Imports SolidWorks.Interop.sldworks
```

```
Imports SolidWorks.Interop.swconst
```

```
Imports System
```

```
Imports System.Diagnostics
```

```
Partial Class SolidWorksMacro
```

```
    Public Sub main()
```

```
        Dim swModel As ModelDoc2
```

```
        Dim swDraw As DrawingDoc
```

```
        Dim vSheetName As Object
```

```
        Dim i As Integer
```

```
        Dim bRet As Boolean
```

```
        swModel = swApp.ActiveDoc
```

```
        swDraw = swModel
```

```
        ' Get the sheets in the drawing document
```

```
        vSheetName = swDraw.GetSheetNames
```

```
        ' Traverse the drawing sheets and determine whether
```

```
        ' they're loaded
```

```
        For i = 0 To UBound(vSheetName)
```

```
            bRet = swDraw.ActivateSheet(vSheetName(i))
```

```
            Dim swSheet As Sheet
```

```
            swSheet = swDraw.Sheet(vSheetName(i))
```

```
            'swSheet = swDraw.GetCurrentSheet
```

```
            If (swSheet.IsLoaded) Then
```

```
                Debug.Print(vSheetName(i) & " is loaded.")
```

```
            Else
```

```
                Debug.Print(vSheetName(i) & " is not loaded.")
```

```
            End If
```

```
        Next i
```

```
    End Sub
```

```
''' <summary>
```

```
''' The SldWorks swApp variable is pre-assigned for you.
```

```
''' </summary>
```

```
Public swApp As SldWorks
```

```
End Class
```

Get Sheet Metal Folder Feature Example (C#)

This example shows how to get the sheet metal folder and its contents.

```
//-----
// Preconditions: Open a multibody sheet metal part that has a folder
// of sheet metal features.
//
// Postconditions: Inspect the Immediate window.
//-----
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System.Runtime.InteropServices;
using System;
using System.Diagnostics;
namespace GetSheetMetalFolder_CSharp.csproj
{
    partial class SolidWorksMacro
    {
        ModelDoc2 myModel;
        FeatureManager featureMgr;
        Feature feat;
        SheetMetalFolder sheetMetalFolder;
        object[] featArray;

        int i;

        public void Main()
        {
            myModel = (ModelDoc2)swApp.ActiveDoc;
            featureMgr = myModel.FeatureManager;

            sheetMetalFolder = (SheetMetalFolder)featureMgr.GetSheetMetalFolder();
            feat = (Feature)sheetMetalFolder.GetFeature();
            Debug.Print("Sheet metal folder name: " + feat.Name);
            Debug.Print(" Number of sheet metal features in the folder: " + sheetMetalFolder.GetSheetMetalCount());
            featArray = (object[])sheetMetalFolder.GetSheetMetals();
            for (i = featArray.GetLowerBound(0); i <= featArray.GetUpperBound(0); i++)
            {
                feat = (Feature)featArray[i];
                Debug.Print("    " + feat.Name);
            }
        }

        public SldWorks swApp;
    }
}
```

Get Sheet Metal Folder Feature Example (VBA)

This example shows how to get the sheet metal folder and its contents.

```
'-----  
' Preconditions: Open a multibody sheet metal part that has a folder  
' of sheet metal features.  
'-----  
' Postconditions: Inspect the Immediate window.  
'-----  
  
Dim swApp As SldWorks.SldWorks  
Dim myModel As SldWorks.ModelDoc2  
Dim featureMgr As SldWorks.FeatureManager  
Dim feat As SldWorks.Feature  
Dim sheetMetalFolder As SldWorks.sheetMetalFolder  
Dim featArray As Variant  
Dim i As Long  
Option Explicit  
  
Sub main()  
    Set swApp = Application.SldWorks  
    Set myModel = swApp.ActiveDoc  
    Set featureMgr = myModel.FeatureManager  
  
    Set sheetMetalFolder = featureMgr.GetSheetMetalFolder  
    Set feat = sheetMetalFolder.GetFeature  
    Debug.Print "Sheet metal folder name: " & feat.Name  
    Debug.Print " Number of sheet metal features in the folder: " & sheetMetalFolder.GetSheetMetalCount  
    featArray = sheetMetalFolder.GetSheetMetals  
    For i = LBound(featArray) To UBound(featArray)  
        Set feat = featArray(i)  
        Debug.Print "    " & feat.Name  
    Next i  
End Sub
```

Get Sheet Metal Folder Feature Example (VB.NET)

This example shows how to get the sheet metal folder and its contents.

```
'-----
' Preconditions: Open a multibody sheet metal part that has a folder
' of sheet metal features.
'
' Postconditions: Inspect the Immediate window.
'-----
```

```
Imports SolidWorks.Interop.sldworks
Imports SolidWorks.Interop.swconst
Imports System.Runtime.InteropServices
Imports System
Imports System.Diagnostics
```

```
Partial Class SolidWorksMacro
```

```
    Dim myModel As ModelDoc2
    Dim featureMgr As FeatureManager
    Dim feat As Feature
    Dim sheetMetalFolder As SheetMetalFolder
    Dim featArray As Object
    Dim i As Integer
```

```
    Sub main()
```

```
        myModel = swApp.ActiveDoc
        featureMgr = myModel.FeatureManager
```

```
        sheetMetalFolder = featureMgr.GetSheetMetalFolder
```

```
        feat = sheetMetalFolder.GetFeature
```

```
        Debug.Print("Sheet metal folder name: " & feat.Name)
```

```
        Debug.Print("  Number of sheet metal features in the folder: " & sheetMetalFolder.GetSheetMetalCount)
```

```
        featArray = sheetMetalFolder.GetSheetMetals
```

```
        For i = LBound(featArray) To UBound(featArray)
```

```
            feat = featArray(i)
```

```
            Debug.Print("      " & feat.Name)
```

```
        Next i
```

```
    End Sub
```

```
    Public swApp As SldWorks
```

```
End Class
```

Get Sheet Numbers and Names Example (VBA)

This example shows how to get the sheet numbers and names for multiple sheets in a drawing document.

```
'-----  
' Preconditions:  
' 1. Open a drawing that has multiple sheets.  
' 2. Open the Immediate window.  
'  
' Postconditions:  
' 1. Gets the number and names of the sheets.  
' 2. Examine the Immediate window.  
'-----  
Option Explicit  
  
Sub main()  
  
    Dim swApp As SldWorks.SldWorks  
    Dim swModel As SldWorks.ModelDoc2  
    Dim swDraw As SldWorks.DrawingDoc  
    Dim swSheet As SldWorks.Sheet  
    Dim vSheetNames As Variant  
    Dim bRet As Boolean  
    Dim i As Long  
  
    Set swApp = Application.SldWorks  
    Set swModel = swApp.ActiveDoc  
    Set swDraw = swModel  
    Set swSheet = swDraw.GetCurrentSheet  
    Debug.Print "FileName = " + swModel.GetPathName  
    Debug.Print " Current sheet = " & swSheet.GetName  
    Debug.Print ""  
    vSheetNames = swDraw.GetSheetNames  
    For i = 0 To UBound(vSheetNames)  
        Debug.Print " SheetName[" & i & "] = " & vSheetNames(i)  
    Next i  
  
End Sub
```

Get Sheet Properties and Insert Object Example (C++ COM)

This example shows how to get a sheet's properties and then insert an object from a file.

```
STDMETHODIMP CDgtlPen::demo1()
{
    HRESULT                hr = S_OK;
    CComPtr <IModelDoc2>    pModel;
    CComQIPtr <IDrawingDoc> pDraw;
    double                 pSheetProps[7];
    CComPtr <ISheet>        pSheet;

    BOOL                   createLink = FALSE;
    double                 x = 0;
    double                 y = 0;
    double                 z = 0;
    VARIANT_BOOL           retval = NULL;
    hr = m_pISldWorks->get_IActiveDoc2(&pModel);
    _ASSERT(pModel);
    pDraw = pModel;
    _ASSERT(pDraw);
    hr = pDraw->IGetCurrentSheet(&pSheet);
    _ASSERT(pSheet);
    ZeroMemory(pSheetProps, 7 * sizeof(double));
    hr = pSheet->IGetProperties(pSheetProps);
    y = pSheetProps[6];
    hr = pModel->InsertObjectFromFile (CComBSTR(_T("C:/Bitmaps/digitalpen.bmp")), createLink, x, y, z, &retval);
    return S_OK;
}
```

Create Shell Feature Example (VBA)

This example shows how to create a shell feature.

```

'-----
' Preconditions:
' 1. Verify that the specified model document exists.
' 2. Open an Immediate window.
'
' Postconditions:
' 1. Selects a face to remove from the model to create the shell.
' 2. Creates Shell1.
' 3. Inspect the Immediate window, graphics area, and
'    FeatureManager design tree.
'
' NOTE: Because the model is used elsewhere, do not save changes.
'-----

Dim swApp                As SldWorks.SldWorks
Dim swModel              As SldWorks.ModelDoc2
Dim swSelMgr             As SldWorks.SelectionMgr
Dim swSelData            As SldWorks.SelectData
Dim swFeat              As SldWorks.Feature
Dim swShell              As SldWorks.ShellFeatureData
Dim vFaceRemArr          As Variant
Dim vFaceRem             As Variant
Dim swFaceRem            As SldWorks.Face2
Dim vMultiFaceArr        As Variant
Dim vMultiFace           As Variant
Dim swMultiFace          As SldWorks.Face2
Dim swEnt                As SldWorks.Entity
Dim i                   As Long
Dim bRet                 As Boolean
Dim longstatus As Long, longwarnings As Long

Option Explicit

Sub main()
    Set swApp = Application.SldWorks

    Set swModel = swApp.OpenDoc6("C:\Users\Public\Documents\SOLIDWORKS\SOLIDWORKS 2018\samples\tutorial\api\block20.sldprt", 1, 0,
    "", longstatus, longwarnings)
    swApp.ActivateDoc2 "block20", False, longstatus
    Set swModel = swApp.ActiveDoc

    bRet = swModel.Extension.SelectByID2("", "FACE", -1.50558029249623E-02, 3.962399999999466E-02, -0.018063862472502, False, 1,
    Nothing, 0)
    swModel.InsertFeatureShell 0.00254, False

    Set swSelMgr = swModel.SelectionManager
    Set swSelData = swSelMgr.CreateSelectData
    Set swFeat = swSelMgr.GetSelectedObject6(1, -1)
    Set swShell = swFeat.GetDefinition

    ' Get shell data
    Debug.Print "File = " & swModel.GetPathName
    Debug.Print " " & swFeat.Name
    Debug.Print "    Direction: " & swShell.Direction
    Debug.Print "    Thickness: " & swShell.Thickness * 1000# & " mm"
    Debug.Print "    Count of faces removed: " & swShell.FacesRemovedCount
    Debug.Print "    Count of faces with alternative thicknesses: " & swShell.GetMultipleThicknessFacesCount

    bRet = swShell.AccessSelections(swModel, Nothing)
    swModel.ClearSelection2 True

    vFaceRemArr = swShell.FacesRemoved

    For Each vFaceRem In vFaceRemArr
        Set swFaceRem = vFaceRem
        Set swEnt = swFaceRem

        bRet = swEnt.Select4(True, swSelData)
    Next

    swModel.ClearSelection2 True
    vMultiFaceArr = swShell.MultipleThicknessFaces

    For Each vMultiFace In vMultiFaceArr
        Set swMultiFace = vMultiFace
        Set swEnt = swMultiFace

        Debug.Print "    Alternative thickness in mm at face (" & i & "): " & swShell.GetMultipleThicknessAtIndex(i) * 1000#
        i = i + 1

        bRet = swEnt.Select4(True, swSelData)
    Next

    swModel.ClearSelection2 True
    swShell.ReleaseSelectionAccess
End Sub

```

Get Shut-Off Surface Data Example (C#)

This example shows how to get a shut-off surface feature's data.

```
//-----
// Preconditions: Open a model that contains a Shut-Off Surface1 feature with
// the patch type of swShutOffSurfacePatchType_e.swPatchTypeTangent = 2.
//
// Postconditions:
// 1. Gets Shut-Off Surface1 data.
// 2. Gets the tangent faces for the specified loops where to create the patches.
// 3. Creates the patches on the opposite tangent faces.
// 4. Examine the graphic area and Immediate window.
//-----

using Microsoft.VisualBasic;
using System;
using System.Collections;
using System.Collections.Generic;
using System.Data;
using System.Diagnostics;
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System.Runtime.InteropServices;

namespace GetShutOffSurfaceEdgeCount_CSharp.csproj
{
    partial class SolidWorksMacro
    {
        ModelDoc2 swModel;
        ModelDocExtension swModelDocExt;
        SelectionMgr swSelMgr;
        Feature swFeat;
        ShutOffSurfaceFeatureData swFeatData;
        Entity swEnt;

        [params]
        bool boolstatus;
        int i;
        int j;

        public void Main()
        {
            swModel = (ModelDoc2)swApp.ActiveDoc;
            swModelDocExt = swModel.Extension;
            swSelMgr = (SelectionMgr)swModel.SelectionManager;

            boolstatus = swModelDocExt.SelectByID2("Shut-Off
Surface1", "REFSURFACE",
0, 0, 0, false,
0, null, (int)swSelectOption_e.swSelectOptionDefault);

            swFeat = (Feature)swSelMgr.GetSelectedObject6(1,
-1);

            swFeatData = (ShutOffSurfaceFeatureData)swFeat.GetDefinition();

            boolstatus = swFeatData.AccessSelections(swModel, null);

            Debug.Print("Status = " + swFeatData.Status());
            Debug.Print("Knit flag = " + swFeatData.Knit);
        }
    }
}
```

```

        Debug.Print("Edge count = " + swFeatData.GetEdgeCount());
        Debug.Print("Loop count = " + swFeatData.GetLoopCount());

        for (i = 0; i <= swFeatData.GetLoopCount() - 1; i++)
        {
            Debug.Print("Loop(" + Convert.ToString(i)
+ ")");
            Debug.Print("    Edge
count = " + swFeatData.GetLoopEdgeCount(i));
            Debug.Print("    Patch
type = " + swFeatData.get_LoopPatchType(i));

            @params = (object[])swFeatData.get_LoopEdges(i);

            for (j = 0; j <= @params.GetUpperBound(0); j++)
            {
                swEnt = (Entity)@params[j];
                boolstatus = swEnt.Select4(false, null);
            }
        }

        @params = (object[])swFeatData.Edges;

        for (i = 0; i <= @params.GetUpperBound(0); i++)
        {
            swEnt = (Entity)@params[i];
            boolstatus = swEnt.Select4(false, null);

            swEnt = (Entity)swFeatData.GetFaceTangentTo(i);

            if ((swEnt != null))
            {
                boolstatus = swEnt.Select4(true, null);
            }

            swFeatData.FlipFaceTangentTo(i);

            swEnt = (Entity)swFeatData.GetFaceTangentTo(i);

            if ((swEnt != null))
            {
                boolstatus = swEnt.Select4(true, null);
            }

            swFeatData.FlipFaceTangentTo(i);
        }

        boolstatus = swFeat.ModifyDefinition(swFeatData, swModel, null);
    }

    public SldWorks swApp;
}
}

```

Get Shut-Off Surface Data Example (VBA)

This example shows how to get a shut-off surface feature's data.

```

'-----
' Preconditions: Open a model that contains a Shut-Off Surface1 feature with
' the patch type of swShutOffSurfacePatchType_e.swPatchTypeTangent = 2.
'
' Postconditions:
' 1. Gets Shut-Off Surface1 data.
' 2. Gets the tangent faces for the specified loops where to create the patches.
' 3. Creates the patches on the opposite tangent faces.
' 4. Examine the graphic area and Immediate window.
'-----
Option Explicit

Dim swApp As SldWorks.SldWorks
Dim swModel As SldWorks.ModelDoc2
Dim swModelDocExt As SldWorks.ModelDocExtension
Dim swSelMgr As SldWorks.SelectionMgr
Dim swFeat As SldWorks.Feature
Dim swFeatData As SldWorks.ShutOffSurfaceFeatureData
Dim swEnt As SldWorks.Entity
Dim params As Variant
Dim boolstatus As Boolean
Dim i As Integer
Dim j As Integer

Sub main()
    Set swApp = Application.SldWorks
    Set swModel = swApp.ActiveDoc
    Set swModelDocExt = swModel.Extension
    Set swSelMgr = swModel.SelectionManager

    boolstatus = swModelDocExt.SelectByID2("Shut-Off Surface1", "REFSURFACE", 0, 0, 0, False, 0, Nothing, swSelectOptionDefault)

    Set swFeat = swSelMgr.GetSelectedObject6(1, -1)
    Set swFeatData = swFeat.GetDefinition

    boolstatus = swFeatData.AccessSelections(swModel, Nothing)

    Debug.Print "Status = " + CStr(swFeatData.Status)
    Debug.Print "Knit flag = " + CStr(swFeatData.Knit)
    Debug.Print "Edge count = " + CStr(swFeatData.GetEdgeCount)
    Debug.Print "Loop count = " + CStr(swFeatData.GetLoopCount)

    For i = 0 To swFeatData.GetLoopCount - 1
        Debug.Print "Loop(" + CStr(i) + ") "
        Debug.Print "    Edge count = " + CStr(swFeatData.GetLoopEdgeCount(i))
        Debug.Print "    Patch type = " + CStr(swFeatData.LoopPatchType(i))

        params = swFeatData.LoopEdges(i)

        For j = 0 To UBound(params)
            Set swEnt = params(j)
            boolstatus = swEnt.Select4(False, Nothing)
        Next j
    Next i

    params = swFeatData.Edges

    For i = 0 To UBound(params)
        Set swEnt = params(i)
        boolstatus = swEnt.Select4(False, Nothing)

        Set swEnt = swFeatData.GetFaceTangentTo(i)

        If Not swEnt Is Nothing Then
            boolstatus = swEnt.Select4(True, Nothing)
        End If

        swFeatData.FlipFaceTangentTo (i)

        Set swEnt = swFeatData.GetFaceTangentTo(i)

        If Not swEnt Is Nothing Then
            boolstatus = swEnt.Select4(True, Nothing)
        End If

        swFeatData.FlipFaceTangentTo (i)
    
```

Next i

boolstatus = swFeat.**ModifyDefinition**(swFeatData, swModel, Nothing)

End Sub

Get Shut-Off Surface Data Example (VB.NET)

This example shows how to get a shut-off surface feature's data.

```
'-----
' Preconditions: Open a model that contains a Shut-Off Surface1 feature with
' the patch type of swShutOffSurfacePatchType_e.swPatchTypeTangent = 2.
'
' Postconditions:
' 1. Gets Shut-Off Surface1 data.
' 2. Gets the tangent faces for the specified loops where to create the patches.
' 3. Creates the patches on the opposite tangent faces.
' 4. Examine the graphic area and Immediate window.
'-----
```

```
Imports SolidWorks.Interop.sldworks
```

```
Imports SolidWorks.Interop.swconst
```

```
Imports System.Runtime.InteropServices
```

```
Imports System
```

```
Imports System.Diagnostics
```

```
Partial Class SolidWorksMacro
```

```
    Dim swModel As ModelDoc2
```

```
    Dim swModelDocExt As ModelDocExtension
```

```
    Dim swSelMgr As SelectionMgr
```

```
    Dim swFeat As Feature
```

```
    Dim swFeatData As ShutOffSurfaceFeatureData
```

```
    Dim swEnt As Entity
```

```
    Dim params As Object
```

```
    Dim boolstatus As Boolean
```

```
    Dim i As Integer
```

```
    Dim j As Integer
```

```
    Sub main()
```

```
        swModel = swApp.ActiveDoc
```

```
        swModelDocExt = swModel.Extension
```

```
        swSelMgr = swModel.SelectionManager
```

```
        boolstatus = swModelDocExt.SelectByID("Shut-Off
Surface1", "REFSURFACE",
0, 0, 0, False,
0, Nothing, swSelectOption_e.swSelectOptionDefault)
```

```
        swFeat = swSelMgr.GetSelectedObject6(1, -1)
```

```
        swFeatData = swFeat.GetDefinition
```

```
        boolstatus = swFeatData.AccessSelections(swModel, Nothing)
```

```
        Debug.Print("Status = " + CStr(swFeatData.Status))
```

```
        Debug.Print("Knit flag = " + CStr(swFeatData.Knit))
```

```
        Debug.Print("Edge count = " + CStr(swFeatData.GetEdgeCount))
```

```
        Debug.Print("Loop count = " + CStr(swFeatData.GetLoopCount))
```

```
        For i = 0 To swFeatData.GetLoopCount - 1
```

```
            Debug.Print("Loop(" + CStr(i) + ")")
```

```
            Debug.Print("    Edge count =
" + CStr(swFeatData.GetLoopEdgeCount(i)))
```

```
            Debug.Print("    Patch type =
" + CStr(swFeatData.LoopPatchType(i)))
```

```

        params = swFeatData.LoopEdges(i)

        For j = 0 To UBound(params)
            swEnt = params(j)
            boolstatus = swEnt.Select4(False, Nothing)
        Next j
    Next i

    params = swFeatData.Edges

    For i = 0 To UBound(params)
        swEnt = params(i)
        boolstatus = swEnt.Select4(False, Nothing)

        swEnt = swFeatData.GetFaceTangentTo(i)

        If Not swEnt Is Nothing Then
            boolstatus = swEnt.Select4(True, Nothing)
        End If

        swFeatData.FlipFaceTangentTo(i)

        swEnt = swFeatData.GetFaceTangentTo(i)

        If Not swEnt Is Nothing Then
            boolstatus = swEnt.Select4(True, Nothing)
        End If

        swFeatData.FlipFaceTangentTo(i)

    Next i

    boolstatus = swFeat.ModifyDefinition(swFeatData, swModel, Nothing)

End Sub

Public swApp As SldWorks

End Class

```

Get Simple Fillet Feature Data Example (C#)

This example shows how to get simple fillet feature data.

```
//-----
// Preconditions:
// 1. Open a model document that contains a simple fillet feature.
// 2. Select the simple fillet feature.
//
// Postconditions: Inspect the Immediate window.
//-----
using Microsoft.VisualBasic;
using System;
using System.Collections;
using System.Collections.Generic;
using System.Data;
using System.Diagnostics;
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System.Runtime.InteropServices;
namespace GetSimpleFilletEdgeCount_CSharp.csproj
{
    partial class SolidWorksMacro
    {
        ModelDoc2 swModel;
        ModelDocExtension swModelDocExt;
        SelectionMgr swSelMgr;
        Feature swFeat;
        SimpleFilletFeatureData2 swFeatData;

        bool boolstatus;

        public void Main()
        {
            swModel = (ModelDoc2)swApp.ActiveDoc;
            swModelDocExt = swModel.Extension;
            swSelMgr = (SelectionMgr)swModel.SelectionManager;

            boolstatus = swModelDocExt.SelectByID2("Fillet2", "BODYFEATURE", 0, 0, 0, false, 0, null, 0);

            swFeat = (Feature)swSelMgr.GetSelectedObject6(1, -1);
            swFeatData = (SimpleFilletFeatureData2)swFeat.GetDefinition();

            Debug.Print("Edge count: " + swFeatData.GetEdgeCount());
        }

        public SldWorks swApp;
    }
}
```

Get Simple Fillet Feature Data Example (VBA)

This example shows how to get simple fillet feature data.

```
'-----  
' Preconditions:  
' 1. Open a model document that contains a simple fillet feature.  
' 2. Select the simple fillet feature.  
'-----  
' Postconditions: Inspect the Immediate window.  
'-----
```

Option Explicit

```
Dim swApp As SldWorks.SldWorks  
Dim swModel As SldWorks.ModelDoc2  
Dim swModelDocExt As SldWorks.ModelDocExtension  
Dim swSelMgr As SldWorks.SelectionMgr  
Dim swFeat As SldWorks.Feature  
Dim swFeatData As SldWorks.SimpleFilletFeatureData2  
Dim boolstatus As Boolean  
  
Sub main()  
    Set swApp = Application.SldWorks  
    Set swModel = swApp.ActiveDoc  
    Set swModelDocExt = swModel.Extension  
    Set swSelMgr = swModel.SelectionManager  
  
    boolstatus = swModelDocExt.SelectByID2("Fillet2", "BODYFEATURE", 0, 0, 0, False, 0, Nothing, 0)  
  
    Set swFeat = swSelMgr.GetSelectedObject6(1, -1)  
    Set swFeatData = swFeat.GetDefinition  
  
    Debug.Print "Edge count: " & swFeatData.GetEdgeCount  
End Sub
```

Get Simple Fillet Feature Data Example (VB.NET)

This example shows how to get simple fillet feature data.

```
'-----  
' Preconditions:  
' 1. Open a model document that contains a simple fillet feature.  
' 2. Select the simple fillet feature.  
'-----  
' Postconditions: Inspect the Immediate window.  
'-----
```

```
Imports SolidWorks.Interop.sldworks  
Imports SolidWorks.Interop.swconst  
Imports System.Runtime.InteropServices  
Imports System  
Imports System.Diagnostics
```

```
Partial Class SolidWorksMacro
```

```
    Dim swModel As ModelDoc2  
    Dim swModelDocExt As ModelDocExtension  
    Dim swSelMgr As SelectionMgr  
    Dim swFeat As Feature  
    Dim swFeatData As SimpleFilletFeatureData2  
    Dim boolstatus As Boolean
```

```
    Sub main()
```

```
        swModel = swApp.ActiveDoc  
        swModelDocExt = swModel.Extension  
        swSelMgr = swModel.SelectionManager
```

```
        boolstatus = swModelDocExt.SelectByID2("Fillet2", "BODYFEATURE", 0, 0, 0, False, 0, Nothing, 0)
```

```
        swFeat = swSelMgr.GetSelectedObject6(1, -1)  
        swFeatData = swFeat.GetDefinition
```

```
        Debug.Print("Edge count: " & swFeatData.GetEdgeCount)
```

```
    End Sub
```

```
    Public swApp As SldWorks
```

```
End Class
```

Get Sketch Arc Data Example (C#)

This example shows how to get sketch arc data.

```
//-----
// Preconditions:
// 1. Verify that the specified document template exists.
// 2. Open an Immediate window.
//
// Postconditions:
// 1. Creates a new model document with a sketch of an open arc.
// 2. Inspect the Immediate window.
//
// NOTE: Because the model is used elsewhere, do not save changes.
//-----
using Microsoft.VisualBasic;
using System;
using System.Collections;
using System.Collections.Generic;
using System.Data;
using System.Diagnostics;
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System.Runtime.InteropServices;

namespace CreateSketchArc_CSharp.csproj
{
    partial class SolidWorksMacro
    {
        SketchArc skArcSegment;
        ModelDoc2 Part;
        SketchPoint centerPt;
        SketchPoint startPt;
        double[] vNormalVector;
        SketchPoint endPt;

        bool boolstatus;

        public void Main()
        {
            Part = (ModelDoc2)swApp.NewDocument("C:\\ProgramData\\SolidWorks\\SOLIDWORKS 2015\\templates\\Part.prtdot", 0, 0, 0);
            Part = (ModelDoc2)swApp.ActiveDoc;

            Part.SketchManager.InsertSketch(true);
            boolstatus = Part.Extension.SelectByID2("Front Plane", "PLANE", -0.0405700227306837, 0.02417279486491, 0.004985607986725,
            Part.ClearSelection2(true);

            skArcSegment = (SketchArc)Part.SketchManager.CreateArc(-0.017822, 0.012246, 0.0, 0.010653, 0.040124, 0.0, -0.011244, -0.
            Part.ClearSelection2(true);
            Part.SketchManager.InsertSketch(true);

            centerPt = (SketchPoint)skArcSegment.GetCenterPoint2();
            Debug.Print("Center point: " + centerPt.X + ", " + centerPt.Y + ", " + centerPt.Z);

            startPt = (SketchPoint)skArcSegment.GetStartPoint2();
            Debug.Print("Start point: " + startPt.X + ", " + startPt.Y + ", " + startPt.Z);

            endPt = (SketchPoint)skArcSegment.GetEndPoint2();
            Debug.Print("End point: " + endPt.X + ", " + endPt.Y + ", " + endPt.Z);

            vNormalVector = (double[])skArcSegment.GetNormalVector();
            Debug.Print("Normal vector: " + vNormalVector[0] + ", " + vNormalVector[1] + ", " + vNormalVector[2]);

            Debug.Print("Radius in mm: " + skArcSegment.GetRadius() * 1000.0);
            Debug.Print("Rotation direction (1 = counterclockwise, -1 = clockwise): " + skArcSegment.GetRotationDir());
            Debug.Print("Complete circle (1 = complete, 0 = partial arc): " + skArcSegment.IsCircle());

        }

        /// <summary>
        /// The SldWorks swApp variable is pre-assigned for you.
        /// </summary>

        public SldWorks swApp;
    }
}
```

Get Sketch Arc Data Example (VBA)

This example shows how to get sketch arc data.

```

'-----
' Preconditions:
' 1. Verify that the specified document template exists.
' 2. Open an Immediate window.
'
' Postconditions:
' 1. Creates a new model document with a sketch of an open arc.
' 2. Inspect the Immediate window.
'
' NOTE: Because the model is used elsewhere, do not save changes.
'-----

Dim swApp As SldWorks.SldWorks
Dim skArcSegment As SldWorks.SketchArc
Dim Part As SldWorks.ModelDoc2
Dim centerPt As SldWorks.SketchPoint
Dim startPt As SldWorks.SketchPoint
Dim vNormalVector As Variant
Dim endPt As SldWorks.SketchPoint
Dim boolstatus As Boolean
Option Explicit

Sub main()
    Set swApp = Application.SldWorks

    Set Part = swApp.NewDocument("C:\ProgramData\SolidWorks\SOLIDWORKS 2015\templates\Part.prtdot", 0, 0, 0)
    Set Part = swApp.ActiveDoc

    Part.SketchManager.InsertSketch True
    boolstatus = Part.Extension.SelectByID2("Front Plane", "PLANE", -4.05700227306837E-02, 0.02417279486491, 4.98560798672921E-03,
    False, 0, Nothing, 0)
    Part.ClearSelection2 True

    Set skArcSegment = Part.SketchManager.CreateArc(-0.017822, 0.012246, 0#, 0.010653, 0.040124, 0#, -0.011244, -0.027057, 0#, -1)
    Part.ClearSelection2 True
    Part.SketchManager.InsertSketch True

    Set centerPt = skArcSegment.GetCenterPoint2
    Debug.Print "Center point: " & centerPt.X & ", " & centerPt.Y & ", " & centerPt.Z

    Set startPt = skArcSegment.GetStartPoint2
    Debug.Print "Start point: " & startPt.X & ", " & startPt.Y & ", " & startPt.Z

    Set endPt = skArcSegment.GetEndPoint2
    Debug.Print "End point: " & endPt.X & ", " & endPt.Y & ", " & endPt.Z

    vNormalVector = skArcSegment.GetNormalVector
    Debug.Print "Normal vector: " & vNormalVector(0) & ", " & vNormalVector(1) & ", " & vNormalVector(2)

    Debug.Print "Radius in mm: " & skArcSegment.GetRadius * 1000#
    Debug.Print "Rotation direction (1 = counterclockwise, -1 = clockwise): " & skArcSegment.GetRotationDir
    Debug.Print "Complete circle (1 = complete, 0 = partial arc): " & skArcSegment.IsCircle

End Sub

```

Get Sketch Arc Data Example (VB.NET)

This example shows how to get sketch arc data.

```

'-----
' Preconditions:
' 1. Verify that the specified document template exists.
' 2. Open an Immediate window.
'
' Postconditions:
' 1. Creates a new model document with a sketch of an open arc.
' 2. Inspect the Immediate window.
'
' NOTE: Because the model is used elsewhere, do not save changes.
'-----

```

```

Imports SolidWorks.Interop.sldworks
Imports SolidWorks.Interop.swconst
Imports System.Runtime.InteropServices
Imports System
Imports System.Diagnostics

```

```
Partial Class SolidWorksMacro
```

```

    Dim skArcSegment As SketchArc
    Dim Part As ModelDoc2
    Dim centerPt As SketchPoint
    Dim startPt As SketchPoint
    Dim vNormalVector As Object
    Dim endPt As SketchPoint
    Dim boolstatus As Boolean

```

```
Sub main()
```

```

    Part = swApp.NewDocument("C:\ProgramData\SolidWorks\SOLIDWORKS 2015\templates\Part.prtdot", 0, 0, 0)
    Part = swApp.ActiveDoc

    Part.SketchManager.InsertSketch(True)
    boolstatus = Part.Extension.SelectByID2("Front Plane", "PLANE", -0.0405700227306837, 0.02417279486491, 0.00498560798672921,
    Part.ClearSelection2(True)

    skArcSegment = Part.SketchManager.CreateArc(-0.017822, 0.012246, 0.0#, 0.010653, 0.040124, 0.0#, -0.011244, -0.027057, 0.0#,
    Part.ClearSelection2(True)
    Part.SketchManager.InsertSketch(True)

    centerPt = skArcSegment.GetCenterPoint2
    Debug.Print("Center point: " & centerPt.X & ", " & centerPt.Y & ", " & centerPt.Z)

    startPt = skArcSegment.GetStartPoint2
    Debug.Print("Start point: " & startPt.X & ", " & startPt.Y & ", " & startPt.Z)

    endPt = skArcSegment.GetEndPoint2
    Debug.Print("End point: " & endPt.X & ", " & endPt.Y & ", " & endPt.Z)

    vNormalVector = skArcSegment.GetNormalVector
    Debug.Print("Normal vector: " & vNormalVector(0) & ", " & vNormalVector(1) & ", " & vNormalVector(2))

    Debug.Print("Radius in mm: " & skArcSegment.GetRadius * 1000.0#)
    Debug.Print("Rotation direction (1 = counterclockwise, -1 = clockwise): " & skArcSegment.GetRotationDir)
    Debug.Print("Complete circle (1 = complete, 0 = partial arc): " & skArcSegment.IsCircle)

```

```
End Sub
```

```

''' <summary>
''' The SldWorks swApp variable is pre-assigned for you.
''' </summary>
Public swApp As SldWorks

```

```
End Class
```

Get Sketch Constraints Example (VBA)

This examples shows how to:

- determine whether a sketch is under-constrained or over-constrained.
- iterate over all the segments in the sketch and get their constraints.

```
'-----
' Preconditions:
' 1. Open public_documents\samples\tutorial\surfaces\nozzle.sldprt.
' 2. Select Sketch14 in the FeatureManager design tree.
' 3. Examine the graphics area.
'
' Postconditions:
' 1. Iterates all constraints in Sketch14 and prints
'    to the Immediate window its constraint status and constraints.
' 2. Examine the Immediate window.
'-----
Option Explicit

Sub main()

    Dim swApp As SldWorks.SldWorks
    Dim swModel As SldWorks.ModelDoc2
    Dim swSelMgr As SldWorks.SelectionMgr
    Dim swFeat As SldWorks.Feature
    Dim swSketch As SldWorks.Sketch
    Dim vSketchSeg As Variant
    Dim vConstraint As Variant
    Dim i As Long
    Dim j As Long
    Dim bRet As Boolean

    Set swApp = Application.SldWorks
    Set swModel = swApp.ActiveDoc
    Set swSelMgr = swModel.SelectionManager
    Set swFeat = swSelMgr.GetSelectedObject6(1, -1)
    Set swSketch = swFeat.GetSpecificFeature2

    Debug.Print "Sketch = " & swFeat.Name
    Debug.Print ""
    Debug.Print " SketchConstraintStatus (3 = fully constrained) = " & swSketch.GetConstrainedStatus
    Debug.Print ""

    ' Put sketch into edit mode to get access
    ' to constraint information for each segment
    ' in the sketch
    swModel.EditSketch

    vSketchSeg = swSketch.GetSketchSegments
    For i = 0 To UBound(vSketchSeg)
        vConstraint = vSketchSeg(i).GetConstraints
        For j = 0 To UBound(vConstraint)
            Debug.Print " SketchSegConstraint[" & i & "] = " & vConstraint(j)
        Next j
    Next i

    ' Exit edit mode and do not rebuild the model
    swModel.InsertSketch2 False

End Sub
```

Get Sketch Contours Example (C#)

This example shows how to get the sketch contours in a model document.

```
//-----
// Preconditions:
// 1. Open a part that contains a Sketch1 feature.
// 2. Open the Immediate window.
//
// Postconditions:
// 1. Gets the number of sketch contours in Sketch1 and whether
//    they are open or closed.
// 2. Gets the number of sketch segments in Sketch1 and their type.
// 3. Gets the number of edges Sketch1.
// 4. Examine the Immediate window.
//-----

using SolidWorks.Interop.sldworks;

using SolidWorks.Interop.swconst;

using System.Runtime.InteropServices;

using System;

using System.Diagnostics;

namespace SelectSketchContour_CSharp.csproj
{
    partial class SolidWorksMacro
    {
        public void Main()
        {
            ModelDoc2 myModel = default(ModelDoc2);
            PartDoc myPart = default(PartDoc);
            SelectionMgr SelMgr = default(SelectionMgr);
            SelectData mySelectData = default(SelectData);
            Feature myFeature = default(Feature);
            Sketch mySketch = default(Sketch);

            int contourCount = 0;

            object[] vSkContours = null;

            SketchContour skContour = default(SketchContour);

            object[] vEdges = null;

            Edge myEdge = default(Edge);

            int skSegCount = 0;

            object[] vSkSegments = null;

            SketchSegment skSegment = default(SketchSegment);

            int skSegType = 0;

            string skSegTypesString = null;

            bool closed = false;

            string closedString = null;

            int i = 0;

            int j = 0;

            int k = 0;

            bool boolstatus = false;

            myModel = (ModelDoc2)swApp.ActiveDoc;

            SelMgr = (SelectionMgr)myModel.SelectionManager;

            mySelectData = (SelectData)SelMgr.CreateSelectData();

            myPart = (PartDoc)myModel;

            myFeature = (Feature)myPart.FeatureByName("Sketch1");
```

```

mySketch = (Sketch)myFeature.GetSpecificFeature2();

//      or
//      Set mySketch = myModel.GetActiveSketch2()
//      Set myFeature = mySketch

if ((mySketch != null))
{
    vSkContours = (object[])mySketch.GetSketchContours();
    contourCount = vSkContours.GetUpperBound(0) -
vSkContours.GetLowerBound(0) + 1;

    Debug.Print("");
    Debug.Print(contourCount + " Contours in sketch " + myFeature.Name);

    for (i = vSkContours.GetLowerBound(0); i <= vSkContours.GetUpperBound(0); i++)
    {
        skContour = (SketchContour)vSkContours[i];

        if ((skContour != null))
        {
            closed = skContour.IsClosed();

            if ((closed == false))
            {
                closedString = "Open";
            }
            else
            {
                closedString = "Closed";
            }

            Debug.Print(" Contour
" + i + ": " + closedString);

            vSkSegments = (object[])skContour.GetSketchSegments();
            skSegCount = vSkSegments.GetUpperBound(0) -
vSkSegments.GetLowerBound(0) + 1;

            for (j = vSkSegments.GetLowerBound(0); j <= vSkSegments.GetUpperBound(0); j++)
            {
                if (j == vSkSegments.GetLowerBound(0))
                {
                    skSegTypesString = "(";
                }

                skSegment = (SketchSegment)vSkSegments[j];

                if ((skSegment != null))
                {
                    skSegType = skSegment.GetType();
                    switch (skSegType)
                    {

                        case (int)swSketchSegments_e.swSketchLINE:
                            skSegTypesString = skSegTypesString + "line";

                            break;

                        case (int)swSketchSegments_e.swSketchARC:
                            skSegTypesString = skSegTypesString + "arc";

```

```

        break;
    case (int)swSketchSegments_e.swSketchELLIPSE:
        skSegTypesString = skSegTypesString + "ellipse";

        break;
    case (int)swSketchSegments_e.swSketchPARABOLA:
        skSegTypesString = skSegTypesString + "parabola";

        break;
    case (int)swSketchSegments_e.swSketchSPLINE:
        skSegTypesString = skSegTypesString + "spline";

        break;
    case (int)swSketchSegments_e.swSketchTEXT:
        skSegTypesString = skSegTypesString + "text";

        break;
    case -1:
        skSegTypesString = skSegTypesString + "unknown";

        break;
    }
}

if (j == vSkSegments.GetUpperBound(0))
{
    skSegTypesString = skSegTypesString + ")";
}
else
{
    skSegTypesString = skSegTypesString + ",";
}
}

Debug.Print("    Sketch
segment count = " + skSegCount + " " + skSegTypesString);

vEdges = (object[])skContour.GetEdges();

if ((vEdges == null))
{
    Debug.Print("    No
edges.");
}
else
{
    for (k = vEdges.GetLowerBound(0); k <= vEdges.GetUpperBound(0); k++)
    {
        myEdge = (Edge)vEdges[k];

        if ((myEdge != null))
        {
            Debug.Print("    Edge

" + k + ": ");
        }
    }
}

```

```
    }

    boolstatus = skContour.Select2(false, mySelectData);
    if (boolstatus == false)
    {
        Debug.Print("    Selection
of contour failed.");
    }

    System.Diagnostics.Debugger.Break();

    }

    }

    }

    }

    public SldWorks swApp;

    }

}
```

Get Sketch Contours Example (VBA)

This example shows how to get the sketch contours in a model document.

```

'-----
' Preconditions:
' 1. Open a part that contains a Sketch1 feature.
' 2. Open the Immediate window.
'
' Postconditions:
' 1. Gets the number of sketch contours in Sketch1 and whether
'    they are open or closed.
' 2. Gets the number of sketch segments in Sketch1 and their type.
' 3. Gets the number of edges Sketch1.
' 4. Examine the Immediate window.
'-----
Option Explicit

Sub main()

    Dim swApp As SldWorks.SldWorks
    Dim myModel As SldWorks.ModelDoc2
    Dim myPart As SldWorks.PartDoc
    Dim SelMgr As SldWorks.SelectionMgr
    Dim mySelectData As SldWorks.SelectData
    Dim myFeature As SldWorks.Feature
    Dim mySketch As SldWorks.Sketch
    Dim contourCount As Integer
    Dim vSkContours As Variant
    Dim skContour As SketchContour
    Dim vEdges As Variant, myEdge As SldWorks.Edge
    Dim skSegCount As Long
    Dim vSkSegments As Variant
    Dim skSegment As SldWorks.SketchSegment
    Dim skSegType As Long, skSegTypesString As String
    Dim closed As Boolean, closedString As String
    Dim i As Integer, j As Integer, k As Integer
    Dim boolstatus As Boolean

    Set swApp = Application.SldWorks
    Set myModel = swApp.ActiveDoc
    Set SelMgr = myModel.SelectionManager
    Set mySelectData = SelMgr.CreateSelectData
    Set myPart = myModel
    Set myFeature = myPart.FeatureByName("Sketch1")
    Set mySketch = myFeature.GetSpecificFeature2()
'
'    or
'    Set mySketch = myModel.GetActiveSketch2()
'    Set myFeature = mySketch

    If Not mySketch Is Nothing Then
        vSkContours = mySketch.GetSketchContours()
        contourCount = UBound(vSkContours) - LBound(vSkContours) + 1

        Debug.Print ""
        Debug.Print contourCount & " Contours in sketch " & myFeature.Name

        For i = LBound(vSkContours) To UBound(vSkContours)
            Set skContour = vSkContours(i)

            If Not skContour Is Nothing Then
                closed = skContour.IsClosed()

                If (closed = 0) Then
                    closedString = "Open"
                Else
                    closedString = "Closed"
                End If

                Debug.Print "  Contour " & i & ": " & closedString
                vSkSegments = skContour.GetSketchSegments()
                skSegCount = UBound(vSkSegments) - LBound(vSkSegments) + 1

                For j = LBound(vSkSegments) To UBound(vSkSegments)
                    If j = LBound(vSkSegments) Then
                        skSegTypesString = "("
                    End If

                    Set skSegment = vSkSegments(j)

                    If Not skSegment Is Nothing Then
                        skSegType = skSegment.GetType()
                        Select Case skSegType

                            Case SwConst.swSketchSegments_e.swSketchLINE
                                skSegTypesString = skSegTypesString & "line"

                            Case SwConst.swSketchSegments_e.swSketchARC
                                skSegTypesString = skSegTypesString & "arc"

                            Case SwConst.swSketchSegments_e.swSketchELLIPSE
                                skSegTypesString = skSegTypesString & "ellipse"

                            Case SwConst.swSketchSegments_e.swSketchPARABOLA
                                skSegTypesString = skSegTypesString & "parabola"

```

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        Case SwConst.swSketchSegments_e.swSketchSPLINE
            skSegTypesString = skSegTypesString & "spline"
        Case SwConst.swSketchSegments_e.swSketchTEXT
            skSegTypesString = skSegTypesString & "text"
        Case Default
            skSegTypesString = skSegTypesString & "unknown"
        End Select
    End If
    If j = UBound(vSkSegments) Then
        skSegTypesString = skSegTypesString & ")"
    Else
        skSegTypesString = skSegTypesString & ", "
    End If
Next j
Debug.Print "    Sketch segment count = " & skSegCount & " " & skSegTypesString
vEdges = skContour.GetEdges()
If IsEmpty(vEdges) Then
    Debug.Print "    No edges."
Else
    For k = LBound(vEdges) To UBound(vEdges)
        Set myEdge = vEdges(k)
        If Not myEdge Is Nothing Then
            Debug.Print "    Edge " & k & ": "
        End If
    Next k
End If
boolstatus = skContour.Select2(False, mySelectData)
If boolstatus = 0 Then
    Debug.Print "    Selection of contour failed."
End If
Stop
End If
Next i
End If
End Sub

```